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Pembrolizumab (Keytruda)

Policy History

Last Review 02/26/2025
Effective: 11/07/2014
Next Review: 07/24/2025

Review History

Definitions

Additional Information

Clinical Policy Bulletin Notes

Number: 0890

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This Clinical Policy Bulletin addresses pembrolizumab (Keytruda) for commercial medical plans. For Medicare criteria, see Medicare Part B Criteria.

Note: Requires Precertification:

Precertification of pembrolizumab (Keytruda) is required of all Aetna participating providers and members in applicable plan designs. For precertification, call (866) 752-7021 or fax (888) 267-3277. For Statement of Medical Necessity (SMN) precertification forms, see Specialty Pharmacy Precertification.

Note: Site of Care Utilization Management Policy applies for pembrolizumab (Keytruda). For information on site of service, see Utilization Management Policy on Site of Care for Specialty Drug Infusions.

I. **Exclusions**

Aetna will not provide coverage for members with *any* of the following exclusions:

A. Pediatric members with TMB-H central nervous system cancers;
B. Members who have experienced disease progression while on programmed death receptor-1 (PD-1) or PD-L1 inhibitor therapy (other than when used as second-line or subsequent therapy for metastatic or unresectable melanoma in combination with ipilimumab following progression on single agent anti-PD-1 immunotherapy).

II. **Criteria for Initial Approval**

Aetna considers pembrolizumab (Keytruda) medically necessary for treatment of the following indications:

A. Ampullary adenocarcinoma - as a single agent for microsatellite instability-high (MSI-H), mismatch repair deficient (dMMR), or tumor mutational burden-high (TMB-H greater than or equal to 10 mutations/megabase [mut/Mb]) ampullary adenocarcinoma;
B. Anal carcinoma - as a single agent for subsequent treatment of metastatic anal carcinoma;
C. Central nervous system (CNS) brain metastases - as a single agent for treatment of CNS brain metastases in members with melanoma or PD-L1 positive non-small cell lung cancer;
D. Cervical cancer - for the treatment of cervical cancer when *any* of the following are met:

1. Persistent, recurrent or metastatic disease in combination with chemotherapy with or without bevacizumab in members whose tumors express PD-L1 (CPS greater than or equal to 1); *or*
2. Recurrent or metastatic disease as single agent subsequent therapy in members whose tumors

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express PD-L1 (CPS ≥1) or are microsatellite instability-high (MSI-H) or mismatch repair deficient (dMMR); *or*

3. Recurrent or metastatic disease and the member has experienced disease progression on or after chemotherapy for tumors that express PD-L1 (defined as a Combined Positive Score (CPS) greater than or equal to 1), as a single agent;
4. FIGO stage III-IVA disease in combination with chemoradiotherapy (CRT);

E. Classical Hodgkin lymphoma - as a single agent or in combination with GVD (gemictabine, vinorelbine, liposomal doxorubicin) or ICE (ifosfamide, carboplatin, etoposide) for treatment of relapsed, refractory, or progressive classical Hodgkin lymphoma;

F. Cutaneous melanoma - for treatment of cutaneous melanoma in *any* of the following settings:

1. For unresectable, recurrent, or metastatic disease as a single agent; *or*
2. As subsequent therapy for disease progression of metastatic or unresectable tumors, as a single agent or in combination with ipilimumab or lenvatinib; *or*
3. As adjuvant treatment following complete lymph node resection or complete resection of stage IIB, IIC, III, or metastatic disease as a single agent; *or*
4. As subsequent or re-induction therapy in combination with trametinib and dabrafenib for metastatic or unresectable disease with a BRAF V600 activating mutation;

G. Cutaneous squamous cell skin carcinoma - as a single agent for treatment of locally advanced, recurrent or metastatic cutaneous squamous cell carcinoma that is not curable by surgery or radiation;

H. Colorectal cancer - as a single agent for the treatment of inoperable, advanced, or metastatic colorectal cancer, including appendiceal carcinoma, for microsatellite instability-high (MSI-H), mismatch repair deficient, (dMMR), or polymerase epsilon/delta (POLE/POLD1) tumors;

I. Epithelial ovarian cancer, fallopian tube cancer, primary peritoneal cancer, carcinosarcoma (malignant mixed Mullerian tumors), clear cell carcinoma of the ovary, mucinous carcinoma of the ovary, grade 1 endometrioid carcinoma, low-grade serous carcinoma -

1. As a single agent for recurrent or persistent microsatellite instability-high (MSI-H) or mismatch repair deficient (dMMR) tumors or tumor mutational burden-high (TMB-H) (tumors greater than or equal to 10 mutations/megabase [mut/Mb]); or
2. In combination with oral cyclophosphamide and bevacizumab for treatment of recurrent or persistent epithelial ovarian cancer, fallopian tube cancer, primary peritoneal cancer, carcinosarcoma (malignant mixed Mullerian tumors), clear cell carcinoma of the ovary, mucinous carcinoma of the ovary, grade 1 endometrioid carcinoma, low-grade serous carcinoma;

J. Endometrial carcinoma -

1. In combination with lenvatinib for treatment of advanced, metastatic, or recurrent endometrial carcinoma when *either* of the following criteria are met:

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carcinoma when *either* of the following criteria are met:

a. The disease is mismatch repair proficient (pMMR); *or*
b. The disease is mismatch repair deficient (dMMR) and has progressed following prior platinum-based chemotherapy;

2. As a single agent for treatment of endometrial carcinoma in members with recurrent unresectable or metastatic microsatellite instability-high (MSI-H), mismatch repair deficient (dMMR), or tumor mutational burden high (TMB-H) (tumors greater than or equal to 10 mutations/megabase [mut/Mb]) tumors;
3. For treatment of endometrial carcinoma in combination with carboplatin and paclitaxel and continued as single agent maintenance therapy (for up to 20 cycles total) in members with stage III-IV or recurrent disease;

K. Esophageal cancer and esophagogastric junction (EGJ) cancer -

1. As a single agent or in combination with platinum and fluoropyrimidine-based chemotherapy for treatment of esophageal and EGJ cancer in members who are surgical candidates when the requested medication will be used to treat microsatellite instability-high (MSI-H) or deficient mismatch repair (dMMR) tumors; *or*
2. In combination with platinum and fluoropyrimidine-based chemotherapy and trastuzumab for treatment of esophageal and EGJ cancer in members who are surgical candidates and have HER2 overexpression positive adenocarcinoma; *or*
3. For treatment of esophageal cancer (including EGJ cancer) in members who are not surgical candidates or have unresectable locally advanced, recurrent, or metastatic disease and when any of the following criteria are met:

a. The requested medication will be used as subsequent therapy as a single agent for microsatellite instability-high (MSI-H), deficient mismatch repair (dMMR) or tumor mutational burden (TMB) high (greater than or equal to 10 mutations/megabase [mut/Mb]) tumors; *or*
b. The requested medication will be used as first line therapy as a single agent or in combination with platinum and fluoropyrimidine-based chemotherapy for microsatellite instability-high or deficient mismatch repair tumors; *or*
c. The requested medication will be used as single agent subsequent therapy with PD-L1 tumor expression by CPS greater than or equal to 10 for squamous cell carcinoma; *or*
d. The requested medication will be used in combination with platinum and fluoropyrimidine-based chemotherapy for squamous cell carcinoma or HER2 overexpression negative adenocarcinoma with PD-L1 tumor expression by CPS greater than or equal to 10; *or*
e. The requested medication will be used in combination with trastuzumab and platinum and fluoropyrimidine-based chemotherapy for HER2 overexpression positive adenocarcinoma;

L. Extranodal NK/T-cell lymphoma - for treatment of extranodal NK/T-cell lymphoma, in members with relapsed or refractory disease;

M. Gastric cancer -

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M. Gastric cancer -

1. For treatment of gastric cancer in members who are not surgical candidates or have unresectable locally advanced, recurrent, or metastatic disease when *any* of the following criteria are met:

    a. The requested medication will be used as subsequent therapy as a single agent for microsatellite instability-high (MSI-H),deficient mismatch repair (dMMR) or tumor mutational burden (TMB) high (greater than or equal to 10 mutations/megabase [mut/Mb]) tumors; *or*
    b. The requested medication will be used as first line therapy as a single agent or in combination with chemotherapy for microsatellite instability-high or deficient mismatch repair tumors; *or*
    c. The requested medication will be used in combination with trastuzumab and chemotherapy for HER2 overexpression positive adenocarcinoma; *or*
    d. The requested medication will be used in combination with chemotherapy for the first-line treatment of HER2-negative adenocarcinoma;
2. For treatment of gastric cancer in members who are surgical candidates when *any* of the following criteria are met:

    a. The requested medication will be used as a single agent or in combination with chemotherapy to treat microsatellite instability-high or deficient mismatch repair tumors; *or*
    b. The requested medication will be used in combination with trastuzumab and chemotherapy to treat early stage or surgically unresectable locoregional disease that is HER2 overexpression positive; *or*
    c. The requested medication will be used in combination with chemotherapy to treat early stage or surgically unresectable locoregional disease that is HER2 overexpression negative with PD-L1 tumor expression by CPS greater than or equal to 10;

N. Gestational trophoblastic neoplasia - as a single agent for treatment of gestational trophoblastic neoplasia for multi-agent chemotherapy-resistant disease when *either* of the following criteria is met:

1. Member has recurrent or progressive intermediate trophoblastic tumor (placental site trophoblastic tumor or epithelioid trophoblastic tumor); *or*
2. Member has high-risk disease;

O. Head and neck squamous cell cancer (HNSCC) - for treatment of members with very advanced head and neck squamous cell carcinoma with mixed subtypes (HNSCC) or nasopharyingeeal cancer when *any* of the following criteria is met:

1. The requested medication will be used as a single agent for first-line treatment in members whose tumors express PD-L1 (CPS greater than or equal to 1), are microsatellite instability-high (MSI-H), mismatch repair deficient (dMMR), or tumor mutational burden high (TMB-H) (tumors greater than or equal to 10 mutations/megabase [mut/Mb]); *or*

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(tumors greater than or equal to 10 mutations/megabase [mut/Mb]); *or*

2. The requested medication will be used as a single agent for subsequent therapy; *or*
3. The requested medication will be used in combination with cetuximab or chemotherapy;

P. Biliary tract cancers -

1. In combination with gemcitabine and cisplatin for locally advanced unresectable, resected gross residual (R2) disease or metastatic biliary tract cancers; *or*
2. As a single agent for unresectable, resected gross residual (R2) disease, or metastatic biliary tract cancers, including intrahepatic and extrahepatic cholangiocarcinoma and gallbladder cancer that is microsatellite instability-high (MSI-H), mismatch repair deficient (dMMR), or tumor mutational burden high (TMB-H) (greater than or equal to 10 mutations/megabase [mut/Mb]);
3. For neoadjuvant treatment of resectable locoregionally advanced gallbladder cancer when either of the following criteria are met:

    a. The requested medication will be used in combination with cisplatin and gemcitabine; or
    b. The requested medication will be used as a single agent and member has microsatellite instability-high (MSI-H) and/or mismatch repair deficient (dMMR) tumors;

Q. Hepatocellular carcinoma - for treatment of hepatocellular carcinoma when *any* of the following criteria are met:

1. The member has disease secondary to hepatitis B and has received prior systemic therapy other than a PD-1/PD-L1- containing regimen and will use the requested medication as a single agent; *or*
2. The member has unresectable or metastatic disease and will use the requested medication as a single agent;

R. Kaposi sarcoma - as a single agent for subsequent treatment of relapsed/refractory endemic or classic Kaposi sarcoma;

S. Merkel cell carcinoma - as a single agent for members with locally advanced, recurrent or metastatic disease;

T. Neuroendocrine tumors and adrenal tumors - for treatment of unresectable, locally advanced or metastatic neuroendocrine and adrenal tumors;

U. Non-small cell lung cancer (NSCLC) -

1. For treatment of recurrent, advanced, or metastatic NSCLC when there are no EGFR exon 19 deletions or L858R mutations or ALK rearrangements (unless testing is not feasible due to insufficient tissue) and *any* of the following criteria are met:

    a. The requested medication will be used as a first-line therapy for PD-L1 positive disease; *or*
    b. The requested medication will be used as single agent or in combination with pemetrexed for maintenance therapy; *or*

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for maintenance therapy; *or*

c. The requested medication will be used in combination with pemetrexed and either carboplatin or cisplatin for nonsquamous cell histology; *or*

d. The requested medication will be used in combination with carboplatin and either paclitaxel or albumin-bound paclitaxel for squamous cell histology; *or*

2. As neoadjuvant treatment when used in combination with platinum containing chemotherapy for resectable (tumors greater than or equal to 4 cm or node positive) NSCLC, and then continued as single agent adjuvant agent therapy after surgery; *or*
3. As a single agent for adjuvant treatment following resection and platinum-based chemotherapy for stage IB (T2a greater than or equal to 4 cm), II, or III NSCLC; *or*
4. For single agent subsequent treatment of PD-L1 positive recurrent, advanced, or metastatic NSCLC;

V. Occult primary cancer - as a single agent for treatment of occult primary cancer in members with microsatellite instability-high (MSI-H) or mismatch repair deficient (dMMR) tumors or tumor mutational burden-high (TMB-H) (greater than or equal to 10 mutations/megabase [mut/Mb]) tumors;

W. Pancreatic adenocarcinoma - as a single agent for treatment of recurrent, locally advanced or metastatic pancreatic adenocarcinoma in members with microsatellite instability-high (MSI-H), mismatch repair deficient (dMMR), or tumor mutational burden high (TMB-H) [greater than or equal to 10 mut/Mb] tumors;

X. Pediatric diffuse high-grade gliomas - as adjuvant treatment for hypermutant tumor pediatric diffuse high-grade glioma or for recurrent or progressive disease;

Y. Primary cutaneous lymphomas - for treatment of primary cutaneous lymphomas when *either* of the following is met:

1. Member has a diagnosis of mycosis fungoides/Sezary syndrome; *or*
2. Member has a diagnosis of relapsed or refractory anaplastic large cell lymphoma (ALCL) and the requested medication will be used as a single agent;

Z. Primary mediastinal large B-cell lymphoma (PMBCL) - as a single agent or in combination with brentuximab vedotin for treatment of primary mediastinal large B-cell lymphoma in members with relapsed or refractory disease;

AA. Prostate cancer - as single agent subsequent therapy for treatment of castration-resistant distant metastatic prostate cancer in members with microsatellite instability-high, mismatch repair deficient, or tumor mutational burden (TMB) greater than or equal to 10 mutations/megabase tumors;

AB. Renal cell carcinoma - for treatment of renal cell carcinoma, when *any* of the following criteria is met:

1. The requested medication will be used as first-line treatment in combination with axitinib or lenvatinib for advanced, relapsed or stage IV disease; *or*
2. The requested medication will be used as subsequent therapy in combination with axitinib or lenvatinib for relapsed or stage IV disease with clear cell histology; *or*

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lenvatinib for relapsed or stage IV disease with clear cell histology; *or*

3. The requested medication will be used as a single agent for relapsed or stage IV disease with non-clear cell histology; *or*
4. The requested medication will be used as a single agent for the adjuvant treatment of members with RCC at intermediate-high or high risk of recurrence following nephrectomy or following nephrectomy and resection of metastatic lesions;

AC. Small bowel adenocarcinoma - as a single agent for treatment of advanced or metastatic small bowel adenocarcinoma for microsatellite instability-high (MSI-H), mismatch repair deficient (dMMR), or polymerase epsilon/delta (POLE/POLD1) tumors;

AD. Soft tissue sarcoma - for treatment of the following types of soft tissue sarcoma when *any* of the following criteria are met:

1. The requested medication will be used as a single agent or in combination with axitinib (Inlyta) for the treatment of alveolar soft part sarcoma (ASPS); *or*
2. The requested medication will be used as a single agent for the treatment of cutaneous angiosarcoma; *or*
3. The requested medication will be used as a single agent for the subsequent treatment of extremity/body wall sarcoma, head/neck sarcoma, retroperitoneal/intra-abdominal sarcoma, and rhabdomyosarcoma;

AE. Solid tumors - as a single agent for the treatment of solid tumors in members with unresectable or metastatic disease that has progressed following prior treatment and who have no satisfactory alternative treatment options when *either* of the following criteria is met:

1. The requested medication will be used for microsatellite instability-high (MSI-H) or mismatch repair deficient (dMMR) solid tumors; *or*
2. The requested medication will be used for tumor mutational burden-high (TMB-H) (greater than or equal to 10 mutations/megabase [mut/Mb]) solid tumors;

AF. Testicular cancer - as a single agent for third-line therapy for treatment of testicular cancer in members with microsatellite instability-high or mismatch repair deficient or tumor mutational burden-high (TMB-H) (greater than or equal to 10 mutations/megabase [mut/Mb]) tumors;

AG. Thymomas and Thymic carcinoma - as a single agent for treatment of thymomas and thymic carcinoma for recurrent, unresectable, locally advanced, or metastatic disease, or as pre or postoperative therapy for in members who cannot tolerate first-line combination regimens;

AH. Thyroid carcinoma (anaplastic) -

1. As a single agent for treatment of metastatic anaplastic thyroid carcinoma for tumor mutational burden-high (TMB-H) (greater than or equal to 10 mutations/megabase [mut/Mb]) tumors; *or*
2. In combination with lenvatinib (Lenvima) for treatment of stage IVC anaplastic thyroid carcinoma;

AI. Thyroid carcinoma (follicular, oncocytic (Hürthle cell), or papillary) - for treatment of

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AI. Thyroid carcinoma (follicular, oncocytic (Hürthle cell), or papillary) - for treatment of unresectable or metastatic follicular, oncocytic (Hürthle cell), or papillary thyroid carcinoma for microsatellite instability-high (MSI-H), mismatch repair deficient (dMMR), or tumor mutational burden-high (TMB-H) (greater than or equal to 10 mutations/megabase [mut/Mb]) tumors not amenable to radioactive iodine therapy;

AJ. Thyroid carcinoma (medullary) - for treatment of unresectable, recurrent, or metastatic medullary thyroid carcinoma for microsatellite instability-high (MSI-H), mismatch repair deficient (dMMR), or tumor mutational burden-high (TMB-H) (greater than or equal to 10 mutations/megabase[mut/Mb]) tumors;

AK. Breast cancer -

1. For treatment in members with no response to preoperative systemic therapy or for recurrent unresectable or metastatic triple-negative breast cancer (TNBC) when *all* of the following criteria are met:

    a. The diagnosis of triple-negative breast cancer is confirmed by the cancer cells testing negative for *all* of the following receptors:

        i. Human epidermal growth factor receptor 2 (HER-2); *and*
        ii. Estrogen; *and*
        iii. Progesterone; *and*

    b. Tumor must express PD-L1; *and*
    c. The requested medication will be used as a single agent or in combination with chemotherapy;

2. For treatment of high-risk early-stage triple-negative breast cancer (TNBC) when *all* of the following criteria are met:

    a. The diagnosis of triple-negative breast cancer is confirmed by the cancer cells testing negative for *all* of the following receptors:

        i. Human epidermal growth factor receptor 2 (HER-2);
        ii. Estrogen;
        iii. Progesterone; *and*

    b. The requested medication will be used as *either*:

        i. Neoadjuvant treatment in combination with chemotherapy; *or*
        ii. Continued adjuvant treatment after surgery, as a single agent;

AL. Urothelial carcinoma -

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1. As a single agent for treatment of urothelial carcinoma when used in *any* of the following subtypes:
   a. Urothelial carcinoma of the bladder in *any* of the following settings:

      i. First line therapy for locally advanced or metastatic disease in members who are not eligible for any platinum containing chemotherapy; or
      ii. Subsequent therapy; *or*
      iii. For the treatment of members with high risk, non-muscle invasive bladder cancer (NMIBC) with carcinoma in situ (CIS) when disease is Bacillus Calmette Guerin (BCG) unresponsive, and member will not undergo cystectomy; *or*

   b. Primary carcinoma of the urethra with locally advanced, recurrent or metastatic disease post-platinum or other chemotherapy or for members who are not eligible for any platinum-containing chemotherapy; or
   c. Urothelial carcinoma of the upper genitourinary tract or urothelial carcinoma of the prostate with metastatic disease post-platinum or other chemotherapy or for members who are not eligible for any platinum-containing chemotherapy;
2. For the treatment of stage II, recurrent, locally advanced or metastatic urothelial carcinoma in combination with enfortumab vedotin-ejfv;

AM. Uveal melanoma - as a single agent for treatment of unresectable or metastatic uveal melanoma;

AN. Vulvar cancer - as a single agent for subsequent treatment of advanced, recurrent or metastatic disease in members with vulvar cancer when *either* of the following criteria is met:

1. Member has microsatellite instability-high (MSI-H), mismatch repair deficient (dMMR), or tumor mutational burden high (TMB-H) (tumors greater than or equal to 10 mutations/megabase [mut/Mb]) tumors; *or*
2. Member has experienced disease progression on or after chemotherapy and whose tumor expresses PD-L1 (Combined Positive Score greater than or equal to 1);

AO. Small cell lung cancer - as a single agent for subsequent therapy of relapsed or progressive disease;

AP. Vaginal cancer - for treatment of vaginal cancer when *any* of the following criteria are met:

1. The requested medication will be used in combination with cisplatin or carboplatin, paclitaxel, and with or without bevacizumab for recurrent or metastatic disease; *or*
2. The requested medication will be used as single agent subsequent treatment for recurrent or metastatic disease that is PD-L1 positive or disease with microsatellite instability-high (MSI-H) or mismatch repair deficient (dMMR) tumors; *or*
3. The requested medication will be used as subsequent treatment for unresectable or metastatic tumor mutational burden-high (TMB-H greater than or equal to 10 mut/Mb) tumors;

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AQ. Pleural or Peritoneal Mesothelioma - for first-line treatment of pleural or peritoneal mesothelioma, including pericardial mesothelioma and tunica vaginalis testis mesothelioma, when used in combination with pemetrexed and platinum chemotherapy;

AR. Histologic (Richter) transformation to diffuse large B-cell lymphoma - for treatment of histologic (Richter) transformation to diffuse large B-cell lymphoma as a single agent or in combination with ibrutinib.

Aetna considers all other indications as experimental, investigational, or unproven.

III. **Continuation of Therapy**

Aetna considers continuation of pembrolizumab (Keytruda) therapy medically necessary for the following indications:

A. Adjuvant treatment of melanoma, high-risk early-stage TNBC, RCC, or NSCLC - for continued treatment (up to 12 months total) in members requesting reauthorization for adjuvant treatment of cutaneous melanoma, high-risk early-stage TNBC, RCC, or NSCLC who have not experienced disease recurrence or an unacceptable toxicity;

B. Non-small cell lung cancer (NSCLC), head and neck squamous cell cancer (HNSCC), classic Hodgkin lymphoma (cHL), primary mediastinal large B-cell lymphoma (PMBCL), microsatellite instability-high (MSI-H) or mismatch repair deficient (dMMR) cancers, gastric cancer, esophageal cancer, cervical cancer, hepatocellular carcinoma (HCC), Merkel cell carcinoma (MCC), renal cell carcinoma (RCC), endometrial carcinoma, cutaneous squamous cell carcinoma (cSCC), recurrent unresectable or metastatic triple-negative breast cancer (TNBC), tumor mutational burden-high (TMB-H) cancer, and biliary tract cancer, pleural or peritoneal mesothelioma, histologic (Richter) transformation to diffuse large B-cell lymphoma - for continued treatment (up to 24 months of continuous use) in members requesting reauthorization for NSCLC, HNSCC, cHL, PMBCL, MSI-H or dMMR cancers, gastric cancer, esophageal cancer, cervical cancer, HCC, MCC, RCC, endometrial carcinoma, cSCC, recurrent unresectable or metastatic TNBC, TMB-H, biliary tract cancers, pleural or peritoneal mesothelioma, including pericardial mesothelioma and tunica vaginalis testis mesothelioma subtypes, and histologic (Richter) transformation to diffuse large B-cell lymphoma who have not experienced disease progression or unacceptable toxicity;

C. Urothelial carcinoma -

1. For continued treatment in members requesting reauthorization for urothelial carcinoma when the requested medication is used in combination with enfortumab vedotin-ejfv who have not experienced disease progression or an unacceptable toxicity; *or*
2. For continued treatment (up to 24 months of continuous use) in members requesting reauthorization for urothelial carcinoma when *both* of the following criteria are met:

   a. Member has not experienced disease progression or unacceptable toxicity; *and*

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b. For high-risk BCG-unresponsive non-muscle invasive bladder cancer only: disease is not persistent or recurrent;

D. All other indications - for continued treatment in members requesting reauthorization for an indication listed in Section II who have not experienced disease progression or an unacceptable toxicity.

IV. **Related Policies**

- CPB 0024 - Interleukin-2 (Aldesleukin, Proleukin, IL-2)
- CPB 0313 - Trastuzumab (Herceptin and biosimilars), Trastuzumab and Hyaluronidase-oysk (Herceptin Hylecta)
- CPB 0404 - Interferons
- CPB 0557 - Cancer Vaccines
- CPB 0715 - Pharmacogenetic and Pharmacodynamic Testing
- CPB 0815 - Ipilimumab (Yervoy)

Dosage and Administration

Note: Approvals may be subject to dosing limits in accordance with FDA-approved labeling, accepted compendia, and/or evidence-based practice guidelines. See Medical Specialty Medication Quantity Limits for more information.

Below includes dosing recommendations per the FDA-approved prescribing information for pembrolizumab (Keytruda).

Pembrolizumab is available as Keytruda as 100 mg/4 mL (25 mg/mL) single-dose vials for intravenous infusion.

- Monotherapy:
    - Adult individuals with unresectable or metastatic melanoma: 200 mg every 3 weeks or 400 mg every 6 weeks until disease progression or unacceptable toxicity
    - Adjuvant treatment of adult patients with melanoma, NSCLC, or RCC: 200 mg every 3 weeks or 400 mg every 6 weeks until disease recurrence, unacceptable toxicity, or up to 12 months
    - Adult individuals with NSCLC, HNSCC, cHL, PMBCL, locally advanced or metastatic urothelial carcinoma, MSI-H or dMMR cancer, MSI-H or dMMR CRC, MSI-H or dMMR endometrial carcinoma, esophageal cancer, cervical cancer, HCC, MCC, TMB-H cancer, or cSCC: 200 mg every 3 weeks or 400 mg every 6 weeks until disease progression, unacceptable toxicity, or up to 24 months
    - Adult individuals with high-risk BCG-unresponsive NMIBC: 200 mg every 3 weeks or 400

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mg every 6 weeks until persistent or recurrent high-risk NMIBC, disease progression, unacceptable toxicity, or up to 24 months

- Pediatric individuals with cHL, PMBCL, MSI-H or dMMR cancer, MCC, or TMB-H cancer: 2 mg/kg every 3 weeks (up to a maximum of 200 mg) until disease progression, unacceptable toxicity, or up to 24 months
- Pediatric individuals (12 years and older) for adjuvant treatment of melanoma: 2 mg/kg every 3 weeks (up to a maximum of 200 mg) until diseae recurrence, unaccpetable toxicity, or up to 12 months

- Combination therapy:

  - Adult individuals with resectable NSCLC; neoadjuvant treatment in combination with chemotherapy for 12 weeks or until disease progression that precludes definitive surgery or unacceptable toxicity, followed by adjuvant treatment with Keytruda as a single agent after surgery for 39 weeks or until disease recurrence or unacceptable toxicity.
  - Adult individuals with NSCLC, MPM*, HNSCC, HER2-negative gastric cancr, esophageal cancer, BTC: 200 mg every 3 weeks or 400 mg every 6 weeks; administer Keytruda prior to chemotherapy when given on the same day; until disease progression, unaccpetable toxicity, or up to 24 months
  - Adult individuals with locally advanced or metastatic urothelial cancer: 200 mg every 3 weeks or 400 mg every 6 weeks; administer Keytruda after enfortumab vedotin when given on the same day; until disease progression, unacceptable toxicity, or up to 24 months
  - Adult individuals with HER2-positive gastric cancer: 200 mg every 3 weeks or 400 mg every 6 weeks; administer Keytruda prior to trastuzumab and chemotherapy when given on the same day; until disease progression, unacceptable toxicity, or up to 24 months
  - Adult individuals with cervical cancer: 200 mg every 3 weeks or 400 mg every 6 weeks; administer Keytruda prior to chemotherapy with or without bevacizumab when given on the same day; until disease progression, unacceptable toxicity, or for Keytruda, up to 24 months
  - Adult individuals with RCC: 200 mg every 3 weeks or 400 mg every 6 weeks; administer Keytruda in combination with axitinib 5 mg orally twice daily or administer Keytruda in combination with lenvatinib 20 mg orally once daily; until disease progression, unacceptable toxicity, or for Keytruda, up to 24 months
  - Adult individuals with endometrial carcinoma: 200 mg every 3 weeks or 400 mg every 6 weeks; administer Keytruda in combination with lenvatinib 20 mg orally once daily; until disease progression, unacceptable toxicity, or for for Keytruda, up to 24 months
  - Adult individuals with high-risk early-stage TNBC: 200 mg every 3 weeks or 400 mg every 6 weeks; administer Keytruda prior to chemotherapy when given on the same day; neoadjuvant treatment in combination with chemotherapy for 24 weeks (8 doses of 200 mg every 3 weeks or 4 dosese of 400 mg every 6 weeks) or until disease progression or unacceptable toxicity, followed by adjuvant treatment with Keytruda as a single agent for up to 27 weeks (9 doses of 200 mg every 3 weeks or 5 doses of 400 mg every 6 weeks) or until disease recurrence or unacceptable toxicity

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or unacceptable toxicity.
- Adult individuals with locally recurrent unresectable or metastatic TNBC: 200 mg every 3 weeks or 400 mg every 6 weeks; administer Keytruda prior to chemotherapy when given on the same day; until disease progression, unacceptable toxicity, or up to 24 months

*MPM = malignant pleural mesothelioma

Keytruda is administered as a 30-minute intravenous infusion. Refer to the full prescribing information for Keytruda for the agents administered in combination with Keytruda for recommended dosing information as appropriate. When axitinib is used in combination with Keytruda, dose escalation of axitinib above the initial 5 mg dose may be considered at intervals of six weeks or longer. Individuals who experience disease progression or unacceptable toxicity related Keytruda with neoadjuvant treatment in combination with chemotherapy should not receive adjuvant single agent Keytruda.

Refer to full prescribing information for Keytruda for dosage modifications and preparation and administration instructions.

Source: Merck, 2024

Experimental, Investigational, or Unproven

Aetna considers pembrolizumab (Keytruda) experimental, investigational, or unproven for all other indications including the following (not an all-inclusive list):

- Central nervous system tumors (other than CNS brain metastases in melanoma or PD-L1 positive non-small cell lung cancer)
- Chronic lymphocytic leukemia / Small lymphocytic lymphoma
- Lacrimal gland cancer
- Multiple myeloma
- Primary cutaneous CD30+ T-cell lymphoproliferative disorders.

Aetna considers combined pembrolizumab (Keytruda) and acalabrutinib experimental, investigational, or unproven for the treatment of bladder cancer.

Aetna considers combined pembrolizumab (Keytruda) and radiotherapy experimental, investigational, or unproven for the treatment of triple-negative breast cancer.

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CPT Codes / HCPCS Codes / ICD-10 Codes

Other CPT codes related to the CPB:

Code	Code Description
***Tumor mutational burden-high (TMB-H), Polymerase delta (POLD1) tumors* - no specific code**	
77427 - 77470	Radiation treatment management
81210	BRAF (B-Raf proto-oncogene, serine/threonine kinase) (eg, colon cancer, melanoma), gene analysis, V600 variants
81235	EGFR (epidermal growth factor receptor) (eg, non-small cell lung cancer) gene analysis, common variants (eg, exon 19 LREA deletion, L858R, T790M, G719A, G719S, L861Q)
81301	Microsatellite instability analysis (eg, hereditary non-polyposis colorectal cancer, Lynch syndrome) of markers for mismatch repair deficiency
88341	Immunohistochemistry or immunocytochemistry, per specimen; each additional single antibody stain procedure (List separately in addition to code for primary procedure)
88342	Immunohistochemistry or immunocytochemistry, per specimen; initial single antibody stain procedure
88360	Morphometric analysis, tumor immunohistochemistry (eg, Her-2/neu, estrogen receptor/progesterone receptor), quantitative or semiquantitative, per specimen, each single antibody stain procedure; manual
88361	using computer-assisted technology
96413	Chemotherapy administration, IV infusion technique; up to 1 hour, single or initial substance/drug
96415	each additional hour (list in addition to code for primary procedure)
96401 – 96411, 96416 – 96417	Injection and intravenous infusion chemotherapy and other highly complex drug or highly complex biologic agent administration
96420 - 96425	Intra-arterial chemotherapy and other highly complex drug or highly complex biologic agent administration
HCPCS codes covered if selection criteria are met:	
J9271	Injection, pembrolizumab, 1 mg
Other HCPCS codes related to the CPB:	
***Dacomitinib, Axitinib, Crizotinib, Entrectinib, Lenvatinib, Ceritinib, Trametinib, Dabrafenib, Sorafenib, ibrutinib* - no specific code**	
J8530	Cyclophosphamide; oral, 25 mg
J8560	Etoposide; oral, 50 mg
J8565	Gefitinib, oral, 250 mg
J9022	Injection, atezolizumab, 10 mg

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J9035	Injection, bevacizumab, 10 mg
J9042	Injection, brentuximab vedotin, 1 mg

Code	Code Description
J9045	Injection, carboplatin, 50 mg
J9055	Injection, cetuximab, 10 mg
J9060	Injection, cisplatin, powder or solution, 10 mg
J9177	Injection, enfortumab vedotin-ejfv, 0.25 mg
J9181	Injection, etoposide, 10 mg
J9201	Injection, gemcitabine hydrochloride, not otherwise specified, 200 mg
J9208	Injection, ifosfamide, 1 gram
J9228	Injection, ipilimumab, 1 mg
J9258	Injection, paclitaxel protein-bound particles (teva) not therapeutically equivalent to j9264, 1 mg
J9259	Injection, paclitaxel protein-bound particles (american regent) not therapeutically equivalent to j9264, 1 mg
J9263	Injection, oxaliplatin, 0.5 mg
J9264	Injection, paclitaxel protein-bound particles, 1 mg
J9267	Injection, paclitaxel, 1 mg
J9292	Injection, pemetrexed (avyxa), not therapeutically equivalent to j9305, 10 mg
J9294	Injection, pemetrexed (hospira) not therapeutically equivalent to j9305, 10 mg
J9296	Injection, pemetrexed (accord) not therapeutically equivalent to j9305, 10 mg
J9297	Injection, pemetrexed (sandoz), not therapeutically equivalent to j9305, 10 mg
J9299	Injection, nivolumab, 1 mg
J9304	Injection, pemetrexed (pemfexy), 10 mg
J9305	Injection, pemetrexed, 10 mg
J9314	Injection, pemetrexed (teva) not therapeutically equivalent to J9305, 10 mg
J9322	Injection, pemetrexed (bluepoint) not therapeutically equivalent to j9305, 10 mg
J9323	Injection, pemetrexed ditromethamine, 10 mg
J9324	Injection, pemetrexed (pemrydi rtu), 10 mg
J9354	Injection, ado-trastuzumab emtansine, 1 mg
J9355	Injection, trastuzumab, excludes biosimilar, 10 mg
J9356	Injection, trastuzumab, 10 mg and Hyaluronidase-oysk
J9358	Injection, fam-trastuzumab deruxtecan-nxki, 1 mg
J9390	Injection, vinorelbine tartrate, 10 mg

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Q2049	Injection, doxorubicin hydrochloride, liposomal, imported lipodox, 10 mg

Code	Code Description
Q2050	Injection, doxorubicin hydrochloride, liposomal, not otherwise specified, 10 mg
Q5112	Injection, trastuzumab-dttb, biosimilar, (Ontruzant), 10 mg
Q5113	Injection, trastuzumab-pkrb, biosimilar, (Herzuma), 10 mg
Q5114	Injection, Trastuzumab-dkst, biosimilar, (Ogivri), 10 mg
Q5116	Injection, trastuzumab-qyyp, biosimilar, (trazimera), 10 mg
Q5117	Injection, trastuzumab-anns, biosimilar, (kanjinti), 10 mg
Q5146	Injection, trastuzumab-strf (hercessi), biosimilar, 10 mg
ICD-10 codes covered if selection criteria are met:	
C00.0 - C14.8	Malignant neoplasm of lip, oral cavity, and pharynx
C15.3 - C15.9	Malignant neoplasm of esophagus [Gastroesophageal junction adenocarcinoma]
C16.0 - C16.9	Malignant neoplasm of stomach
C17.0 - C17.9	Malignant neoplasm of small intestine including duodenum [small bowel adenocarcinoma]
C18.0 - C20	Malignant neoplasm of colon and rectum
C21.0 - C21.8	Malignant neoplasm of anus and anal canal [anal adenocarcinoma]
C22.0	Liver cell carcinoma [hepatocellular carcinoma]
C22.1	Intrahepatic bile duct carcinoma [cholangiocarcinoma]
C23	Malignant neoplasm of gallbladder
C24.0 - C24.9	Malignant neoplasm of other and unspecified parts of biliary tract
C25.0 - C25.9	Malignant neoplasm of pancreas
C26.0	Malignant neoplasm of intestinal tract, part unspecified
C30.0 - C30.1	Malignant neoplasm of nasal cavity and middle ear
C31.0 - C31.9	Malignant neoplasms of accessory sinuses
C32.0 - C32.9	Malignant neoplasm of larynx
C34.00 - C34.92	Malignant neoplasm of bronchus and lung
C37	Malignant neoplasm of thymus
C40.00 - C41.9	Malignant neoplasm of bone and articular cartilage [osteosarcoma, chordoma, ewing sarcoma, osteosarcoma] [tumor mutational burden-high disease]
C43.0 - C43.9	Malignant melanoma of skin [unresectable, recurrent]
C44.00 - C44.99	Other and unspecified malignant neoplasm of skin [cutaneous angiosarcoma] [recurrent, advanced, or metastatic and tumor mutational burden-high (TMB-H)]
C4A.0 - C4A.9	Merkel cell carcinoma

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C45.0 - C45.9	Mesothelioma
C46.0 – C46.9	Kaposi's sarcoma

Code	Code Description
C48.1 - C48.2	Malignant neoplasm of peritoneum [primary peritoneal cancer]
C49.0 - C49.9	Malignant neoplasm of other connective and soft tissue [alveolar soft part sarcoma] [recurrent, advanced, or metastatic and tumor mutational burden-high (TMB-H)] [extremity/body wall sarcoma, head/neck sarcoma, retroperitoneal/intra-abdominal sarcoma and rhabdomyosarcoma]
C49.A0 - C49.A9	Gastrointestinal stromal tumor
C50.011 - C50.929	Malignant neoplasm of breast [tumor mutational burden-high disease] [triple-negative breast cancer (TNBC)]
C51.0 - C51.9	Malignant neoplasm of vulva
C52	Malignant neoplasm of vagina
C53.0 - C53.9	Malignant neoplasm of cervix uteri
C54.0 – C54.9	Malignant neoplasm of corpus uteri
C55	Malignant neoplasm of uterus, part unspecified [uterine sarcoma]
C56.1 - C56.9	Malignant neoplasm of ovary
C57.00 - C57.02	Malignant neoplasm of fallopian tube
C58	Malignant neoplasm of placenta [gestational trophoblastic neoplasia]
C60.1 - C60.9	Malignant neoplasm of penis [tumor mutational burden-high disease]
C61	Malignant neoplasm of prostate
C62.90 - C62.92	Malignant neoplasm of testis, unspecified whether descended or undescended
C64.1 - C64.9	Malignant neoplasm of kidney, except renal pelvis
C65.1 - C65.9	Malignant neoplasm of renal pelvis
C66.1 - C66.9	Malignant neoplasm of ureter
C67.0 - C67.9	Malignant neoplasm of bladder
C68.0	Malignant neoplasm of urethra
C69.40 - C69.42	Malignant neoplasm of ciliary body [uveal melanoma]
C71.0 - C71.9	Malignant neoplasm of brain [pediatric diffuse high-grade gliomas]
C73	Malignant neoplasm of thyroid gland [Anaplastic]
C74.00 - C74.92	Malignant neoplasm of adrenal gland
C76.0	Malignant neoplasm of head, face and neck
C7A.010 - C7A.8	Malignant neuroendocrine tumors
C7B.8	Other secondary neuroendocrine tumors

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C80.1	Malignant (primary) neoplasm, unspecified [occult primary cancer]
C81.10 - C81.79	Hodgkin lymphoma

Code	Code Description
C83.30 – C83.398	Diffuse large B-cell lymphoma [histologic (Richter) transformation]
C84.00 - C84.09	Mycosis fungoides [non-Hodgkin's lymphoma] [not covered for Stage III mycosis fungoides]
C84.10 - C84.19	Sezary disease [non-Hodgkin's lymphoma] [not covered for Stage IV Sezary syndrome]
C84.70 – C84.7A	Anaplastic large cell lymphoma, ALK-negative [relapsed or refractory]
C84.90 – C84.99	Mature T/NK-cell lymphomas, unspecified
C84.A0 - C84.A9	Cutaneous T-cell lymphoma, unspecified [primary cutaneous lymphomas]
C84.Z0 - C84.Z9	Other mature T/NK-cell lymphomas
C85.20 - C85.29	Mediastinal (thymic) large B-cell lymphoma
C86.00 - C86.01	Extranodal NK/T-cell lymphoma, nasal type
C86.60 - C86.61	Primary cutaneous CD30-positive T-cell proliferations
D03.0 - D03.9	Melanoma in situ
D09.0	Carcinoma in situ of bladder
ICD-10 codes not covered for indications listed in the CPB:	
C22.0 - C22.9	Malignant neoplasm of liver and intrahepatic bile ducts
C69.50 - C69.52	Malignant neoplasm of lacrimal gland and duct
C90.00 - C90.02	Multiple myeloma
C91.10 - C91.12	Chronic lymphocytic leukemia of B-cell type [small lymphocytic lymphoma]
C92.10 - C92.12	Chronic myeloid leukemia, BCR/ABL-positive

Background

U.S. Food and Drug Administration (FDA)-Approved Indications

- *Melanoma*

  Keytruda (pembrolizumab) is indicated for the treatment of patients with unresectable or metastatic melanoma.

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Keytruda is indicated for the adjuvant treatment of adult and pediatric (12 years and older) patients with Stage IIB, IIC, or III melanoma following complete resection.

- *Non-Small Cell Lung Cancer*

  Keytruda, in combination with pemetrexed and platinum chemotherapy, is indicated for the first-line treatment of patients with metastatic nonsquamous non-small cell lung cancer (NSCLC), with no EGFR or ALK genomic tumor aberrations.

  Keytruda, in combination with carboplatin and either paclitaxel or paclitaxel protein-bound, is indicated for the first-line treatment of patients with metastatic squamous NSCLC.

  Keytruda, as a single agent, is indicated for the first-line treatment of patients with NSCLC expressing PD-L1 [Tumor Proportion Score (TPS) ≥1%] as determined by an FDA-approved test, with no EGFR or ALK genomic tumor aberrations, and is:

  - stage III where patients are not candidates for surgical resection or definitive chemoradiation, or
  - metastatic.

  Keytruda, as a single agent, is indicated for the treatment of patients with metastatic NSCLC whose tumors express PD-L1 (TPS ≥1%) as determined by an FDA approved test, with disease progression on or after platinum-containing chemotherapy. Patients with EGFR or ALK genomic tumor aberrations should have disease progression on FDA-approved therapy for these aberrations prior to receiving Keytruda.

  Keytruda, in combination with platinum-containing chemotherapy, is indicated for the treatment of patients with resectable (tumors ≥ 4 cm or node positive) NSCLC as neoadjuvant treatment, and then continued as a single agent as adjuvant treatment after surgery.

  Keytruda, as a single agent, is indicated for adjuvant treatment following resection and platinum-based chemotherapy for adult patients with stage 1B (T2a ≥ 4cm), II, or IIIA NSCLC

- *Head and Neck Squamous Cell Cancer*

  Keytruda, in combination with platinum and fluorouracil (FU), is indicated for the first-line treatment of patients with metastatic or with unresectable, recurrent head and neck squamous cell carcinoma (HNSCC).

  Keytruda, as a single agent, is indicated for the first line treatment of patients with metastatic or with unresectable, recurrent HNSCC whose tumors express PD-L1 [Combined Positive Score (CPS) ≥1] as determined by an FDA-approved test.

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Keytruda, as a single agent, is indicated for the treatment of patients with recurrent or metastatic HNSCC with disease progression on or after platinum-containing chemotherapy.

- *Classical Hodgkin Lymphoma*

  Keytruda is indicated for the treatment of adult patients with relapsed or refractory classical Hodgkin lymphoma (cHL).

  Keytruda is indicated for the treatment of pediatric patients with refractory cHL, or cHL that has relapsed after 2 or more prior lines of therapy.

- *Primary Mediastinal Large B-cell Lymphoma*

  Keytruda is indicated for the treatment of adult and pediatric patients with refractory primary mediastinal large B-cell lymphoma (PMBCL), or who have relapsed after 2 or more prior lines of therapy.

  Limitations of Use:

  Keytruda is not recommended for treatment of patients with PMBCL who require urgent cytoreductive therapy.

- *Urothelial Carcinoma*

  Keytruda, as a single agent, is indicated for the treatment of patients with locally advanced or metastatic urothelial carcinoma:

  - who are not eligible for any platinum-containing chemotherapy, or
  - who have disease progression during or following platinum-containing chemotherapy or within 12 months of neoadjuvant or adjuvant treatment with platinum-containing chemotherapy.

  Keytruda, as a single agent, is indicated for the treatment of patients with Bacillus Calmette-Guerin (BCG)-unresponsive, high-risk, non-muscle invasive bladder cancer (NMIBC) with carcinoma in situ (CIS) with or without papillary tumors who are ineligible for or have elected not to undergo cystectomy.

  Keytruda, in combination with enfortumab vedotin, is indicated for the treatment of adult patients with locally advanced or metastatic urothelial carcinoma.

- *Microsatellite Instability-High Cancer or Mismatch Repair Deficient Cancer*

  Keytruda is indicated for the treatment of adult and pediatric patients with unresectable or

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Keytruda is indicated for the treatment of adult and pediatric patients with unresectable or metastatic, microsatellite instability-high (MSI-H) or mismatch repair deficient (dMMR) solid

tumors, as determined by an FDA-approved test, that have progressed following prior treatment and who have no satisfactory alternative treatment options.

- *Microsatellite Instability-High or Mismatch Repair Deficient Colorectal Cancer (CRC)*

  Keytruda is indicated for the treatment of patients with unresectable or metastatic MSI-H or dMMR colorectal cancer (CRC) as determined by an FDA-approved test.

- *Gastric Cancer*

  Keytruda, in combination with trastuzumab, fluoropyrimidine- and platinum-containing chemotherapy, is indicated for the first-line treatment of adults with locally advanced unresectable or metastatic HER2-positive gastric or gastroesophageal junction (GEJ) adenocarcinoma whose tumors express PD-L1 (CPS ≥ 1) as determined by an FDA-approved test.

  Keytruda, in combination with fluoropyrimidine- and platinum-containing chemotherapy is indicated for the first-line treatment of adults with locally advanced unresectable or metastatic HER2-negative gastric or gastroesophageal junction (GEJ) adenocarcinoma.

- *Esophageal Cancer*

  Keytruda is indicated for the treatment of patients with locally advanced or metastatic esophageal or gastroesophageal junction (GEJ) (tumors with epicenter 1 to 5 centimeters above the GEJ) carcinoma that is not amenable to surgical resection or definitive chemoradiation *either*:

  - In combination with platinum- and fluoropyrimidine-based chemotherapy, or
  - As a single agent after one or more prior lines of systemic therapy for patients with tumors of squamous cell histology that express PD-L1 (CPS ≥ 10) as determined by an FDA-approved test.

- *Cervical Cancer*

  Keytruda in combination with chemotherapy, with or without bevacizumab, is indicated for the treatment of patients with persistent, recurrent, or metastatic cervical cancer whose tumors express PD-L1 (CPS ≥1) as determined by an FDA-approved test.

  Keytruda, as a single agent, is indicated for the treatment of patients with recurrent or metastatic cervical cancer with disease progression on or after chemotherapy whose tumor express PD-L1 (CPS ≥ 1) as determined by an FDA-approved test.

  Keytruda, in combination with chemoradiotherapy (CRT), is indicated for the treatment of patients

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Keytruda, in combination with chemoradiotherapy (CRT), is indicated for the treatment of patients with FIGO 2014 Stage III-IVA cervical cancer.

- *Hepatocellular Carcinoma*

  Keytruda is indicated for the treatment of patients with hepatocellular carcinoma (HCC) secondary to hepatitis B who have received prior systemic therapy other than a PD-1/PD-L1-containing regimen.

- *Biliary Tract Cancer*

  Keytruda, in combination with gemcitabine and cisplatin is indicated for the treatment of patients with locally advanced unresectable or metastatic biliary tract cancer (BTC),

- *Merkel Cell Carcinoma*

  Keytruda is indicated for the treatment of adult and pediatric patients with recurrent locally advanced or metastatic Merkel cell carcinoma (MCC).

- *Renal Cell Carcinoma*

  Keytruda, in combination with axitinib, is indicated for the first-line treatment of adult patients with advanced renal cell carcinoma (RCC).

  Keytruda, in combination with lenvatinib, is indicated for the first-line treatment of adult patients with advanced RCC.

  Keytruda is indicated for the adjuvant treatment of patients with RCC at intermediate-high or high risk of recurrence following nephrectomy, or following nephrectomy and resection of metastatic lesions.

- *Endometrial Carcinoma*

  Keytruda, in combination with carboplatin and paclitaxel, followed by Keytruda as a single agent, for the treatment of adult patients with primary advanced or recurrent endometrial carcinoma.

  Keytruda, in combination with lenvatinib, is indicated for the treatment of patients with advanced endometrial carcinoma that is mismatch repair proficient (pMMR) as determined by an FDA-approved test or not MSI-H, who have disease progression following prior systemic therapy in any setting and are not candidates for curative surgery or radiation.

  Keytruda, as a single agent, is indicated for the treatment of patients with advanced endometrial carcinoma that is MSI-H or dMMR, as determined by an FDA-approved test, who

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endometrial carcinoma that is MSI-H or dMMR, as determined by an FDA-approved test, who have disease progression following prior systemic therapy in any setting and are not candidates for curative surgery or radiation.

- *Tumor Mutational Burden-High Cancer*

  Keytruda is indicated for the treatment of adult and pediatric patients with unresectable or metastatic tumor mutational burden-high (TMB-H) [≥10 mutations/megabase (mut/Mb)] solid tumors, as determined by an FDA-approved test, that have progressed following prior treatment and who have no satisfactory alternative treatment options.

  Limitations of use:

  The safety and effectiveness of Keytruda in pediatric patients with TMB-H central nervous system cancers have not been established.

- *Cutaneous Squamous Cell Carcinoma*

  Keytruda is indicated for the treatment of patients with recurrent or metastatic cutaneous squamous cell carcinoma (cSCC) or locally advanced cSCC that is not curable by surgery or radiation.

- *Triple-Negative Breast Cancer*

  Keytruda in combination with chemotherapy, is indicated for the treatment of patients with locally recurrent unresectable or metastatic triple-negative breast cancer (TNBC) whose tumors express PD-L1 [Combined Positive Score (CPS) ≥10] as determined by an FDA approved test.

  Keytruda is indicated for the treatment of patients with high-risk early-stage triple-negative breast cancer (TNBC) in combination with chemotherapy as neoadjuvant treatment, and then continued as a single agent as adjuvant treatment after surgery.

- *Adult Classical Hodgkin Lymphoma and Adult Primary Mediastinal Large B-Cell Lymphoma: Additional Dosing Regimen of 400mg Every 6 Weeks*

  Keytruda is indicated for use at an additional recommended dosage of 400mg every 6 weeks for classical Hodgkin lymphoma and primary mediastinal large B-cell lymphoma in adults.

- *Malignant Pleural Mesothelioma*

  Keytruda, in combination with pemetrexed and platinum chemotherapy, is indicated for the first-line treatment of adult patients with unresectable advanced or metastatic malignant pleural mesothelioma.

Compendial Uses

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Compendial Uses

- Cutaneous melanoma
- Non-small cell lung cancer
- Head and neck squamous cell cancer
- Classical Hodgkin Lymphoma
- Urothelial carcinoma
    - Bladder cancer
    - Primary carcinoma of the urethra
    - Upper genitourinary tract tumors
    - Urothelial carcinoma of the prostate
- Anaplastic thyroid carcinoma
- Follicular, oncocytic (Hürthle cell), or papillary thyroid carcinoma
- Medullary thyroid carcinoma
- Colorectal cancer
- Small bowel adenocarcinoma
- Gastric cancer and esophagogastric junction cancer
- Esophageal cancer
- Cervical cancer
- Epithelial ovarian cancer/fallopian tube cancer/primary peritoneal cancer
- Uveal melanoma
- Testicular cancer
- Endometrial carcinoma
- Anal carcinoma
- Central nervous system (CNS) brain metastases
- Primary mediastinal large B-cell lymphoma
- Pancreatic adenocarcinoma
- Biliary tract cancers
- Vulvar cancer
- Renal cell carcinoma
- Thymomas and Thymic carcinoma
- Primary cutaneous lymphomas
    - Mycosis fungoides/Sezary syndrome
    - Anaplastic large cell lymphoma (ALCL)
- Extranodal NK/T-cell lymphoma
- Gestational trophoblastic neoplasia
- Neuroendocrine and adrenal tumors

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    - Well differentiated Grade 3 tumors
    - Adrenal gland tumors
    - Extrapulmonary poorly differentiated/large or small cell carcinoma
    - Adrenocortical carcinoma

- Soft tissue sarcomas:

    - Alveolar soft part sarcoma (ASPS)
    - Cutaneous angiosarcoma
    - Extremity/body wall sarcoma
    - Head/neck sarcoma
    - Retroperitoneal/intra-abdominal sarcoma
    - Rhabdomyosarcoma

- Occult primary cancer
- Prostate cancer
- Bone cancer:

    - Chondrosarcoma
    - Chordoma
    - Ewing sarcoma
    - Osteosarcoma

- Breast cancer
- Salivary gland tumors
- Merkel cell carcinoma
- Penile cancer
- Uterine sarcoma
- Small cell lung cancer
- Ampullary adenocarcinoma
- Pediatric diffuse high-grade gliomas
- Cutaneous squamous cell skin carcinoma
- Nasopharyngeal cancer
- Kaposi sarcoma
- Vaginal cancer
- Pleural or Peritoneal mesothelioma
- Histologic (Richter) transformation to diffuse large B-cell lymphoma

Pembrolizumab is available as Keytruda (Merck & Co., Inc.) and is a programmed death receptor-1 (PD-1) blocking monoclonal antibody that works by preventing the interaction between PD-1 and the PD-L1 and PD-L2 ligands. It is indicated for multiple types of cancers, including certain types of melanoma, non-small cell lung cancer (NSCLC), gastric cancer, Hodgkin lymphoma, head and neck

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melanoma, non-small cell lung cancer (NSCLC), gastric cancer, Hodgkin lymphoma, head and neck cancer, hepatocellular cancer, urothelial carcinoma, microsatellite instability-high (MSI-H) or mismatch repair deficient solid tumors, cervical cancer, primary mediastinal large B-cell lymphoma (PMBCL), and Merkel cell carcinoma.

Per the prescribing information, pembrolizumab (Keytruda) carries the following warnings and precautions:

- Immune-Mediated Pneumonitis: Pneumonitis occurred in 12 (2.9%) of 411 melanoma patients, including Grade 2 or 3 cases in 8 (1.9%) and 1 (0.2%) patients, respectively, receiving Keytruda (pembrolizumab) in Trial 1. The median time to development of pneumonitis was 5 months (range 0.3 weeks-9.9 months). The median duration was 4.9 months (range 1 week-14.4 months). Five of eight patients with Grade 2 and the one patient with Grade 3 pneumonitis required initial treatment with high-dose systemic corticosteroids (greater than or equal to 40 mg prednisone or equivalent per day) followed by a corticosteroid taper. The median initial dose of high-dose corticosteroid treatment was 63.4 mg/day of prednisone or equivalent with a median duration of treatment of 3 days (range 1-34) followed by a corticosteroid taper. Pneumonitis led to discontinuation of Keytruda (pembrolizumab) in 3 (0.7%) patients. Pneumonitis completely resolved in seven of the nine patients with Grade 2-3 pneumonitis.
- Immune-Mediated Colitis: Colitis (including microscopic colitis) occurred in 4 (1%) of 411 patients, including Grade 2 or 3 cases in 1 (0.2%) and 2 (0.5%) patients, respectively, receiving Keytruda (pembrolizumab) in Trial 1. The median time to onset of colitis was 6.5 months (range 2.3-9.8). The median duration was 2.6 months (range 0.6 weeks-3.6 months). All three patients with Grade 2 or 3 colitis were treated with high-dose corticosteroids (greater than or equal to 40 mg prednisone or equivalent per day) with a median initial dose of 70 mg/day of prednisone or equivalent; the median duration of initial treatment was 7 days (range 4-41), followed by a corticosteroid taper. One patient (0.2%) required permanent discontinuation of Keytruda (pembrolizumab) due to colitis. All four patients with colitis experienced complete resolution of the event.
- Immune-Mediated Hepatitis: Hepatitis (including autoimmune hepatitis) occurred in 2 (0.5%) of 411 patients, including a Grade 4 case in 1 (0.2%) patient, receiving Keytruda (pembrolizumab) in Trial 1. The time to onset was 22 days for the case of Grade 4 hepatitis which lasted 1.1 months. The patient with Grade 4 hepatitis permanently discontinued Keytruda (pembrolizumab) and was treated with high-dose (greater than or equal to 40 mg prednisone or equivalent per day) systemic corticosteroids followed by a corticosteroid taper. Both patients with hepatitis experienced complete resolution of the event.
- Immune-Mediated Hypophysitis: Hypophysitis occurred in 2 (0.5%) of 411 patients, consisting of one Grade 2 and one Grade 4 case (0.2% each), in patients receiving Keytruda (pembrolizumab) in Trial 1. The time to onset was 1.7 months for the patient with Grade 4 hypophysitis and 1.3 months for the patient with Grade 2 hypophysitis. Both patients were treated with high-dose (greater than or equal to 40 mg prednisone or equivalent per day) corticosteroids followed by a corticosteroid taper and remained on a physiologic replacement dose.

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taper and remained on a physiologic replacement dose.

- Renal Failure and Immune-Mediated Nephritis: Nephritis occurred in 3 (0.7%) patients, consisting of one case of Grade 2 autoimmune nephritis (0.2%) and two cases of interstitial nephritis with renal failure (0.5%), one Grade 3 and one Grade 4. The time to onset of autoimmune nephritis was 11.6 months after the first dose of Keytruda (pembrolizumab) (5 months after the last dose) and lasted 3.2 months; this patient did not have a biopsy. Acute interstitial nephritis was confirmed by renal biopsy in two patients with Grades 3-4 renal failure. All three patients fully recovered renal function with treatment with high-dose corticosteroids (greater than or equal to 40 mg prednisone or equivalent per day) followed by a corticosteroid taper.
- Immune-Mediated Hyperthyroidism and Hypothyroidism: Hyperthyroidism occurred in 5 (1.2%) of 411 patients, including Grade 2 or 3 cases in 2 (0.5%) and 1 (0.2%) patients, respectively, receiving Keytruda (pembrolizumab) in Trial 1. The median time to onset was 1.5 months (range 0.5-2.1). The median duration was 2.8 months (range 0.9 to 6.1). One of two patients with Grade 2 and the one patient with Grade 3 hyperthyroidism required initial treatment with high-dose corticosteroids (greater than or equal to 40 mg prednisone or equivalent per day) followed by a corticosteroid taper. One patient (0.2%) required permanent discontinuation of Keytruda (pembrolizumab) due to hyperthyroidism. All five patients with hyperthyroidism experienced complete resolution of the event.
- Hypothyroidism occurred in 34 (8.3%) of 411 patients, including a Grade 3 case in 1 (0.2%) patient, receiving Keytruda (pembrolizumab) in Trial 1. The median time to onset of hypothyroidism was 3.5 months (range 0.7 weeks-19 months). All but two of the patients with hypothyroidism were treated with long-term thyroid hormone replacement therapy. The other two patients only required short-term thyroid hormone replacement therapy. No patient received corticosteroids or discontinued Keytruda (pembrolizumab) for management of hypothyroidism.
- Immune-mediated skin adverse reactions, including SJS, TEN (some cases with fatal outcome), exfoliative dermatitis, and bullous pemphigoid: , can occur. Monitor patients for suspected severe skin reactions and exclude other causes. Based on the severity of the adverse reaction, withhold or permanently discontinue pembrolizumab and administer corticosteroids. For signs or symptoms of SJS or TEN, withhold pembrolizumab and refer the patient for specialized care for assessment and treatment. If SJS or TEN is confirmed, permanently discontinue pembrolizumab.
- Other Immune-Mediated Adverse Reactions: The following clinically significant, immune-mediated adverse reactions occurred in less than 1% (unless otherwise indicated) of patients treated with Keytruda (pembrolizumab) in clinical trial: exfoliative dermatitis, uveitis, arthritis (1.5%), myositis, pancreatitis, hemolytic anemia, partial seizures arising in a patient with inflammatory foci in brain parenchyma, adrenal insufficiency, Guillain-Barré syndrome, myasthenia gravis, vasculitis, sarcoidosis and encephalitis. In addition, myelitis and myocarditis were reported in other clinical trials, including cHL, and post-marketing use. Solid organ transplant rejection has been reported in the post-marketing setting in patients treated with pembrolizumab . Treatment with pembrolizumab may increase the risk of rejection in solid organ transplant recipients. Consider the benefit of treatment with pembrolizumab versus the risk of possible organ rejection in these patients.
- In two randomized clinical trials in patients with multiple myeloma, the addition of pembrolizumab to a thalidomide analogue plus dexamethasone, a use for which no PD-1 or PD-L1 blocking antibody is indicated, resulted in increased mortality. Treatment of patients with multiple myeloma

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antibody is indicated, resulted in increased mortality. Treatment of patients with multiple myeloma with a PD-1 or PDL1 blocking antibody in combination with a thalidomide analogue plus dexamethasone is not recommended outside of controlled clinical trials.

- Pregnancy—Category D. Based on its mechanism of action, Keytruda (pembrolizumab) may cause fetal harm when administered to a pregnant woman. Animal models link the PD-1/PDL-1 signaling pathway with maintenance of pregnancy through induction of maternal immune tolerance to fetal tissue. If this drug is used during pregnancy, or if the patient becomes pregnant while taking this drug, apprise the patient of the potential hazard to a fetus. Human IgG4 (immunoglobulins) are known to cross the placenta; therefore, pembrolizumab has the potential to be transmitted from the mother to the developing fetus. Based on its mechanism of action, fetal exposure to pembrolizumab may increase the risk of developing immune-mediated disorders or of altering the normal immune response.

Pembrolizumab (Keytruda) carries the following adverse reactions of which the most common (reported in 20% or more of patients) are (Merck, 2024):

- Keytruda as a single agent: fatigue, musculoskeletal pain, decreased appetite, pruritus, diarrhea, nausea, rash, pyrexia, cough, dyspnea, constipation, pain, and abdominal pain.
- Keytruda in combination with chemotherapy: fatigue/asthenia, nausea, constipation, diarrhea, decreased appetite, rash, vomiting, cough, dyspnea, pyrexia, alopecia, peripheral neuropathy, mucosal inflammation, stomatitis, headache, and weight loss.
- Keytruda in combination with axitinib: diarrhea, fatigue/asthenia, hypertension, hepatotoxicity, hypothyroidism, decreased appetite, palmar-plantar erythrodysesthesia, nausea, stomatitis/mucosal inflammation, dysphonia, rash, cough, and constipation.
- Keytruda in combination with lenvatinib: fatigue, hypertension, musculoskeletal pain, diarrhea, decreased appetite, hypothyroidism, nausea, stomatitis, vomiting, weight loss, abdominal pain, headache, constipation, urinary tract infection, dysphonia, hemorrhagic events, hypomagnesemia, palmar-plantar erythrodysesthesia, dyspnea, cough, and rash.

On April 28, 2020, the Food and Drug Administration granted accelerated approval to a new dosing regimen of 400 mg every six weeks for pembrolizumab (Keytruda, Merck) across all currently approved adult indications, in addition to the current 200 mg every three weeks dosing regimen. The approval was based on pharmacokinetic modeling and exposure-response analyses that compared the predicted exposure of pembrolizumab 400 mg every six weeks to observed exposures of pembrolizumab in patients who received pembrolizumab at 2 mg/kg every three weeks, 200 mg every three weeks, and 10 mg/kg administered every two weeks. The pharmacokinetic modeling were supported by additional exposure-response analyses across the pembrolizumab development program and an interim analysis of pharmacokinetics and overall response rate (ORR) in a cohort of patients (Cohort B) enrolled in Study KEYNOTE-555 (NCT03665597). Cohort B of Study KEYNOTE-555 was an international, single-arm, multi-center study that enrolled 101 patients with advanced or metastatic melanoma who had not received prior PD-1, PD-L1, or CTLA-4 inhibitors (other than CTLA-4 inhibitors in the adjuvant setting). The ORR was 39% (95% CI: 24, 55) in the first 44 patients enrolled in KEYNOTE-555. This new dosing regimen is approved under accelerated approval based on pharmacokinetic data, the

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new dosing regimen is approved under accelerated approval based on pharmacokinetic data, the relationship of exposure to efficacy, and the relationship of exposure to safety.

Biliary Tract Cancer

On October 31, 2023, the U.S. Food and Drug Administration (FDA) approved pembrolizumab (Keytruda) to be used with gemcitabine and cisplatin for locally advanced unresectable or metastatic biliary tract cancer (BTC). The FDA approval was based on supporting data from KEYNOTE-966 study, a multicenter randomized, double-blind, placebo-controlled trial. Investigators evaluated the efficacy of Keytruda in 1,069 patients with locally advanced unresectable or metastatic BTC who had not received prior systemic therapy for advanced disease. Patients were randomized (1:1) to receive either Keytruda on Day 1 plus gemcitabine and cisplatin on Day 1 and Day 8 every 3 weeks, or placebo on Day 1 plus gemcitabine and cisplatin on the above schedule. Treatment continued until unacceptable toxicity or disease progression. Cisplatin administration was for a maximum of 8 cycles; gemcitabine was continued at the physician's discretion. Keytruda or placebo were continued until disease progression, unacceptable toxicity, or a maximum of 2 years (FDA, 2023c).

The major efficacy measure was overall survival (OS). Keytruda plus chemotherapy demonstrated a statistically significant improvement in OS compared to placebo plus chemotherapy with a hazard ratio of 0.83 (95% Confidence Interval [CI]: 0.72, 0.95); one-sided p-value=0.0034). The median OS was 12.7 months (95% CI: 11.5, 13.6) and 10.9 months (95% CI: 9.9, 11.6) in the respective arms (FDA, 2023c).

The "NCCN Guidelines Version 3.2023 Biliary Tract Cancers" discuss adjuvant chemotherapy and chemoradiation for patients with gallbladder cancer or extrahepatic cholangiocarcinoma (CCA) with resected, positive margins (R1) or gross residual local disease (R2) or those with intrahepatic CCA with residual local disease (R2) after resection. A multidisciplinary team should review the available treatment options on a case-by-case basis for these patients. Furthermore, the evaluation and treatment of gross residual disease (R2) should be consistent with that for unresectable disease. Although the optimal treatment strategy has not been established for patients with R1 margins or positive regional nodes, available options include systemic therapy (preferred), clinical trial (preferred), or fluoropyrimidine-based chemoradiation, with or without fluoropyrimidine-based or gemcitabine-based chemotherapy.

Breast Cancer

In November 2020, the U.S. FDA granted accelerated approval to pembrolizumab in combination with chemotherapy for the treatment of patients with locally recurrent unresectable or metastatic triple-negative breast cancer (TNBC) whose tumors express PD-L1 (CPS greater than or equal to 10) as determined by an FDA approved test (FDA, 2020). Approval was based on KEYNOTE-355 (NCT02819518), a multicenter, double-blind, randomized, placebo-controlled trial in patients with locally recurrent unresectable or metastatic TNBC, who had not been previously treated with

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chemotherapy in the metastatic setting. Patients with active autoimmune disease that required systemic therapy within 2 years of treatment or a medical condition that required immunosuppression were

ineligible. Randomization was stratified by chemotherapy treatment (paclitaxel or paclitaxel protein-bound vs. gemcitabine and carboplatin), tumor PD-L1 expression (CPS ≥1 vs. CPS <1) according to the PD-L1 IHC 22C3 pharmDx kit, and prior treatment with the same class of chemotherapy in the neoadjuvant setting (yes vs. no). Patients (n=847) were randomized (2:1) to receive pembrolizumab 200 mg on day 1 every 3 weeks or placebo in combination with different chemotherapy treatments (paclitaxel protein-bound, or paclitaxel, or gemcitabine plus carboplatin) via intravenous infusion. Assessment of tumor status was performed at Weeks 8, 16, and 24, then every 9 weeks for the first year, and every 12 weeks thereafter. The main efficacy outcome measure was progression-free survival (PFS) as assessed by blinded independent review according to RECIST 1.1, tested in the subgroup of patients with CPS ≥10. Median PFS was 9.7 months (95% CI: 7.6, 11.3) in the pembrolizumab plus chemotherapy arm and 5.6 months in the placebo arm (HR 0.65; 95% CI: 0.49, 0.86; one-sided p=0.0012) (FDA, 2020, Merck, 2020).

On July 27, 2021, the U.S. Food and Drug Administration (FDA) approved Keytruda for the treatment of patients with high-risk early-stage triple-negative breast cancer (TNBC) in combination with chemotherapy as neoadjuvant tretment and then continued as a single agent as adjuvant treatment after surgery. This approval was based on supporting data from the Phase 3 KEYNOTE-522 study (Merck, 2021a).

Bone Cancers

George et al (2017) stated that response to immune checkpoint blockade in mesenchymal tumors is poorly characterized, but immunogenomic dissection of these cancers could inform immunotherapy mediators. These researchers identified a treatment-naive patient who has metastatic uterine leiomyosarcoma and has experienced complete tumor remission for greater than 2 years on programmed cell death protein 1 (PD-1) (pembrolizumab) monotherapy. They analyzed the primary tumor, the sole treatment-resistant metastasis, and germline tissue to explore mechanisms of immunotherapy sensitivity and resistance. Both tumors stained diffusely for PD-L2 and showed sparse PD-L1 staining. PD-1+ cell infiltration significantly decreased in the resistant tumor (p = 0.039). Genomically, the treatment-resistant tumor uniquely harbored bi-allelic PTEN loss and had reduced expression of two neo-antigens that demonstrated strong immuno-reactivity with patient T cells in-vitro, suggesting long-lasting immunological memory. In this near-complete response to PD-1 blockade in a mesenchymal tumor, the authors identified PTEN mutations and reduced expression of genes encoding neo-antigens as potential mediators of resistance to immune checkpoint therapy.

In an open-label, multi-center, phase II clinical trial, Toulmonde et al (2017) evaluated the safety and effectiveness of PD-1 targeting in combination with metronomic chemotherapy in sarcomas. This study consisted of 4 cohorts of patients with advanced soft-tissue sarcoma (STS), including leiomyosarcoma (LMS), undifferentiated pleomorphic sarcoma (UPS), other sarcomas (others), and gastro-intestinal

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stromal tumor (GIST). All patients received 50-mg twice-daily cyclophosphamide 1 week on and 1 week off, and 200-mg of intravenous pembrolizumab every 3 weeks. There was a dual primary end-

point, encompassing both the non-progression and objective responses at 6 months per Response Evaluation Criteria in Solid Tumors (RECIST) v1.1 for LMS, UPS, and others and 6-month non-progression for GIST. An objective response rate of 20 % and/or a 6-month non-progression rate of 60 % were determined as reasonable objectives for treatment with meaningful effect. Correlative studies of immune biomarkers were planned from patient tumor and plasma samples. Between June 2015 and July 2016, a total of 57 patients were included (median [range] age of 59.5 [18.5 to 84.0] years; 24 women [42 %]); 50 patients were assessable for the efficacy end-point; 3 patients experienced tumor shrinkage, resulting in a partial response (PR) in a single solitary fibrous tumor. The 6-month non-progression rates were 0 %, 0 %, 14.3 % (95 % CI: 1.8 % to 42.8 %) for LMS, UPS, and others, respectively, and 11.1 % (95 % CI: 2.8 % to 48.3 %) for GIST. The most frequent adverse events (AEs) were grade 1 or 2 fatigue, diarrhea, and anemia. The only patient who experienced PR was the only one with strong PD-1 ligand 1-positive staining in immune cell. Strong infiltration by macrophage expressing the inhibitory enzyme indoleamine 2,3-dioxygenase 1 (IDO1) was observed in the majority of cases. Moreover, a significant increase in the kynurenine to tryptophan ratio was observed in patient plasma samples during the study treatment. The authors found that PD-1 inhibition has limited activity in selected STS and GIST. This may be explained by an immunosuppressive tumor microenvironment resulting from macrophage infiltration and IDO1 pathway activation.

Le et al (2017) stated the genomes of cancers deficient in mismatch repair contain exceptionally high numbers of somatic mutations. In a proof-of-concept study, the authors previously showed that colorectal cancers with mismatch repair deficiency were sensitive to immune checkpoint blockade with antibodies to programmed death receptor-1 (PD-1). The authors expanded this study to evaluate the efficacy of PD-1 blockade in patients with advanced mismatch repair-deficient cancers across 12 different tumor types. Objective radiographic responses were observed in 53% of patients, and complete responses were achieved in 21% of patients. Responses were durable, with median progression-free survival and overall survival still not reached. Functional analysis in a responding patient demonstrated rapid in vivo expansion of neoantigen-specific T cell clones that were reactive to mutant neopeptides found in the tumor. These data support the hypothesis that the large proportion of mutant neoantigens in mismatch repair-deficient cancers make them sensitive to immune checkpoint blockade, regardless of the cancers' tissue of origin.

Tawbi et al (2017) stated patients with advanced sarcomas have a poor prognosis and few treatment options that improve overall survival. Chemotherapy and targeted therapies offer short-lived disease control. The authors assessed pembrolizumab, an anti-PD-1 antibody, for safety and activity in patients with advanced soft-tissue sarcoma or bone sarcoma. In this two-cohort, single-arm, open-label, phase 2 study, The authors enrolled patients with soft-tissue sarcoma or bone sarcoma from 12 academic centres in the USA that were members of the Sarcoma Alliance for Research through Collaboration (SARC). Patients with soft-tissue sarcoma had to be aged 18 years or older to enroll; patients with bone sarcoma could enroll if they were aged 12 years or older. Patients had histological evidence of metastatic or surgically unresectable locally advanced sarcoma, had received up to three previous lines of systemic

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surgically unresectable locally advanced sarcoma, had received up to three previous lines of systemic anticancer therapy, had at least one measurable lesion according to the Response Evaluation Criteria In Solid Tumors version 1.1, and had at least one lesion accessible for biopsy. All patients were treated with 200 mg intravenous pembrolizumab every 3 weeks. The primary endpoint was investigator-assessed objective response. Patients who received at least one dose of pembrolizumab were included in the safety analysis and patients who progressed or reached at least one scan assessment were included in the activity analysis. Accrual is ongoing in some disease cohorts. Between March 13, 2015, and Feb 18, 2016, 86 patients were enrolled, 84 of whom received pembrolizumab (42 in each disease cohort) and 80 of whom were evaluable for response (40 in each disease cohort). Median follow-up was 17·8 months (IQR 12·3-19·3). Seven (18%) of 40 patients with soft-tissue sarcoma had an objective response, including four (40%) often patients with undifferentiated pleomorphic sarcoma, two (20%) of ten patients with liposarcoma, and one (10%) of ten patients with synovial sarcoma. No patients with leiomyosarcoma (n=10) had an objective response. Two (5%) of 40 patients with bone sarcoma had an objective response, including one (5%) of 22 patients with osteosarcoma and one (20%) of five patients with chondrosarcoma. None of the 13 patients with Ewing's sarcoma had an objective response. The most frequent grade 3 or worse adverse events were anemia (six [14%]), decreased lymphocyte count (five [12%]), prolonged activated partial thromboplastin time (four [10%]), and decreased platelet count (three [7%]) in the bone sarcoma group, and anemia, decreased lymphocyte count, and prolonged activated partial thromboplastin time in the soft-tissue sarcoma group (three [7%] each). Nine (11%) patients (five [12%] in the bone sarcoma group and four [10%] in the soft-tissue sarcoma group) had treatment-emergent serious adverse events (SAEs), five of whom had immune-related SAEs, including two with adrenal insufficiency, two with pneumonitis, and one with nephritis. The authors concluded that the primary endpoint of overall response was not met for either cohort. However, pembrolizumab showed encouraging activity in patients with undifferentiated pleomorphic sarcoma or dedifferentiated liposarcoma. Enrolment to expanded cohorts of those subtypes is ongoing to confirm and characterize the activity of pembrolizumab.

Central Nervous System Tumors

Blumenthal and colleagues (2016) noted that patients with progressive primary brain tumors (PBT) are attracted to promising new treatments, even prior to convincing data. Anti-PD1 immunotherapies have been in the spotlight since publication of ground-breaking results for metastatic melanoma with pembrolizumab. These investigators reported on the response and toxicity of pembrolizumab in patients with advanced PBT. They retrospectively reviewed the charts of 22 patients (17 adults and 5 children) with recurrent central nervous system (CNS) tumors treated with pembrolizumab. The authors analyzed prior antineoplastic therapies, steroid usage, and outcomes. Patients received a median of 2 neoplastic therapies prior to pembrolizumab, and a median of 3 infusions of pembrolizumab in adults and 4 in children; 12 patients (9 adults and 3 children) started pembrolizumab on steroids (median dose in adults 4 mg; range of 2 to 8, and in children 1.5 mg, range of 0.5 to 4) and 5 patients received steroids later during pembrolizumab treatment. A total of 12 patients (10 adults and 2 children) received concomitant bevacizumab with pembrolizumab; side effects were minimal. All patients showed progressive tumor growth during therapy. Median OS from the start of pembrolizumab was 2.6 months in adults and 3.2

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months in children; 2 glioblastoma patients underwent tumor resection following treatment with pembrolizumab. Tumor-lymphocytic response in these cases was unremarkable, and PD-L1 immuno-

staining was negative. The authors concluded that in this series of 22 patients with recurrent primary brain tumors, pembrolizumab showed no clinical or histologic effectiveness. These investigators do not recommend further use of pembrolizumab for recurrent PBT unless convincing prospective clinical trial data are published.

Reardon and colleagues (2021) noted that current treatments for recurrent glioblastoma offer limited benefit. These investigators reported the anti-tumor activity and safety of pembrolizumab in PD-L1-positive, recurrent glioblastoma. Adult patients with PD-L1-positive tumors were enrolled in the recurrent glioblastoma cohort of the multi-cohort, phase-Ib KEYNOTE-028 Trial and received pembrolizumab 10 mg/kg every 2 weeks for up to 2 years. The primary endpoint was investigator-assessed ORR according to the RECIST, version 1.1. Archival tumor samples were assessed for PD-L1 expression levels (prospectively) and T-cell-inflamed gene expression profile score (retrospectively). After a median follow-up of 14 months (range of 2 to 55 months) among the 26 enrolled patients, the ORR was 8 % (95 % CI: 1 % to 26 %). Two PRs, lasting 8.3 and 22.8 months, occurred; PFS (median of 2.8 months; 95 % CI: 1.9 to 8.1 months) rate at 6 months was 37.7 %, and the OS (median of 13.1 months; 95 % CI: 8.0 to 26.6 months) rate at 12 months was 58 %. Correlation of therapeutic benefit to level of PD-L1 expression, gene expression profile score, or baseline steroid use could not be established. Treatment-related AEs occurred in 19 patients (73 %) -- 5 s experienced grade-3 or grade-4 events (there were no grade-5 events). Immune-mediated AEs and infusion reactions occurred in 7 patients (27 %). The authors concluded that pembrolizumab monotherapy demonstrated durable anti-tumor activity in a subset of patients with manageable toxicity in this small, signal-finding, recurrent glioblastoma cohort. Moreover, these researchers stated that future studies examining rationally designed pembrolizumab combination regimens may improve outcomes in patients with recurrent glioblastoma.

Cervical Cancer

On June 12, 2018, the Food and Drug Administration approved pembrolizumab for patients with recurrent or metastatic cervical cancer with disease progression on or after chemotherapy whose tumors express PD-L1 (CPS ≥1) as determined by an FDA-approved test. The FDA also concurrently approved PD-L1 IHC 22C3 pharmDx (Dako North America Inc.) as a companion diagnostic.

Pembrolizumab was investigated in 98 patients with recurrent or metastatic cervical cancer enrolled in a single cohort (Cohort E) in Study KEYNOTE-158 (NCT02628067), a multicenter, non-randomized, open-label, multi-cohort trial. The trial excluded patients with autoimmune disease or a medical condition that required immunosuppression. Patients were treated with pembrolizumab intravenously at a dose of 200 mg every 3 weeks until unacceptable toxicity or documented disease progression. Patients with initial radiographic disease progression could receive additional doses of treatment during confirmation of progression unless disease progression was symptomatic, was rapidly progressive,

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required urgent intervention, or occurred with a decline in performance status. Patients without disease progression could be treated for up to 24 months. Assessment of tumor status was performed every 9

weeks for the first 12 months, and every 12 weeks thereafter. The major efficacy outcome measures were ORR according to RECIST 1.1, as assessed by blinded independent central review, and duration of response.

Among the 98 patients in Cohort E, 77 (79%) had tumors that expressed PD-L1 with a CPS ≥ 1 and received at least one line of chemotherapy in the metastatic setting. PD-L1 status was determined using the PD-L1 IHC 22C3 pharmDx Kit. The baseline characteristics of these 77 patients were: median age was 45 years (range: 27 to 75 years); 81% were White, 14% Asian, 3% Black; ECOG PS was 0 (32%) or 1 (68%); 92% had squamous cell carcinoma, 6% adenocarcinoma, and 1% adenosquamous histology; 95% had M1 disease and 5% had recurrent disease; 35% had one and 65% had two or more prior lines of therapy in the recurrent or metastatic setting. No responses were observed in patients whose tumors did not have PD-L1 expression (CPS ˂1).

Of the 77 patients with measurable endpoints, objective response rates were seen in 14.3% (95% CI: 7.4, 24.1). Complete response was seen in 2.6% and partial response was seen in 11.7% of patients. A response duration greater than or equal to 6 months was seen in 91% of patients.

On October 13, 2021, the U.S. Food and Drug Administration (FDA) approved pembrolizumab (Keytruda) in combination with chemotherapy, with or without bevacizumab, for individuals with persistent, recurrent or metastatic cervical cancer whose tumors express PD-L1 (CPS≥1), as determined by FDA-approved test. The FDA approval was based on supporting data from the KEYNOTE-826 study, a multicenter, randomized, placebo-controlled phase 3 trial (FDA, 2021). Colombo and colleagues (2021) concluded that pembrolizumab significantly extended progression-free and overall survival in comparison to placebo among individuals with persistent, recurrent, or metastatic cervical cancer who were also receiving chemotherapy with or without bevacizumab.

On January 12, 2024, the U.S. Food and Drug Administration approved pembrolizumab (Keytruda) with chemoradiotherapy (CRT) for patients with FIGO 2014 Stage III-IVA cervical cancer. The FDA approval was based on supporting data from the KEYNOTE-A18 study (FDA, 2024a).

The KEYNOTE-A18 study, a multicenter, randomized, double-blind, placebo-controlled trial, investigated the efficacy of pembrolizumab (Keytruda) in 1,060 patients with cervical cancer who had not previously received definitive surgery, radiation, or systemic therapy. The study included 596 patients with FIGO 2014 Stage III-IVA disease and 462 patients with FIGO 2014 IB2-IIB, node-positive disease. Patient randomization was 1:1 to receive Keytruda 200 mg or placebo every 3 weeks for 5 cycles + CRT, followed by Keytruda 400 mg or placebo every 6 weeks for 15 cycles. The CRT regimen included cisplatin 40 mg/m2 IV weekly for 5 cycles (with an optional 6th cycle) and external beam radiation therapy (EBRT) followed by brachytherapy. Randomization was stratified by planned type of EBRT, stage, and planned total radiotherapy dose. The major efficacy outcome measures were progression-free survival (PFS) assessed by the investigator according to RECIST v1.1, or

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progression-free survival (PFS) assessed by the investigator according to RECIST v1.1, or histopathologic confirmation, and overall survival (OS). The study showed a statistically significant improvement in PFS in the overall population. In an exploratory subgroup analysis in the 596 patients with FIGO 2014 Stage III-IVA-disease, the PFS hazard ratio (HR) estimate was 0.59 (95% confidence interval [CI]: 0.43, 0.82), and 21% of patients in the pembrolizumab arm experienced a PFS event compared to 31% of patients in the placebo arm. However, in an exploratory subgroup analysis in the 462 patients with FIGO 2014 Stage IB2-IIB disease, the PFS HR estimate was 0.91 (95% CI: 0.63, 1.31), indicating that the PFS improvement in the overall population was primarily attributed to patients with FIGO 2014 Stage III-IVA disease. OS data were not mature at the time of PFS analysis (FDA, 2024a).

Chronic Lymphocytic Leukemia

Ding et al (2017) stated chronic lymphocytic leukemia (CLL) patients progressed early on ibrutinib often develop Richter transformation (RT) with a short survival of about 4 months. Preclinical studies suggest that programmed death 1 (PD-1) pathway is critical to inhibit immune surveillance in CLL. This phase 2 study was designed to test the efficacy and safety of pembrolizumab, a humanized PD-1-blocking antibody, at a dose of 200 mg every 3 weeks in relapsed and transformed CLL. Twenty-five patients including 16 relapsed CLL and 9 RT (all proven diffuse large cell lymphoma) patients were enrolled, and 60% received prior ibrutinib. Objective responses were observed in 4 out of 9 RT patients (44%) and in 0 out of 16 CLL patients (0%). All responses were observed in RT patients who had progression after prior therapy with ibrutinib. After a median follow-up time of 11 months, the median overall survival in the RT cohort was 10.7 months, but was not reached in RT patients who progressed after prior ibrutinib. Treatment-related grade 3 or above adverse events were reported in 15 (60%) patients and were manageable. Analyses of pretreatment tumor specimens from available patients revealed increased expression of PD-ligand 1 (PD-L1) and a trend of increased expression in PD-1 in the tumor microenvironment in patients who had confirmed responses. Overall, pembrolizumab exhibited selective efficacy in CLL patients with RT. The results of this study are the first to demonstrate the benefit of PD-1 blockade in CLL patients with RT, and could change the landscape of therapy for RT patients if further validated.

An UpToDate review on “Overview of the treatment of chronic lymphocytic leukemia” (Rai and Stilgenbauer, 2023) does not mention pembrolizumab as a therapeutic option.

Colorectal Cancer

The FDA approved pembrolizumab for the treatment of adult and pediatric patients with unresectable or metastatic, microsatellite instability-high (MSI-H) or mismatch repair deficient colorectal cancer that has progressed following treatment with a fluoropyrimidine, oxaliplatin, and irinotecan. This indication was approved under accelerated approval based on tumor response rate and durability of response. Continued approval for this indication may be contingent upon verification and description of clinical benefit in the confirmatory trials.

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Endometrial Carcinoma

In September 2019, the FDA announced an accelerated approval to the combination of pembrolizumab (Keytruda) and lenvatinib (Lenvima) for the treatment of patients with advanced endometrial cancer who have disease progression following prior systemic therapy. The indication applies to patients who are not candidates for curative surgery or radiation and who have disease that is not microsatellite instability–high (MSI-H) or mismatch repair deficient (dMMR). The approval was supported by findings from the single-arm, multicenter, open-label, multi-cohort phase Ib/II trial (Study 111; KEYNOTE-146; NCT02501096), which evaluated 108 patients with metastatic endometrial carcinoma whose tumors had progressed following at least one prior systemic therapy in any setting and were not MSI-H or dMMR. Patients with active autoimmune disease or a medical condition that required immunosuppression were ineligible. Patients received 20 mg lenvatinib orally once daily plus 200 mg pembrolizumab given intravenously every 3 weeks until disease progression or unacceptable toxicity as determined by the investigator. The median duration of study treatment was 7 months (range: 0.03 to 37.8 months). The major efficacy outcome measures were objective response rate (ORR) and duration of response (DOR) by independent radiologic review committee (IRC) using RECIST 1.1. Administration of Pembrolizumab and lenvatinib was permitted beyond RECIST-defined disease progression if the patient was clinically stable and considered by the investigator to be deriving clinical benefit. Pembrolizumab dosing was continued for a maximum of 24 months; however, treatment with lenvatinib could be continued beyond 24 months. Assessment of tumor status was performed at baseline and then every 6 weeks until week 24, followed by every 9 weeks thereafter. Overall, 87% (n=94 out of 108) of patients had non–MSI-H/dMMR tumors, whereas 10% (n=11) did, and the remaining patients had unknown MSI/MMR status. Tumor MSI status was determined through polymerase chain reaction testing, and MMR status was assessed with immunohistochemistry testing. The baseline characteristics of the 94 patients with tumors that were not MSI-H or dMMR were: median age of 66 years, 62% age 65 or older; and 86% White, 6% Black, 4% Asian, and 3% other races; and ECOG PS of 0 (52%) or 1 (48%). All 94 of these patients received prior systemic therapy for endometrial carcinoma: 51% had one, 38% had two, and 11% had three or more prior systemic therapies. The objective response rate (ORR), as assessed by independent radiologic review committee using RECIST 1.1 criteria, was 38.3% (95% CI, 29%-49%) in the patients with non–MSI-H/dMMR tumors. The complete response rate was 10.6% and the partial response rate was 27.7%. The median duration of response (DOR) was not reached at the time of data cutoff; 69% of responders had DORs of at least 6 months. The most common adverse events (≥20%) observed with the pembrolizumab/lenvatinib combination in patients with endometrial cancer were fatigue, hypertension, musculoskeletal pain, diarrhea, decreased appetite, hypothyroidism, nausea, stomatitis, vomiting, decreased weight, abdominal pain, headache, constipation, urinary tract infection, dysphonia, hemorrhagic events, hypomagnesemia, palmar-plantar erythrodysesthesia, dyspnea, cough, and rash.

In a prior interim analysis of the study published in Lancet Oncology of 53 evaluable patients at a median of 13.3 months of follow-up, the ORR by investigator review was 39.6% (95% CI, 26.5%-54.0%) at 24 weeks. By independent review, the ORR at 24 weeks was 45.3% (95% CI, 31.6%-59.6%). Patients in the interim analysis population had a mean age of 64 years and 62% had an

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51.6%-59.6%). Patients in the interim analysis population had a mean age of 64 years and 62% had an ECOG performance status of 1. The histology was endometrioid adenocarcinoma in 41% and serous adenocarcinoma in 38%. Most patients had received either 1 or 2 prior systemic therapies (43% each), but 13% had received at least 3 prior regimens. Prior chemotherapy consisted of a platinum-based doublet for 98% of patients and 57% of patients also received prior radiotherapy. Twenty-five percent of patients had PD-L1–positive tumors, whereas the status was unknown in 55%; 85% of patients had microsatellite stable tumors and 8% had MSI-H tumors. By investigator review, 64.5% of patients had responses lasting at least 12 months compared with 79.3% by independent review. The median time to response was 2.7 months by investigator review and 2.6 months by independent review. The most common treatment-related adverse events (TRAEs) of any grade included hypertension (58%), fatigue (55%), diarrhea (51%), and hypothyroidism (47%). The most common grade 3 TRAEs were hypertension in 34% of patients and diarrhea in 8%. No grade 4 TRAEs were observed. Immune-mediated adverse events were observed in 55.6% of patients, including skin, endocrine, gastrointestinal, pulmonary, hepatic, and renal events. Ten percent of these patients required high-dose glucocorticoids. Serious TRAEs were reported in 30% of patients, and one patient died from treatment-related intracranial hemorrhage. The other 4 deaths that occurred in the study were considered to be due to progressive disease. Nine percent of patients discontinued treatment due to TRAEs.

On March 21, 2022, the U.S. Food and Drug Administration (FDA) approved pembrolizumab (Keytruda), as a single agent, for the treatment of patients with advanced endometrial carcinoma that is microsatellite instability-high (MSI-H) or mismatch repair deficient (dMMR), as determined by an FDA-approved test, who have disease progression following prior systemic therapy in any setting and who are not candidates for curative surgery or radiation. The FDA approval was based on supporting data from the KEYNOTE-158 study.

In the KEYNOTE-158 study, a nonrandomized, open-label, multi-cohort, phase II clinical trial, O'Malley and colleagues (2022) evaluated the efficacy and safety of pembrolizumab (Keytruda) treatment in patients from cohorts D (endometrial cancer, irrespective of MSI-H/dMMR status) and K (any MSI-H/dMMR solid tumor, except colorectal) with previously treated, advanced MSI-H/dMMR. Patients received Keytruda 200 mg intravenously once every 3 weeks until unacceptable toxicity or documented disease progression. Treatment in patients without disease progression could be up to 24 months. The primary endpoint was objective response rate (ORR) and secondary endpoints included duration of response, progression-free survival, overall survival, and safety. Among the 90 patients (in cohorts D and K), 20% (18 of 90) completed 35 cycles of Keytruda and 58% (52 of 90) had discontinued treatment. In the efficacy group (n=79), the objective response rate was 48% (95% confidence interval [CI], 37 to 60), and median duration of response was not reached (2.9-49.7 + months). The median progression-free survival was 13.1 (95% CI, 4.3 to 34.4) months, and median overall survival was not reached (95% CI, 27.2 months to not reached). Out of all treated patients, 76% had ≥ 1 treatment-related adverse event (grades 3-4, 12%). No fatal treatment-related events occurred. Immune-mediated adverse events or infusion reactions were noted in 28% of patients (grades 3-4, 7%; no fatal events).

Esophageal Cancer and Esophagogastric Junction Cancer

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Esophageal Cancer and Esophagogastric Junction Cancer

On March 23, 2021, the U.S. Food and Drug Administration (FDA) granted approval to Keytruda (pembrolizumab) for the treatment of patients with locally advanced or metastatic esophageal or gastroesophageal junction (GEJ) (tumors with epicenter 1 to 5 centimeters above the GEJ) carcinoma that is not amenable to surgical resection or definitive chemoradiation in combination with platinum- and fluoropyrimidine-based chemotherapy. The FDA approval was based on data supported by the Phase 3 KEYNOTE-590 trial (Merck, 2021f).

The Phase 3 KEYNOTE -590 study is an ongoing randomized, double-blind, placebo-controlled trial where Kato and colleagues (2019) evaluated the efficacy of Keytruda in 749 enrolled patients. the patients were randomly assigned in a 1:1 ratio to receive Keytruda 200 mg or placebo (normal saline) by intravenous infusion every 3 weeks in combination with chemotherapy. The chemotherapy regimen in both arms consisted of cisplatin 80 mg/m2 intravenous infusion every 3 weeks (maximum six doses) plus 5-fluorouracil 800 mg/m2 continuous intravenous infusion on days 1-5 every 3 weeks. Treatment continued until confirmatory radiographic progression, unacceptable toxicity, investigator or patient decision to withdraw, nonadherence to treatment or trial procedures or completion of 35 cycles of Keytruda or placebo (approximately 2 years). Primary outcome measures were progression-free survival (PFS) and overall survival (OS) in all patients and in the subgroup of patients with a PD-L1 combined positive score (CPS) ≥ 10. During pre-specified formal test of OS in patients with PD-L1 CPS ≥ 10 (n=383) the median was 13.5 months (95% CI: 11.1, 15.6) for the Keytruda arm and 9.4 months (95% CI: 8.0, 10.7) for the placebo arm (Merck, 2021f).

Gastric Cancer and Esophagogastric Junction Cancer

On September 22, 2017, The U.S. Food and Drug Administration (FDA) granted accelerated approval of Keytruda (pembrolizumab) for the treatment of persons with recurrent locally advanced or metastatic gastric or gastroesophageal junction (GEJ) adenocarcinoma whose tumors express PD-L1 [Combined Positive Score (CPS) ≥1] as determined by an FDA-approved test, with disease progression on or after two or more prior lines of therapy including fluoropyrimidine- and platinum-containing chemotherapy and if appropriate, HER2/neu-targeted therapy. This indication is approved under accelerated approval based on tumor response rate and durability of response. Continued approval for this indication may be contingent upon verification and description of clinical benefit in the confirmatory trials (FDA, 2017).

The accelerated approval for Keytruda was based on results from the KEYNOTE 059 (NCT02335411) study, a global, multicenter, non-randomized, non-comparative, open-label, multi-cohort trial , in which 259 individuals were enrolled with gastric or GEJ adenocarcinoma that progressed on at least two prior systemic treatments for advanced disease. Among the 259 patients, 55 percent (n=143, median age 64 years) had tumors that expressed PD-L1 with a CPS ≥1, and microsatellite stable (MSS) tumor status or undetermined microsatellite instability (MSI) or mismatch repair (MMR) status. Previous treatment must have included a fluoropyrimidine and platinum doublet. HER2/neu-positive patients must have

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previously received treatment with approved HER2/neu-targeted therapy. Patients with active autoimmune disease or a medical condition that required immunosuppression or with clinical evidence

of ascites by physical exam were ineligible. Patients received Keytruda at a dose of 200 mg every 3 weeks until unacceptable toxicity or disease progression that was symptomatic, rapidly progressive, required urgent intervention, occurred with a decline in performance status, or was confirmed at least 4 weeks later with repeat imaging. Patients without disease progression were treated for up to 24 months. Assessment of tumor status was performed every six to nine weeks. The major efficacy outcome measures were objective response rate (ORR) according to the Response Evaluation Criteria In Solid Tumors (RECIST) 1.1, as assessed by blinded independent central review, and duration of response (FDA, 2017).

For the 143 patients with tumors expressing PD-L1 and who were either MSS or had unknown MSI or dMMR status, the objective response rate was 13.3% (95% CI: 8.2, 20.0); 1.4% had complete responses and 11.9% had partial responses. Among the 19 responding patients, the response duration ranged from 2.8+ to 19.4+ months, with 11 patients (58%) having response durations of 6 months or longer and 5 patients (26%) having response durations of 12 months or longer (FDA, 2017).

Among the 259 patients, 7 (3%) had tumors that were determined to be MSI-High. An objective response was observed in 4 patients, including 1 complete response. The duration of response ranged from 5.3+ to 14.1+ months (FDA, 2017).

Adverse reactions occurring in patients with gastric cancer were similar to those occurring in patients with melanoma or non-small cell lung cancer (NSCLC). The most common adverse reactions for Keytruda (pembrolizumab) (reported in ≥20% of patients) were fatigue, musculoskeletal pain, decreased appetite, pruritis, diarrhea, nausea, rash, pyrexia, cough, dyspnea, and constipation (FDA, 2017).

On May 5, 2021, the U.S. Food and Drug Administration approved Keytruda in combination with trastuzumab, fluoropyrimidine- and platinum-containing chemotherapy, for the first-line treatment of patients with locally advanced unresectable or metastatic HER2-positive gastric or gastroesophageal junction (GEJ) adenocarcinoma. This accelerated approval was based on supporting data from the ongoing Phase 3 KEYNOTE -811 study (Merck, 2021e).

The KEYNOTE-811 study consisted of multicenter, randomized, double-blind, placebo-controlled trial that enrolled 692 patients with HER2-positive advanced gastric or gastroesophageal junction (GEJ) adenocarcinoma who had not previously received systemic therapy for metastatic disease. Patients were randomized (1:1) into one of the following treatment arms: a) Keytruda 200 mg, trastuzumab 8 mg/kg on first infusion and 6 mg/kg in subsequent cycles, followed by investigator's choice of combination chemotherapy of cisplatin 80 mg/m^2 for up 6 cycles and 5-FU 800 mg/m^2/day for 5 days (FP) or oxaliplatin 130 mg/m^2 up to 6-8 cycles and capecitabine 1000 mg/m^2 bid for 14 days (CAPOX) or b) placebo, trastuzumab 8 mg/kg on first infusion and 6 mg/kg in subsequent cycles, followed by investigator's choice of combination chemotherapy of cisplatin 80 mg/m^2 for up to 6 cycles and 5-FU 800 mg/m^2/day for 5 days (FP) or oxaliplatin 130 mg/m2 up to 6-8 cycles and capecitabine 1000 mg/m^2

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800 mg/m2/day for 5 days (FP) or oxaliplatin 130 mg/m2 up to 6-8 cycles and capecitabine 1000 mg/m2 bid for 14 days (CAPOX). A statistically significant improvement in objective response rate was

evidenced in patients randomized to Keytruda in combination with trastuzumab and chemotherapy compared with placebo in combination with trastuzumab and chemotherapy (Merck, 2021e).

Gentzler et al (2016) stated that immune checkpoint inhibitors have been identified as breakthrough treatment in melanoma given its dramatic response to PD-1/PD-L1 blockade. This is likely to extend to many other cancers as hundreds of clinical trials are being conducted or proposed using this exciting modality of therapy in a variety of malignancies. While immune checkpoint inhibitors have been extensively studied in melanoma and more recently in lung cancer, little is known regarding immune checkpoint blockade in other cancers. The authors focused on the tumor immune microenvironment, the expression of PD-1/PD-L1 and the effect of immune modulation using PD-1 or PD-L1 inhibitors in patients with head and neck, prostate, urothelial, renal, breast, gastro-intestinal and lung cancers.

On November 16, 2023, the U.S. Food and Drug Administration (FDA) approved pembrolizumab (Keytruda) with fluoropyrimidine- and platinum-containing chemotherapy for the first-line treatment of adults with locally advanced unresectable or metastatic HER2-negative gastric or gastroesophageal junction (GEJ) adenocarcinoma. The FDA approval was based on supporting data from KEYNOTE-859 study (FDA, 2023d).

The KEYNOTE-859 study, a multicenter, randomized, double-blind, placebo-controlled trial, investigated the efficacy of pembrolizumab (Keytruda) in 1, 579 patients with HER2-negative advanced gastric or GEJ adenocarcinoma who had not previously received systemic therapy for metastatic disease. Patient randomization was 1:1 to receive Keytruda 200 mg or placebo with investigator's choice of combination chemotherapy consisting either of cisplatin 80 mg/m2 plus 5-FU 800 mg/m2/day for 5 days (FP) or oxaliplatin 130 mg/m2 on Day 1 plus capecitabine 1000 mg/m2 twice a day for 14 days (CAPOX) of each 21-day cycle. The major efficacy outcome was overall survival (OS) and secondary efficacy outcomes included progression free survival (PFS), overall response rate (ORR), and duration of response (DOR) assessed by blinded independent central review (BCIR) according to RECIST v1.1, modified to follow a maximum of 10 target lesions and a maximum of 5 target lesions per organ. Treatment with Keytruda and chemotherapy showed a statistically significant improvement in OS, PFS, and ORR. Median OS was 12.9 months (95% confidence interval [CI]: 11.9, 14.0) in the Keytruda arm compared 11.5 months (95% CI: 10.6, 12.1) in the placebo arm (hazard ratio [HR] 0.78 [95% CI: 0.70, 0.87]; $p < 0.0001$). Median PFS was 6.9 months (95% CI: 6.3, 7.2) and 5.6 months (95% CI: 5.5, 5.7) (HR 0.76 [95% CI: 0.67, 0.85]; $p < 0.0001$] in the respective arms. ORR was 51% (95% CI; 48, 55) and 42% (95% CI: 38, 45) ($p < 0.0001$) in the respective arms; median DOR was 8 months (95% CI: 7.0, 9.7) in the Keytruda arm and 5.7 months (95% CI: 5.5, 6.9) in the placebo arm. In addition, a pre-specified analysis revealed a statistically significant improvement in OS, PFS, and ORR in patients receiving Keytruda based on tumors expressing PD-L1 CPS ≥ 1 and CPC ≥ 10 (FDA, 2023d).

Head and Neck Squamous Cell Cancer (HNSCC)

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In an open-label, multi-center, phase Ib trial, Seiwert et al (2016) examined the effects of pembrolizumab in patients with recurrent or metastatic HNSCC. Patients were eligible for enrolment if they were aged 18 years or older, had a confirmed diagnosis of recurrent or metastatic HNSCC, and had any level of PD-L1 expression (i.e., at least 1 % of tumor cells or stroma that were PD-L1-positive by immunohistochemistry). Patients received pembrolizumab 10 mg/kg intravenously every 2 weeks. Primary outcomes were safety in the per-protocol population and the proportion of patients with centrally reviewed overall response per Response Evaluation Criteria In Solid Tumors (RECIST, version 1.1). Overall response was analyzed in the full analysis set, which was defined as all patients who had received at least 1 dose of pembrolizumab, had measurable disease at baseline, and 1 post-baseline scan or patients without a post-baseline scan who discontinued therapy because of disease progression or a drug-related adverse event. Of the 104 patients screened between June 7, 2013, and October 3, 2013, 81 (78 %) were PD-L1-positive. Of these, 60 patients with PD-L1-positive HNSCC were enrolled and treated: 23 (38 %) were HPV-positive and 37 (62 %) were HPV-negative. Pembrolizumab was well-tolerated, with 10 (17 %) of 60 patients having grade 3 to 4 drug-related adverse events, the most common of which were increases in alanine aminotransferase and in aspartate aminotransferase, and hyponatremia, each occurring in 2 of 60 patients; 1 patient developed a grade 3 drug-related rash. 27 (45 %) of 60 patients experienced a serious adverse event. There were no drug-related deaths. The proportion of patients with an overall response by central imaging review was 18 % (8 of 45 patients; 95 % CI: 8 to 32) in all patients and was 25 % (4 of 16 patients; 7 to 52) in HPV-positive patients and 14 % (4 of 29 patients; 4 to 32) in HPV-negative patients. The authors concluded that pembrolizumab was well-tolerated and demonstrated clinically meaningful anti-tumor activity in recurrent or metastatic HNSCC, supporting further study of pembrolizumab as anti-cancer therapy for advanced head and neck cancers.

On August 5, 2016, the FDA granted accelerated approval to pembrolizumab for the treatment of patients with recurrent or metastatic HNSCC with disease progression on or after platinum-containing chemotherapy. The approval was based on demonstration of a durable ORR in a subgroup of patients in an international, multi-center, non-randomized, open-label, multi-cohort study. This subgroup included 174 patients with recurrent or metastatic HNSCC who had disease progression on or after platinum-containing chemotherapy. Patients received intravenous pembrolizumab 10 mg/kg every 2 weeks or 200 mg every 3 weeks; ORR was determined by an independent review committee according to (RECIST 1.1. The ORR for these 174 patients was 16 % (95 % CI: 11 to 22). The median response duration had not been reached at the time of analysis. The range for duration of response was 2.4 months to 27.7 months (response ongoing). Among the 28 responding patients, 23 (82 %) had responses of 6 months or longer. Safety data was evaluated in 192 patients with HNSCC receiving at least 1 dose of pembrolizumab 10 mg/kg every 2 weeks or 200 mg every 3 weeks. The most common (greater than or equal to 20 %) AEs were fatigue, decreased appetite, and dyspnea. Adverse events occurring in patients with HNSCC were similar to those occurring in patients with melanoma or NSCLC, with the exception of an increased incidence of facial edema (10 % all grades, 2.1 % grades 3 to 4) and new or worsening hypothyroidism (14.6 % all grades). The most frequent (greater than or equal to 2 %) serious AEs were

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hypothyroidism (14.6 % all grades). The most frequent (greater than or equal to 2 %) serious AEs were pneumonia, dyspnea, confusional state, vomiting, pleural effusion, and respiratory failure. Clinically

significant immune-mediated AEs included pneumonitis, colitis, hepatitis, adrenal insufficiency, diabetes mellitus, skin toxicity, myositis, and thyroid disorders.

As a condition of the accelerated approval, Merck is required to conduct a multi-center, randomized trial establishing the superiority of pembrolizumab over standard therapy to verify and describe the clinical benefit of pembrolizumab. Merck has an ongoing multicenter, randomized trial (KEYNOTE 040) in patients with recurrent or met HNSCC with disease progression on or after platinum-containing chemotherapy with a primary end-point of OS. The recommended dose and schedule of pembrolizumab for this indication is 200 mg administered as an intravenous infusion over 30 minutes every 3 weeks (FDA, 2016).

Hepatocellular Carcinoma (HCC)

On November 9, 2018, the U.S. Food and Drug Administration approved pembrolizumab (Keytruda) for the treatment of patients with hepatocellular carcinoma (HCC) who have been previously treated with sorafenib (Nexavar). This approval was based on data from the KEYNOTE-224 study (Zhu et al 2018), a single-arm trial in patients with HCC who had disease progression on or after sorafenib or were intolerant to sorafenib; had measurable disease; and had Child-Pugh class A liver impairment.

Zhu et al 2018 (KEYNOTE-224; NCT02702414) stated immune checkpoint blockade therapy has shown promising results in patients with advanced hepatocellular carcinoma. The authors aimed to assess the efficacy and safety of pembrolizumab in this patient population. KEYNOTE-224 is a non-randomised, multicentre, open-label, phase 2 trial that is set in 47 medical centres and hospitals across ten countries. Eligible patients had pathologically confirmed hepatocellular carcinoma; had previously been treated with sorafenib and were either intolerant to this treatment or showed radiographic progression of their disease after treatment; an Eastern Cooperative Oncology Group performance status of 0-1; adequate organ function, and were Child-Pugh class A. Participants received 200 mg pembrolizumab intravenously every 3 weeks for about 2 years or until disease progression, unacceptable toxicity, patient withdrawal, or investigator decision. The primary endpoint was objective response, defined as the proportion of patients with complete or partial response in all patients who received at least one dose of pembrolizumab, which was radiologically confirmed by use of the Response Evaluation Criteria in Solid Tumors version 1.1 by central review. Safety was also assessed in all treated patients. This trial is ongoing but closed to enrollment. Between June 7, 2016, and Feb 9, 2017, 169 patients with advanced hepatocellular carcinoma were screened, of whom 104 eligible patients were enrolled and treated. As of data cutoff on Feb 13, 2018, 17 (16%) patients were still receiving pembrolizumab. The authors recorded an objective response in 18 (17%; 95% CI 11-26) of 104 patients. The best overall responses were one (1%) complete and 17 (16%) partial responses; meanwhile, 46 (44%) patients had stable disease, 34 (33%) had progressive disease, and six (6%) patients who did not have a post-baseline assessment on the cutoff date were considered not to be assessable. Treatment-

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related adverse events occurred in 76 (73%) of 104 patients, which were serious in 16 (15%) patients. Grade 3 treatment-related events were reported in 25 (24%) of the 104 patients; the most common were

increased aspartate aminotransferase concentration in seven (7%) patients, increased alanine aminotransferase concentration in four (4%) patients, and fatigue in four (4%) patients. One (1%) grade 4 treatment-related event of hyperbilirubinemia occurred. One death associated with ulcerative esophagitis was attributed to treatment. Immune-mediated hepatitis occurred in three (3%) patients, but there were no reported cases of viral flares. The authors concluded that pembrolizumab was effective and tolerable in patients with advanced hepatocellular carcinoma who had previously been treated with sorafenib. These results indicate that pembrolizumab might be a treatment option for these patients. This drug is undergoing further assessment in two phase 3, randomised trials as a second-line treatment in patients with hepatocellular carcinoma.

Ma et al (2018) stated the remarkable clinical activity of PD-1 antibody in advanced hepatocellular carcinoma (HCC) highlights the importance of PD-1/PD-L1-mediated immune escape as therapeutic target in HCC. However, the frequency and prognostic significance of PD-Ls genetic alterations in HCC remain unknown. Methods: Fluorescence in situ hybridization were used to determine PD-Ls genetic alterations, and qPCR data coupled with immunofluorescence were used to measure the mRNA and protein levels of PD-Ls. Clinical relevance and prognostic value of 9p24.1 genetic alterations were investigated on tissue microarray containing three independent cohorts of 578 HCC patients. The results were further validated in an independent cohort of 442 HCC patients from The Cancer Genome Atlas (TCGA) database. In total, 7.1%-15.0% for amplification and 15.8%-31.3% for polysomy of 9p24.1 were revealed in three cohorts of HCC patients, similar to the objective response rate of PD-1 antibody in HCC. Patients with 9p24.1 genetic alterations significantly and independently correlated with unfavorable outcomes than those without. FISH and qPCR data coupled with immunofluorescence revealed that genetic alterations of 9p24.1 robustly contributed to PD-L1 and PD-L2 upregulation. In addition, increased expression of PD-L1 instead of PD-L2 also predicted poor survival by multivariate analyses. Meanwhile, high infiltration of PD-1+ immune cells also indicated dismal survival in HCC. The authors concluded that amplification or higher expression of PD-L1 significantly and independently correlated with unfavorable survival in HCC patients, authenticating the PD-1/PD-L1 axis as rational immunotherapeutic targets for HCC.

Finn et al 2017 (KEYNOTE-240; NCT0270240) states there are no approved therapies for patients with hepatocellular carcinoma (HCC) after disease progression on sorafenib, or for patients with intolerance to sorafenib. HCC often arises in the background of chronic inflammation and is also associated with an immunosuppressed microenvironment, providing a strong rationale to evaluate immunotherapy in HCC. The randomized, double-blind, placebo-controlled phase 3 KEYNOTE-240 was designed to compare the efficacy and safety of the anti–PD-1 antibody pembrolizumab + best supportive care (BSC) vs placebo + BSC in patients with previously treated advanced HCC. Methods: Eligibility criteria include age ≥18 years, confirmed diagnosis of Barcelona Clinic Liver Cancer (BCLC) stage C disease or BCLC stage B disease not amenable to or refractory to locoregional therapy, disease not amenable to a curative treatment approach (e.g., transplantation, surgery, or ablation), documented progression after treatment with sorafenib or intolerance to sorafenib, Child-Pugh liver score A, and ECOG performance status 0-1

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with sorafenib or intolerance to sorafenib, Child-Pugh liver score A, and ECOG performance status 0-1. ~408 patients will be randomly assigned 2:1 to receive pembrolizumab 200 mg IV every 3 weeks (Q3W) + BSC or placebo Q3W + BSC for up to 35 cycles or until disease progression, unacceptable toxicity, patient withdrawal of consent, or investigator decision. Randomization will be stratified by geographic region, macrovascular invasion, and α-fetoprotein. BSC will be provided by the investigator per local treatment practices. Response will be assessed every 6 weeks per RECIST v1.1 by central imaging vendor review. Adverse events (AEs) will be assessed throughout treatment and for 30 days thereafter (90 days for serious AEs) and graded per NCI CTCAE v4.0. Primary objectives are comparison of progression-free survival per RECIST v1.1 by central imaging vendor review and overall survival between treatment arms. Secondary objectives are comparison of objective response rate, duration of response, disease control rate, and time to progression per RECIST v1.1 by central imaging vendor review, and evaluation of safety and tolerability. Enrollment in KEYNOTE-240 is ongoing.

Hodgkin's Lymphoma

Kroemer and Galluzzi (2015) noted that the blockade of immunological checkpoints has been studied for the treatment of various solid neoplasms including melanoma, mesothelioma, non-small cell lung carcinoma, and renal cell carcinoma. A recent study indicated that the vast majority of patients with advanced, heavily pretreated Hodgkin's lymphoma (HL) also respond to a monoclonal antibody targeting programmed cell death 1 (PDCD1, best known as PD-1). Thus, checkpoint blockers may soon become part of the therapeutic armamentarium against hematological tumors. The authors anticipated that the realm of immunotherapy will eventually conquer vast portions of the territory that now belongs to hematological malignancies.

The FDA approved pembrolizumab for adults and children with classical Hodgkin lymphoma who are refractory to treatment or who have relapsed after three or more prior lines of therapy (Merck, 2017). The product labeling states that this indication was approved under accelerated approval based on tumor response rate and durability of response. The labeling states that continued approval for this indication may be contingent upon verification and description of clinical benefit in the confirmatory trials.

The accelerated FDA approval was based on data in 210 patients with relapsed or refractory classical Hodgkin lymphoma who were enrolled in the multicenter, nonrandomized, open-label KEYNOTE-087 study (Merck, 2017). Patients with active, non-infectious pneumonitis, an allogeneic hematopoietic stem cell transplant (HSCT) within the past five years (or greater than five years but with symptoms of graft-versus-host disease (GVHD)), active autoimmune disease, a medical condition that required immunosuppression, or an active infection requiring systemic therapy were ineligible for the trial. Patients received pembrolizumab at a dose of 200 mg every three weeks until unacceptable toxicity or documented disease progression, or for up to 24 months in patients who did not progress. The major efficacy outcome measures (overall response rate (ORR), complete remission rate (CRR), and duration of response) were assessed by blinded independent central review according to the 2007 revised International Working Group (IWG) criteria. Fifty-eight percent (58%) of patients were refractory to the last prior therapy, including 35 percent with primary refractory disease and 14 percent whose disease

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was chemorefractory to all prior regimens. Additionally, 61 percent of patients had undergone prior auto-HSCT, 17 percent had no prior brentuximab use, and 36 percent had prior radiation therapy. The

median follow-up time was 9.4 months. Efficacy analysis showed an ORR of 69 percent (95% CI: 62, 75), which included a CRR of 22 percent and a partial remission rate (PRR) of 47 percent. Median duration of response among the 145 responders was 11.1 months (range, 0+ to 11.1).

Pembrolizumab was discontinued due to adverse reactions in five percent of 210 patients with classical Hodgkin lymphoma and treatment was interrupted due to adverse reactions in 26 percent of patients (Merck, 2017). Fifteen percent (15%) of patients had an adverse reaction requiring systemic corticosteroid therapy. Serious adverse reactions occurred in 16 percent of patients. The most frequent serious adverse reactions (≥1%) included pneumonia, pneumonitis, pyrexia, dyspnea, GVHD, and herpes zoster. Two patients died from causes other than disease progression; one from GVHD after subsequent allogeneic HSCT and one from septic shock. The most common adverse reactions (occurring in ≥20% of patients) were fatigue (26%), pyrexia (24%), cough (24%), musculoskeletal pain (21%), diarrhea (20%), and rash (20%).

Immune-mediated complications, including fatal events, occurred in patients with classical Hodgkin lymphoma who underwent allogeneic hematopoietic stem cell transplantation (HSCT) after being treated with pembrolizumab. (Merck, 2017) The labeling recommends following patients closely for early evidence of transplant-related complications, and intervene promptly.

Data on pediatric patients are limited and efficacy was extrapolated from the results in the adult trial (Merck, 2017).

In refractory or relapsed classical Hodgkin lymphoma, pembrolizumab is approved for use in adult patients at a fixed dose of 200 mg and in pediatric patients at a dose of 2 mg/kg (up to a maximum of 200 mg). Pembrolizumab is administered intravenously every three weeks until disease progression or unacceptable toxicity, or up to 24 months in patients without disease progression.

On October 14, 2020, the Food and Drug Administration extended the approval of pembrolizumab (Keytruda) for the following indications:

- adult patients with relapsed or refractory classical Hodgkin lymphoma (cHL) and
- pediatric patients with refractory cHL, or cHL that has relapsed after 2 or more lines of therapy.

Previously, pembrolizumab was approved for the treatment of adult and pediatric patients with refractory cHL, or who have relapsed after 3 or more prior lines of therapy.

Approval was based on KEYNOTE-204 (NCT02684292), a phase 3, randomized, open-label trial in 304 adult patients with relapsed or refractory cHL after at least one multiagent regimen. Patients were randomized (1:1) to receive either pembrolizumab 200 mg every 3 weeks or brentuximab vedotin (BV) 1.8 mg/kg every 3 weeks for up to 2 years. Efficacy was based on progression-free survival (PFS) per

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1.8 mg/kg every 3 weeks for up to 2 years. Efficacy was based on progression-free survival (PFS) per blinded independent central review assessment. PFS was statistically significantly longer in the pembrolizumab arm. The median PFS was 13.2 months (95% CI: 10.9, 19.4) in the pembrolizumab arm and 8.3 months (95% CI: 5.7, 8.8) in the BV arm, with a hazard ratio of 0.65 (95% CI: 0.48, 0.88; p=0.0027).

Serious adverse reactions occurred in 30% of the patients who received pembrolizumab. Serious adverse reactions in ≥1% of patients included pneumonitis, pneumonia, pyrexia, myocarditis, acute kidney injury, febrile neutropenia, and sepsis. Adverse reactions in ≥20% of pembrolizumab recipients included upper respiratory tract infection, musculoskeletal pain, diarrhea, cough, pyrexia, fatigue, and rash. Thirty-eight percent of patients had adverse reactions requiring systemic corticosteroids, including pneumonitis in 11%.

The recommended pembrolizumab dose for patients with lymphoma is 200 mg every 3 weeks or 400 mg every 6 weeks intravenously for adults, or 2 mg/kg (up to 200 mg) every 3 weeks intravenously for pediatric patients, for up to 2 years.

Melanoma

Melanoma is a serious form of skin cancer that arises from melanocytes; however, in rare instances, it can originate in the eye or other non-skin organs. Risk factors for melanoma entail freckling, genetic factors, history of sunburns, light skin or eye color, poor tanning ability, sun exposure, as well as tanning bed use. About 80 % of melanomas are detected in a localized stage. When detected early, the 5-year survival rate of melanoma is 98.5 %. However, when melanoma is diagnosed after distant metastasis, the 5-year survival rate decreases to 15 % with a median survival between 8 and 9 months. Tumor thickness, along with nodal involvement, is a prognostic factor for melanoma. As tumor thickness increases to greater than 1.0 mm, the survival rate is reduced by 50 %. The incidence of melanoma is rapidly increasing. According to the National Cancer Institute (NCI), melanoma is the leading cause of death from skin disease. Approximately 76.100 new cases of melanoma will be diagnosed in the United States in 2014 and about 9,700 people will die from the disease. Treatments of melanoma include chemotherapy, immunotherapy, radiation therapy, surgery, as well as vaccine therapy (NCI, 2014). Moreover, the National Comprehensive Cancer Network (NCCN)'s clinical practice guideline on Melanoma: Cutaneous" (Version 3.2023) states that little consensus exists regarding the optimal standard chemotherapy for patients with metastatic melanoma. Enrollment in a clinical trial, if available, is the preferred choice.

The past several years were marked by the emergence of new molecules for the treatment of metastatic cutaneous melanoma with a significant benefit on the survival -- ipilimumab (Yervoy), peginterferon alfa-2b (PEG-Intron) and vemurafenib (Zelboraf) were approved by the Food and Drug Administration (FDA) in 2011; dabrafenib (Tafinlar) and trametinib (Mekinist) were approved in 2013. More recently, much attention has been devoted to the blockade of programmed death 1 (PD-1) signaling to activate an immune response to cancer. PD-1, a protein expressed on T cells, is a member of the CD28 superfamily,

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and it transmits co-inhibitory signals upon engagement with its ligands PD-L1 and PD-L2. Accumulating evidence suggests that the PD-1 system plays pivotal roles in the regulation of

autoimmunity, transplantation immunity, infectious immunity, and tumor immunity. Because the interaction of PD-1 with its ligands occurs in the effector phase of killer T cell responses in peripheral blood, anti-PD-1 and anti-PD-L1 monoclonal antibodies are ideal as specific agents to augment T cell responses to tumors with fewer adverse events than with the inhibition of CTLA-4, because the interaction of CTLA-4 with its ligands occurs in the priming phase of T cell responses within lymph nodes. In recent phase I clinical trials, objective responses were observed in patients with melanoma, renal cell carcinoma, and non-small cell lung cancer (NSCLC) who underwent immunotherapy with an anti-PD-1 monoclonal antibody. In addition, the anti-tumor activity of an anti-PD-L1 monoclonal antibody was observed in patients with melanoma, renal cell carcinoma, NSCLC, and ovarian cancer (Tanaka and Okamura, 2013).

Pembrolizumab (formerly known as lambrolizumab; MK-3475), a negative regulator of T-cell effector mechanisms that limits immune responses against cancer, is an agent that targets the PD-1 receptor. Blockade of interactions between PD-1 and PD-L1 enhances immune function in-vitro and mediates anti-tumor activity in pre-clinical models. Moreover, pembrolizumab has shown potent anti-tumor activity at different doses and schedules in patients with melanoma.

Keytruda (pembrolizumab) is indicated for the treatment of patients with unresectable or metastatic melanoma and for metastatic non-small cell lung cancer where disease progression on or following chemotherapy and patients with EGFR or ALK genomic tumor aberrations should have disease progression after failure of prior approved therapy for these aberrations.

Binding of the programmed death-1 (PD-1) ligands, PD-L1 and PD-L2, to the PD-1 receptor found on T cells, inhibits T cell proliferation and cytokine production. Upregulation of PD-1 ligands occurs in some tumors and signaling through this pathway can contribute to inhibition of active T-cell immune surveillance of tumors. Keytruda (pembrolizumab) is a monoclonal antibody that binds to the PD-1 receptor and blocks its interaction with PD-L1 and PD-L2, releasing PD-1 pathway-mediated inhibition of the immune response, including the anti-tumor immune response. In syngeneic mouse tumor models, blocking PD-1 activity resulted in decreased tumor growth.

Hamid et al (2013) tested the anti-PD-1 antibody lambrolizumab (previously known as MK-3475) in patients with advanced melanoma. These investigators administered lambrolizumab intravenously at a dose of 10 mg/kg body weight every 2 or 3 weeks or 2 mg/kg every 3 weeks in patients with advanced melanoma, both those who had received prior treatment with ipilimumab and those who had not. Tumor responses were assessed every 12 weeks. A total of 135 patients with advanced melanoma were treated. Common adverse events (AEs) attributed to treatment were fatigue, rash, pruritus, and diarrhea; most of the AEs were low grade. The confirmed response rate across all dose cohorts, evaluated by central radiologic review according to the Response Evaluation Criteria in Solid Tumors (RECIST), version 1.1, was 38 % (95 % confidence interval [CI]: 25 to 44), with the highest confirmed response rate observed in the cohort that received 10 mg/kg every 2 weeks (52 %; 95 % CI: 38 to 66). The response rate did not

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in the cohort that received 10 mg/kg every 2 weeks (52 %, 95 % CI: 38 to 66). The response rate did not differ significantly between patients who had received prior ipilimumab treatment and those who had not (confirmed response rate, 38 % [95 % CI: 23 to 55] and 37 % [95 % CI: 26 to 49], respectively). Responses were durable in the majority of patients (median follow-up of 11 months among patients who had a response); 81 % of the patients who had a response (42 of 52) were still receiving treatment at the time of analysis in March 2013. The overall median progression-free survival (PFS) among the 135 patients was longer than 7 months. The authors concluded that in patients with advanced melanoma, including those who had had disease progression while they had been receiving ipilimumab, treatment with lambrolizumab resulted in a high rate of sustained tumor regression, with mainly grade 1 or 2 toxic effects.

In an open-label, international, multi-center, expansion cohort of a phase 1 clinical trial, Robert and colleagues (2014) compared the safety and effectiveness of pembrolizumab at doses of 2 mg/kg and 10 mg/kg every 3 weeks in patients with ipilimumab-refractory advanced melanoma. Patients (aged greater than or equal to 18 years) with advanced melanoma whose disease had progressed after at least 2 ipilimumab doses were randomly assigned with a computer-generated allocation schedule (1:1 final ratio) to intravenous pembrolizumab at 2 mg/kg every 3 weeks or 10 mg/kg every 3 weeks until disease progression, intolerable toxicity, or consent withdrawal. Primary end-point was overall response rate (ORR) assessed with RECIST, version 1.1 by independent central review. Analysis was done on the full-analysis set (all treated patients with measurable disease at baseline). A total of 173 patients received pembrolizumab 2 mg/kg (n = 89) or 10 mg/kg (n = 84). Median follow-up duration was 8 months. Overall response rate was 26 % at both doses: 21 of 81 patients in the 2 mg/kg group and 20 of 76 in the 10 mg/kg group (difference 0 %, 95 % CI: -14 to 13; p = 0·96). Treatment was well-tolerated, with similar safety profiles in the 2 mg/kg and 10 mg/kg groups and no drug-related deaths. The most common drug-related AEs of any grade in the 2 mg/kg and 10 mg/kg groups were fatigue (29 [33 %] versus 31 [37 %]), pruritus (23 [26 %] versus 16 [19 %]), and rash (16 [18 %] versus 15 [18 %]). Grade-3 fatigue, reported in 5 (3 %) patients in the 2 mg/kg pembrolizumab group, was the only drug-related grade 3 to 4 AE reported in more than 1 patient. The authors concluded that these findings suggested that pembrolizumab at a dose of 2 mg/kg or 10 mg/kg every 3 weeks might be an effective treatment in patients for whom there are few effective treatment options.

On September 4, 2014, the FDA approved pembrolizumab (Keytruda) for the treatment of patients with advanced or unresectable melanoma who are no longer responding to other drugs. Keytruda is the first approved drug that blocks the PD-1 pathway, which restricts the body's immune system from attacking melanoma cells. Keytruda is intended for use following treatment with ipilimumab. For melanoma patients whose tumors express a gene mutation called BRAF V600, Keytruda is intended for use after treatment with ipilimumab and a BRAF inhibitor, a therapy that blocks activity of BRAF gene mutations. The FDA granted Keytruda breakthrough therapy designation because the sponsor demonstrated through preliminary clinical evidence that the drug may offer a substantial improvement over available therapies. It also received priority review and orphan product designation. Priority review is granted to drugs that have the potential, at the time the application was submitted, to be a significant improvement in safety or effectiveness in the treatment of a serious condition. Orphan product designation is given to drugs intended to treat rare diseases. The effectiveness of Keytruda was

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product designation is given to drugs intended to treat rare diseases. The effectiveness of Keytruda was established in 173 subjects with advanced melanoma whose disease progressed after prior treatment. All participants were treated with Keytruda, either at the recommended dose of 2 mg/kg or at a higher dose of 10 mg/kg. In the half of the participants who received Keytruda at the recommended dose of 2 mg/kg, approximately 24 % had their tumors shrink. This effect lasted at least 1.4 to 8.5 months and continued beyond this period in most patients. A similar percentage of patients had their tumor shrink at the 10 mg/kg dose. The most common side effects of Keytruda were fatigue, cough, nausea, pruritus, rash, decreased appetite, constipation, arthralgia and diarrhea. Keytruda also has the potential for severe immune-mediated side effects involving healthy organs, including the lung, colon, hormone-producing glands and liver, occurred uncommonly.

On February 15, 2019, the Food and Drug Administration approved pembrolizumab (KEYTRUDA, Merck) for the adjuvant treatment of patients with melanoma with involvement of lymph node(s) following complete resection. Approval was based on EORTC1325/KEYNOTE-054 (Eggermont, 2018), a randomized, double-blind, placebo-controlled, trial in 1019 patients with completely resected, stage IIIA (>1 mm lymph node metastasis), IIIB or IIIC melanoma (AJCC 7th ed). Patients with mucosal or ocular melanoma were not eligible. Eggermont et al (2018; NCT02362594) stated that the programmed death 1 (PD-1) inhibitor pembrolizumab has been found to prolong progression-free and overall survival among patients with advanced melanoma. This was a multicenter, randomized, double-blind, placebo-controlled trial to evaluate pembrolizumab as adjuvant therapy in patients with completely resected stage IIIA (>1 mm lymph node metastasis), IIIB or IIIC melanoma. Patients were randomly assigned (with stratification according to cancer stage and geographic region) to receive 200 mg of pembrolizumab (514 patients) or placebo (505 patients) intravenously every 3 weeks for a total of 18 doses (approximately 1 year) or until disease recurrence or unacceptable toxic effects occurred. Recurrence-free survival (RFS) in the overall intention-to-treat population and in the subgroup of patients with cancer that was positive for the PD-1 ligand (PD-L1) were the primary end points. RFS was defined as the time between the date of randomization and first recurrence (local, regional, or distant metastasis) or death from any cause, whichever occurred first. Safety was also evaluated. At a median follow-up of 15 months, pembrolizumab was associated with significantly longer recurrence-free survival than placebo in the overall intention-to-treat population (1-year rate of recurrence-free survival, 75.4% [95% confidence interval {CI}, 71.3 to 78.9] vs. 61.0% [95% CI, 56.5 to 65.1]; hazard ratio for recurrence or death, 0.57; 98.4% CI, 0.43 to 0.74; P<0.001) and in the subgroup of 853 patients with PD-L1-positive tumors (1-year rate of recurrence-free survival, 77.1% [95% CI, 72.7 to 80.9] in the pembrolizumab group and 62.6% [95% CI, 57.7 to 67.0] in the placebo group; hazard ratio, 0.54; 95% CI, 0.42 to 0.69; P<0.001). Adverse events of grades 3 to 5 that were related to the trial regimen were reported in 14.7% of the patients in the pembrolizumab group and in 3.4% of patients in the placebo group. There was one treatment-related death due to myositis in the pembrolizumab group. The authors concluded that as adjuvant therapy for high-risk stage III melanoma, 200 mg of pembrolizumab administered every 3 weeks for up to 1 year resulted in significantly longer recurrence-free survival than placebo, with no new toxic effects identified.

According to the Prescribing Information, Keytruda (pembrolizumab) is indicated for the treatment of patients with unresectable or metastatic melanoma and for the adjuvant treatment of patients with

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patients with unresectable or metastatic melanoma and for the adjuvant treatment of patients with melanoma with involvement of lymph node(s) following complete resection.

On December 3, 2021, the U.S. Food and Drug Administration (FDA) approved Keytruda for the adjuvant treatment of adult and pediatric (12 years and older) patients with stage IIB or IIC melanoma following complete resection. The FDA approval was based on supporting data from the first interim analysis of the KEYNOTE-716 study, a multicenter, randomized (1:1), double-blind, placebo-controlled phase 3 trial. The study investigators concluded that Keytruda demonstrated a statistically significant improvement in recurrence-free survival, reducing the risk of disease recurrence or death compared to placebo (Merck, 2021d).

Merkel Cell Carcinoma

Winkler et al (2016) reported the use of an anti-PD-1 antibody for treatment of a patient with metastatic MCC. An 80-yearold patient with metastatic MCC received off-label treatment with the anti-PD-1 antibody pembrolizumab after the disease had progressed during therapy with oral etoposide. A PET-CT scan performed after 3 cycles of pembrolizumab revealed response to therapy with reduced size of the adrenal gland metastases and less PET activity in the adrenal gland and lymph node metastases. Further on, treatment was resumed due to disease progression in a treatment-free interval of more than 4 months. During subsequent months of treatment size of the metastases stabilized and uptake of nuclide by all tumor sites once again decreased. The authors concluded that these results revealed the potential effectiveness of an anti-PD-1 antibody for treatment of metastatic MCC. They thus contribute to currently limited data on use of anti-PD-1 antibodies for treatment of MCC. This is, moreover, the 1st report of successful resumption of treatment of metastatic MCC with an anti-PD-1 antibody. Moreover, they stated that results from ongoing trials will contribute to determination of the relevance of PD-1 blockade in metastatic MCC.

Nghiem et al (2016) stated that MCC is an aggressive skin cancer that is linked to exposure to ultraviolet light and the Merkel-cell polyomavirus (MCPyV). Advanced MCC often responds to chemotherapy, but responses are transient. Blocking the programmed death 1 (PD-1) immune inhibitory pathway is of interest, because these tumors often express PD-L1, and MCPyV-specific T cells express PD-1. In this multi-center, phase II, non-controlled study, these researchers assigned adults with advanced MCC who had received no previous systemic therapy to receive pembrolizumab (anti-PD-1) at a dose of 2 mg/kg of body weight every 3 weeks. The primary end-point was the objective response rate according to RECIST, version 1.1. Efficacy was correlated with tumor viral status, as assessed by serologic and immune-histochemical testing. A total of 26 patients received at least 1 dose of pembrolizumab. The objective response rate among the 25 patients with at least 1 evaluation during treatment was 56 % (95 % confidence interval [CI]: 35 to 76); 4 patients had a complete response, and 10 had a partial response. With a median follow-up of 33 weeks (range of 7 to 53), relapses occurred in 2 of the 14 patients who had had a response (14 %). The response duration ranged from at least 2.2 months to at least 9.7 months. The rate of progression-free survival (PFS) at 6 months was 67 % (95 % CI: 49 to 86). A total of 17 of the 26 patients (65 %) had virus-positive tumors. The response rate was 62 % among patients

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with MCPyV-positive tumors (10 of 16 patients) and 44 % among those with virus-negative tumors (4 of 9 patients). Drug-related grade 3 or 4 adverse events occurred in 15 % of the patients. The authors

concluded that in this study, 1st-line therapy with pembrolizumab in patients with advanced MCC was associated with an objective response rate of 56 %. Responses were observed in patients with virus-positive tumors and those with virus-negative tumors. Also, the authors stated that "As of now, none of the predictive tests are sufficiently robust to be used in clinical decision making regarding whether to use or not to use PD-1 blockers in Merkel-cell carcinoma". Additional investigation with longer follow-up and larger patient cohorts are needed to ascertain the effectiveness of pembrolizumab for the treatment of Merkel-cell carcinoma.

In a phase II single-arm study, the PD-1 checkpoint inhibitor pembrolizumab effectively shrank tumors and prolonged survival in more than 50 %of patients with MCC. Responses were significantly more durable than those typically seen with chemotherapy (No authors listed, 2016).

On December 19, 2018, the U.S. Food and Drug Administration (FDA) approved pembrolizumab (Keytruda) for adult and pediatric patients with recurrent locally advanced or metastatic Merkel cell carcinoma (MCC). The efficacy of pembrolizumab was investigated in KEYNOTE-017 (NCT02267603), a multicenter, nonrandomized, open-label trial that enrolled 50 patients with recurrent locally advanced or metastatic MCC who had not received prior systemic therapy for their advanced disease. Patients with active autoimmune disease or a medical condition that required immunosuppression were ineligible. Patients received pembrolizumab 2 mg/kg every 3 weeks until unacceptable toxicity or disease progression that was symptomatic, rapidly progressive, required urgent intervention, occurred with a decline in performance status, or was confirmed at least 4 weeks later with repeat imaging. Patients without disease progression were treated for up to 24 months. Assessment of tumor status was performed at 13 weeks followed by every 9 weeks for the first year and every 12 weeks thereafter. The study population characteristics were: median age of 71 years (range: 46 to 91), 80% age 65 or older; 68% male; 90% White; and 48% ECOG PS of 0 and 52% ECOG PS of 1. Fourteen percent had stage IIIB disease and 86% had stage IV. Eighty-four percent of patients had prior surgery and 70% had prior radiation therapy. The major efficacy outcome measures were overall response rate (ORR) and response duration assessed by blinded independent central review per RECIST 1.1. The ORR was 56% (95% CI: 41, 70) with a complete response rate of 24%. The median response duration was not reached. Among the 28 patients with responses, 96% had response durations of greater than 6 months and 54% had response durations of greater than 12 months. The most common adverse reactions of pembrolizumab reported in at least 20% of patients who received pembrolizumab as a single agent were fatigue, musculoskeletal pain, decreased appetite, pruritus, diarrhea, nausea, rash, pyrexia, cough, dyspnea, constipation, pain, and abdominal pain. The recommended pembrolizumab dose for MCC is 200 mg administered as a 30-minute intravenous infusion every 3 weeks for adults; 2 mg/kg (to a maximum of 200 mg) administered as a 30-minute intravenous infusion every 3 weeks for patients less than 18 years of age (pediatric patients).

Mesothelioma

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An UpToDate review on “Systemic treatment for unresectable malignant pleural mesothelioma” (Tsao, 2023) states that “Immunotherapy using checkpoint inhibitors offers a more promising direction for disease control. In a separate study, treatment of 25 mesothelioma patients (88 percent with previous systemic therapy) with pembrolizumab resulted in five partial responses and 13 patients with stable disease, for an overall disease control rate of 72 percent. All responding patients remained on therapy at the time of analysis. Additional clinical studies with pembrolizumab are being initiated”.

Neuroendocrine Tumors

Mehnert and colleagues (2020) noted that despite a protracted disease course and multiple available therapies, patients with well-differentiated neuroendocrine tumors (NETs) inevitably experience disease progression; and PD-L1 has been associated with NET progression and prognosis. The multi-cohort, phase-I KEYNOTE-028 Trail examined the activity and safety pembrolizumab in patients with well-differentiated or moderately-differentiated NETs. Patients with PD-L1-positive, locally advanced or metastatic carcinoid or well-differentiated or moderately-differentiated pancreatic NETs (pNETs) were enrolled into separate cohorts and received pembrolizumab at a dose of 10 mg/kg every 2 weeks for up to 2 years. The primary endpoint was ORR (as per RECIST version 1.1, by investigator review); secondary endpoint was safety. Of 170 and 106 patients, respectively, who had evaluable samples among those screened for the carcinoid and pNET cohorts, 21 % and 25 %, respectively, had PD-L1-positive tumors; of these, 25 and 16 patients, respectively, were eligible and treated. The median follow-up was 20 months (range of 2 to 35 months) and 21 months (range of 5 to 32 months), respectively. The ORR was 12.0 % (95 % CI: 2.5 % to 31.2 %) and 6.3 % (95 % CI: 0.2 % to 30.2 %), respectively; 3 PRs occurred among the carcinoid cohort and 1 among the pNET cohort. The median duration of response in the carcinoid cohort was 9.2 months (range of 6.9 to 11.1 months); and was not reached in the pNET cohort; no CRs occurred. Treatment-related AEs occurred in 68 % and 69 % of patients, respectively, most often diarrhea (7 patients in the carcinoid cohort and 4 patients in the pNET cohort) and fatigue (6 patients in each cohort). Hypothyroidism was the most common immune-mediated AE (5 patients in the carcinoid cohort and 2 patients in the pNET cohort). The authors concluded that this study of anti-PD-1 therapy for the treatment of NETs demonstrated anti-tumor activity and manageable safety in a subset of patients; thus, supporting further evaluation of immune checkpoint blockers in NETs in larger studies. Evaluation of the primary tumor site, differentiation, Ki-67 index, and biomarkers within the tumor microenvironment in patients with carcinoid and pNETs is needed to identify those patients who are more likely to respond to pembrolizumab or to have prolonged stable disease (SD). The phase-II KEYNOTE-158 Trial is evaluating pembrolizumab in larger patient cohorts with advanced solid tumors, including NETs, and additional biomarkers.

The authors stated that this study had several drawbacks. As a phase-Ib study, the population size in each cohort was relatively small, and there was no comparator group. The KEYNOTE-028 Trial was initiated prior to the availability of evidence suggesting a potential role for anti-PD-1 therapy

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irrespective of PD-L1 expression, and was designed to identify a signal for anti-tumor activity in numerous solid tumor cohorts and thus included a patient population enriched for potential response to

treatment. Specifically, KEYNOTE-028 enrolled patients with PD-L1-positive, advanced or metastatic carcinoid or well-differentiated or moderately-differentiated pNETs who had no available standard therapy (the majority had received multiple prior lines of therapy); therefore, the results should not be generalized beyond these populations. Furthermore, because patients were not required to have evidence of disease progression at baseline and these tumors can remain stable for long periods, it is possible that some patients entered the study with SD, and during the study either continued to have SD despite treatment or, as noted above, had a prolonged period of SD resulting from study treatment. It should be noted that the potential role of anti-PD-1 therapy in patients with more aggressive, poorly differentiated NETs, which often are more responsive to chemotherapy, is unknown. Specific information regarding tumor grade, mitotic rate, and Ki-67 index, which could influence response to pembrolizumab, was not collected in the present study and it is unclear whether some patients in the pNET cohort may have had higher grade tumors falling into the recently defined grade-3 category that included both well-differentiated and poorly-differentiated tumors, which can be challenging to distinguish. Finally, these investigators used a prototype immunohistochemistry assay for PD-L1 testing because this study began prior to FDA approval of the companion diagnostic assay in certain tumor types.

Non-Small Cell Lung Cancer (NSCLC)

Langer (2015) noted that immune checkpoint inhibition as a new treatment approach is undergoing extensive investigation in NSCLC and other malignancies. Unlike standard chemotherapy or targeted agents, which act directly on the tumor cells, immune checkpoint inhibitors work by restoring the immune system's capacity to eradicate tumors. Agents currently in active clinical development for lung cancer include ipilimumab, which modulates the cytotoxic T-lymphocyte-associated antigen 4 pathway, and multiple agents targeting the PD-1 pathway, both anti-PD-1 compounds (nivolumab, pembrolizumab [MK-3475]) and those that target PD-L1, a key ligand for PD-1 (BMS-936559, MPDL3280A). The authors stated that preliminary evidence showed activity for these agents in NSCLC as monotherapy or in combination with chemotherapy.

Asmar and Rizvi (2015) stated that lung cancers are immunogenic tumors that manage to evade the immune system by exploiting checkpoint pathways that render effector T cells anergic. Inhibition of these checkpoints can restore and invigorate endogenous antitumor T-cell responses. The immunotherapeutic approach of checkpoint inhibition has become an important treatment option for patients with advanced NSCLC, playing a role that will continue to evolve over the coming years. The PD-1 inhibitors nivolumab and pembrolizumab have both been shown to induce durable responses and improve survival in a subset of patients with platinum-refractory metastatic HSCLC.

Sgambato and colleagues (2016) noted that the better understanding of immunology and anti-tumor immune responses have prompted the development of novel immunotherapy agents like PD-1

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checkpoint inhibitors (anti-PD-1 and anti-PDL-1 antibodies) that improve the capacity of the immune system to acknowledge and delete tumors, including lung cancer. Currently, 2 anti-PD-1 (nivolumab

and pembrolizumab) and 1 anti- PD-L1 (MPDL-3280A) agents are in advanced stages of development in advanced or metastatic NSCLC. Among these, nivolumab demonstrated a survival benefit versus docetaxel in refractory squamous NSCLC, reporting 41 % reduction in risk of death (median overall survival [OS]: 9.2 versus 6.0 months; ORR: 20 % versus 9 %), and better safety profile than standard-of-care chemotherapy (grade 3 to 4 adverse events: 7 % versus 55 %). However, the enhancement of immune response to cancer targeting specific immune regulatory checkpoints is associated with a toxicity profile different from that related to traditional chemotherapeutic agents and molecularly targeted therapies.

In a phase I clinical trial, Garon et al (2015) evaluated the safety and effectiveness of programmed cell death 1 (PD-1) inhibition with pembrolizumab in patients with advanced NSCLC. These researchers also sought to define and validate an expression level of the PD-1 ligand 1 (PD-L1) that is associated with the likelihood of clinical benefit. They assigned 495 patients receiving pembrolizumab (at a dose of either 2 mg or 10 mg per kilogram of body weight every 3 weeks or 10 mg per kilogram every 2 weeks) to either a training group (182 patients) or a validation group (313 patients). They assessed PD-L1 expression in tumor samples using immunohistochemical analysis, with results reported as the percentage of neoplastic cells with staining for membranous PD-L1 (proportion score). Response was assessed every 9 weeks by central review. Common side effects that were attributed to pembrolizumab were fatigue, pruritus, and decreased appetite, with no clear difference according to dose or schedule. Among all the patients, the objective response rate was 19.4 %, and the median duration of response was 12.5 months. The median duration of progression-free survival (PFS) was 3.7 months, and the median duration of overall survival (OS) was 12.0 months. PD-L1 expression in at least 50 % of tumor cells was selected as the cut-off from the training group. Among patients with a proportion score of at least 50 % in the validation group, the response rate was 45.2 %. Among all the patients with a proportion score of at least 50 %, median PFS was 6.3 months; median OS was not reached. The authors concluded that pembrolizumab had an acceptable side-effect profile and showed anti-tumor activity in patients with advanced NSCLC. PD-L1 expression in at least 50 % of tumor cells correlated with improved efficacy of pembrolizumab.

On October 2, 2015, the FDA granted accelerated approval for Keytruda (pembrolizumab) to treat patients with advanced (metastatic) NSCLC whose disease has progressed after other treatments and with tumors that express a protein called PD-L1. Keytruda is approved for use with a companion diagnostic, the PD-L1 IHC 22C3 pharmDx test, the first test designed to detect PD-L1 expression in NSCLC. The safety of Keytruda was studied in 550 patients with advanced NSCLC. The most common side effects of Keytruda included fatigue, decreased appetite, shortness of breath or impaired breathing (dyspnea) and cough. Keytruda also has the potential to cause severe side effects that result from the immune system effect of Keytruda. The effectiveness of Keytruda for this use was demonstrated in a subgroup of 61 patients enrolled within a larger multi-center, open-label, multi-part study. The subgroup consisted of patients with advanced NSCLC that progressed following platinum-based chemotherapy or, if appropriate, targeted therapy for certain genetic mutations (ALK or EGFR)

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based chemotherapy or, if appropriate, targeted therapy for certain genetic mutations (ALK or EGFR). This subgroup also had PD-L1 positive tumors based on the results of the 22C3 pharmDx diagnostic test. Study participants received 10 mg/kg of Keytruda every 2 or 3 weeks. The major outcome measure was ORR (percentage of patients who experienced complete and partial shrinkage of their tumors). Tumors shrank in 41 % of patients treated with Keytruda and the effect lasted between 2.1 and 9.1 months. In the 550 study participants with advanced NSCLC, severe immune-mediated side effects occurred involving the lungs, colon and hormone-producing glands. Other uncommon immune-mediated side effects were rash and inflammation of blood vessels (vasculitis). Women who are pregnant or breast-feeding should not take Keytruda because it may cause harm to a developing fetus or newborn baby. Across clinical studies, a disorder in which the body's immune system attacks part of the peripheral nervous system (Guillain-Barre syndrome) also occurred.

The FDA approved pembrolizumab in combination with pemetrexed (Alimta) and carboplatin (pem/carbo) for the first-line treatment of metastatic nonsquamous NSCLC, irrespective of PD-L1 expression (Merck, 2017). Under the FDA's accelerated approval regulations, this indication was approved based on tumor response rate and progression-free survival (PFS). Continued approval for this indication may be contingent upon verification and description of clinical benefit in the confirmatory trials.

The FDA approval was based on data from KEYNOTE-021, Cohort G1, in 123 previously untreated patients with metastatic nonsquamous NSCLC with no EGFR or ALK genomic tumor aberrations and irrespective of PD-L1 expression (Merck, 2017). The key eligibility criteria for this cohort were locally advanced or metastatic nonsquamous NSCLC, regardless of tumor PD-L1 expression status, and no prior systemic treatment for metastatic disease. Patients with autoimmune disease that required systemic therapy within two years of treatment; a medical condition that required immunosuppression; or who had received more than 30 Gy of thoracic radiation within the prior 26 weeks were ineligible. Patients in KEYNOTE-021G1 were randomized to receive pembrolizumab + pem/carbo (n=60) or pem/carbo alone (n=63). Patients in the pembrolizumab combination arm received pembrolizumab (200 mg), pemetrexed (500 mg/m2), and carboplatin (AUC 5 mg/mL/min) every three weeks for four cycles followed by pembrolizumab every three weeks. In the control arm, patients received pemetrexed (500 mg/m2) and carboplatin (AUC 5 mg/mL/min) alone for four cycles. At the investigator's discretion, maintenance pemetrexed (500 mg/m2) every three weeks was permitted in both treatment arms. Treatment with pembrolizumab continued until Response Evaluation Criteria in Solid Tumors (RECIST) 1.1-defined progression of disease as determined by blinded independent central review (BICR), unacceptable toxicity, or a maximum of 24 months. Administration of pembrolizumab was permitted beyond RECIST-defined disease progression if the patient was clinically stable and deriving clinical benefit as determined by the investigator.

The major efficacy outcome measure was ORR as assessed by BICR using RECIST 1.1 (Merck, 2017). Additional efficacy outcome measures were PFS as assessed by BICR using RECIST 1.1, duration of response, and overall survival (OS).

Findings from this cohort demonstrated an ORR with pembrolizumab + pem/carbo of 55 percent (95%

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Findings from this cohort demonstrated an ORR with pembrolizumab + pem/carbo of 55 percent (95% CI: 42, 68) compared to 29 percent (95% CI: 18, 41) for pem/carbo alone; all responses were partial responses (Merck, 2017). Among patients who received pembrolizumab + pem/carbo, 93 percent had a duration of response of six months or more (range 1.4+ to 13.0+ months) compared to 81 percent who received pem/carbo alone (range 1.4+ to 15.2+ months). In addition, findings demonstrated an improvement in PFS (HR 0.53 [95% CI, 0.31-0.91; p=0.0205]), with a median PFS of 13.0 months (95% CI, 8.3-not estimable) for patients treated with pembrolizumab + pem/carbo compared to 8.9 months (95% CI, 4.4-10.3) with pem/carbo alone.

Exploratory analyses found similar results in patients with or without PD-L1 expression, with an ORR in patients whose tumors did not express PD-L1 (TPS <1%) of 57 percent with pembrolizumab + pem/carbo compared to 13.0 percent with pem/carbo alone; in patients with PD-L1 TPS ≥1%, the ORR was 54 percent with pembrolizumab + pem/carbo compared to 38 percent with pem/carbo alone (Merck, 2017).

The product labeling state that, when administering pembrolizumab in combination with pem/carbo, pembrolizumab should be administered first prior to chemotherapy when given on the same day (Merck, 2017). In metastatic NSCLC, pembrolizumab is approved at a fixed dose of 200 mg administered as an intravenous infusion over 30 minutes every three weeks until disease progression, unacceptable toxicity, or up to 24 months in patients without disease progression; pemetrexed and carboplatin should be administered according to their FDA-approved labels.

In the KEYNOTE-021G1 trial, safety was evaluated in 59 patients who received pembrolizumab + pem/carbo and 62 patients who received pem/carbo alone (Merck, 2017). When pembrolizumab was administered in combination with pemetrexed and carboplatin, pembrolizumab was discontinued in 10% of 59 patients. The most common adverse reaction resulting in discontinuation of pembrolizumab (≥2%) was acute kidney injury (3.4%). Adverse reactions leading to interruption of pembrolizumab occurred in 39% of patients; the most common (≥2%) were fatigue (8%), neutrophil count decreased (8%), anemia (5%), dyspnea (3.4%), and pneumonitis (3.4%).The most common adverse reactions (≥20%) with pembrolizumab compared to carbo/pem alone were fatigue (71% vs 50%), nausea (68% vs 56%), constipation (51% vs 37%), rash (42% vs 21%), vomiting (39% vs 27%), dyspnea (39% vs 21%), diarrhea (37% vs 23%), decreased appetite (31% vs. 23%), headache (31% vs 16%), cough (24% vs 18%), dizziness (24%vs 16%), insomnia (24% vs 15%), pruritus (24% vs 4.8%), peripheral edema (22% vs 18%), dysgeusia (20% vs 11%), alopecia (20% vs 3.2%), upper respiratory tract infection (20% vs 3.2%), and arthralgia (15% vs 24%). The study was not designed to demonstrate a statistically significant difference in adverse reaction rates for pembrolizumab plus chemotherapy, as compared to chemotherapy alone, for any specified adverse reaction.

On January 26, 2023, the U.S. Food and Drug Administration (FDA) approved pembrolizumab (Keytruda) for adjuvant treatment following resection and platinum-based chemotherapy for stage IB (T2a ≥ 4 cm), II, or IIIA non-small cell lung cancer (NSCLC). The FDA approval was based on supporting data from the KEYNOTE-091 study, a multicenter, randomized, triple-blind, placebo-controlled trial. The investigators assessed the efficacy of Keytruda by conducting this study of 1177

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controlled trial. The investigators assessed the efficacy of Keytruda by conducting this study of 1177 patients with completely resected stage IB (T2a ≥ 4 cm), II, or IIIA NSCLC. The patients had not received neoadjuvant radiotherapy or chemotherapy and randomization was 1:1 where patients received Keytruda 200 mg or placebo intravenously every 3 weeks. Randomization was stratified by stage of disease, receipt of adjuvant chemotherapy, PD-L1 status, and geographic region. Treatment continued until Response Evaluation Criteria in Solid Tumors (RECIST) v1.1-defined disease recurrence as determined by the investigator, unacceptable toxicity or up to one year. Of the 1177 patients randomized, 1010 (86%) received adjuvant platinum-based chemotherapy following complete resection. The major efficacy outcome was investigator-assessed disease-free survival (DFS). The trial met its primary endpoint, demonstrating a statistically significant improvement in DFS in the overall population for patients randomized to the Keytruda arm versus patients randomized to the placebo arm. In an exploratory subgroup analysis of the 167 patients who did not receive adjuvant chemotherapy, the DFS hazard ratio was 1.25 (95% Confidence Interval [CI]: 0.76, 2.05). For patients who received adjuvant chemotherapy, median DFS was 58.7 months in the Keytruda arm (95% CI: 39.2, not reached) and 34.9 months in the placebo arm (95% CI: 28.6, not reached)(hazard ratio = 0.73; 95% CI: 0.60, 0.89]).The adverse reactions observed in this study were generally similar to those occurring in other patients with NSCLC receiving Keytruda as a single agent, with the exception of hypothyroidism (22%), hyperthyroidism (11%), and pneumonitis (7%). Two fatal adverse reactions of myocarditis were reported (FDA, 2023b; Merck, 2023).

On October 16, 2023, the U.S. Food and Drug Administration (FDA) approved pembrolizumab (Keytruda) with platinum-containing chemotherapy as neoadjuvant treatment, and with continuation of single-agent Keytruda as post-surgical adjuvant treatment for resectable (tumors ≥4 cm or node positive) non-small cell lung cancer (NSCLC). The FDA approval was based on supporting data from the KEYNOTE-671 study, a multicenter, randomized, double-blind, placebo-controlled trial. Investigators evaluated the efficacy of Keytruda in 797 patients with previously untreated and resectable Stage II, IIIA, or IIIB (N2) by AJCC 8th edition. Patients were randomized (1:1) to either Keytruda or placebo, with platinum-based chemotherapy, every 3 weeks for 4 cycles (neoadjuvant treatment) followed by either continued single-agent Keytruda or placebo, every 3 weeks for up to 13 cycles (adjuvant treatment). The major efficacy out measures were overall survival (OS) and investigator-assessed event-free survival (EFS). Median OS was not reached in the Keytruda arm (95% CI: not estimable [NE], NE) and 52.4 months for those receiving placebo (95% CI: 45.7, NE) (hazard ratio [HR] 0.72 [95% CI: 0.56, 0.93]; p-value=0.0103). Median EFS was not reached in the pembrolizumab arm (95% CI: 34.1 months, NE) and 17 months in the placebo arm (95% CI: 14.3, 22.0) (HR 0.58 [95% CI: 0.46, 0.72]; p-value= <0.0001) (FDA, 2023a; Merck, 2024).

Pleural Mesothelioma

On September 17, 2024, the U.S. Food and Drug Administration (FDA) approved pembrolizumab (Keytruda) with pemetrexed and platinum chemotherapy as first-line treatment of unresectable advanced or metastatic malignant pleural mesothelioma (MPM). The FDA approval was based on supporting data from the KEYNOTE-483 study (FDA, 2024b).

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In the KEYNOTE-483 study, a multicenter, randomized, open-label, active-controlled trial, investigators evaluated the efficacy of pembrolizumab (Keytruda) in 440 enrolled patients with unresectable advanced or metastatic MPM and no prior systemic therapy for advanced/metastatic disease. Patient enrollment was regardless of PD-L1 expression. Additionally, patients with autoimmune disease that required systemic therapy within 3 years of treatment or a medical condition that required immunosuppression were ineligible. Patients randomization, stratified by histological subtype (epithelioid vs. non-epithelioid), was 1:1 to receive either Keytruda for up to 2 years in combination with pemetrexed and platinum-based chemotherapy for up to 6 cycles (n=222) or pemetrexed and platinum-based chemotherapy for up to 6 cycles (n=218). All study medications were given by intravenous infusion in the following treatment arms (FDA, 2024b; Merck, 2024):

- Keytruda 200 mg with pemetrexed 500 mg/m2 and cisplatin 75 mg/m2 or carboplatin area under the curve (AUC) 5-6 mg/mL/min on Day 1 of each 21-day cycle for up to 6 cycles, followed by Keytruda 200 mg every 3 weeks. Keytruda was given prior to chemotherapy on Day 1.
- Pemetrexed 500 mg/m2 and cisplatin 75 mg/m2 or carboplatin AUC 5-6 mg/mL/min on Day 1 of each 21-day cycle for up to 6 cycles.

Treatment with Keytruda was given until disease progression as determined by investigator according to modified Response Evaluation Criteria In Solid Tumors version 1.1 (RECIST 1.1) for mesothelioma (mRECIST), unacceptable toxicity, or a maximum of 24 months. Tumor status was assessed and performed every 6 weeks for 18 weeks, followed by every 12 weeks thereafter (Merck, 2024).

The primary efficacy outcome measure was overall survival (OS) and secondary efficacy outcome measures included progression-free survival (PFS), objective response rate (ORR), and duration of response (DoR), as assessed by blinded independent central review, according to modified RECIST 1.1 for mesothelioma. The trial showed a statistically significant improvement in OS for patients receiving Keytruda with chemotherapy compared to those receiving chemotherapy alone. Median OS was 17.3 months (95% confidence interval [CI]: 14.4, 21.3) compared to 16.1months (95% CI: 13.1, 18.2) (hazard ratio [HR] 0.79 [95% CI: 0.64, 0.98]; p=0.0162) (FDA, 2024b; Merck, 2024).

The median PFS was 7.1 months (95% CI: 6.9, 8.1) compared to 7.1 months (95% CI: 6.8, 7.7) in Keytruda plus chemotherapy and the chemotherapy alone arm, respectively (HR 0.80 [95% CI: 0.65, 0.99]; p=0.0194). Confirmed ORR was 52% (95% CI: 45.5, 59.0) in the Keytruda plus chemotherapy arm and 29% (95% CI: 23.0, 35.4) in the chemotherapy alone arm. Median DoR was 6.9 months (95% CI: 5.8, 8.3) and 6.8 months (95% CI: 5.5, 8.5), respectively (FDA, 2024b; Merck, 2024).

Primary Mediastinal Large B-cell Lymphoma (PMBCL)

On June 13, 2018, the Food and Drug Administration granted accelerated approval to pembrolizumab

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(Keytruda, Merck) for the treatment of adult and pediatric patients with refractory primary mediastinal large B-cell lymphoma (PMBCL), or who have relapsed after two or more prior lines of therapy. The

efficacy of pembrolizumab was investigated in 53 patients with relapsed or refractory PMBCL enrolled in a multicenter, open-label, single-arm trial (Study KEYNOTE-170; NCT02576990). Patients were not eligible if they had active non-infectious pneumonitis, allogeneic HSCT within the past 5 years (or greater than 5 years but with symptoms of GVHD), active autoimmune disease, a medical condition that required immunosuppression, or an active infection requiring systemic therapy. The patients were treated with pembrolizumab 200 mg intravenously every 3 weeks until unacceptable toxicity or documented disease progression, or for up to 24 months for patients who did not progress. Disease assessments were performed every 12 weeks and assessed by blinded independent central review according to the 2007 revised IWG criteria.

Among the 53 patients accrued, the baseline characteristics were: median age 33 years (range: 20 to 61 years), 43% male; 92% White; 43% had an ECOG performance status (PS) of 0 and 57% had an ECOG PS of 1. The median number of prior lines of therapy administered for the treatment of PMBCL was 3 (range 2 to 8). Thirty-six percent had primary refractory disease, 49% had relapsed disease refractory to the last prior therapy, and 15% had untreated relapse. Twenty-six percent of patients had undergone prior autologous HSCT, and 32% of patients had prior radiation therapy. All patients had received rituximab as part of a prior line of therapy.

An overall response rate (ORR) was seen in 45% of patients (95% CI: 32, 60); complete response was seen in 11% of patients; and partial response in 34% of patients. For the 24 responders, the median time to first objective response (complete or partial response) was 2.8 months (range 2.1 to 8.5 months).

Prostate Cancer

Hansen et al (2018) stated patients with castration-resistant prostate cancer derive only modest clinical benefit from available therapies. Blockade of the inhibitory programmed death 1 (PD-1) receptor by monoclonal antibodies has been effective in several malignancies. Results from the prostate adenocarcinoma cohort of the nonrandomized phase Ib KEYNOTE-028 trial of pembrolizumab in advanced solid tumors are presented. Key eligibility criteria included advanced prostate adenocarcinoma, unsuccessful standard therapy, measurable disease per Response Evaluation Criteria in Solid Tumors, version 1.1 (RECIST v1.1), and PD-1 ligand (PD-L1) expression in ≥1% of tumor or stromal cells. Patients received pembrolizumab 10 mg/kg every 2 weeks until disease progression or intolerable toxicity for up to 24 months. Primary end point was objective response rate (ORR) per RECIST v1.1 by investigator review. The median patient age in this cohort (n = 23) was 65 years; 73.9% of patients received at least two prior therapies for metastatic disease. There were four confirmed partial responses, for an ORR of 17.4% [95% confidence interval (CI) 5.0%-38.8%]; 8 of 23 (34.8%) patients had stable disease. Median duration of response was 13.5 months. Median progression-free survival (PFS) and overall survival (OS) were 3.5 and 7.9 months, respectively; 6-month PFS and OS rates were 34.8% and 73.4%, respectively. One patient remained on treatment at data cutoff. After a median follow-

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up of 7.9 months, 14 (60.9%) patients experienced treatment-related adverse events (TRAEs), most commonly nausea (n = 3, 13.0%). Four (17.3%) experienced grade 3/4 TRAEs: grade 3 peripheral

neuropathy, grade 3 asthenia, grade 3 fatigue, and grade 4 lipase increase. No pembrolizumab-related deaths or discontinuations occurred. The authors concluded that pembrolizumab resulted in durable objective response in a subset of patients with heavily pretreated, advanced PD-L1-positive prostate cancer, and its side effect profile was favorable.

Renal Cell Carcinoma (RCC)

On April 19, 2019, the Food and Drug Administration approved pembrolizumab plus axitinib (Inlyta) for the first-line treatment of patients with advanced renal cell carcinoma (RCC). The efficacy of pembrolizumab in combination with axitinib was investigated in KEYNOTE-426 (NCT02853331), a randomized, multicenter, open-label trial conducted in 861 patients who had not received systemic therapy for advanced RCC (Rini 2019). Patients were enrolled regardless of PD-L1 tumor expression status. Patients with active autoimmune disease requiring systemic immunosuppression within the last 2 years were ineligible. Randomization was stratified by International Metastatic RCC Database Consortium (IMDC) risk categories (favorable versus intermediate versus poor) and geographic region (North America versus Western Europe versus “Rest of the World”).

Rini et al (2019) stated the combination of pembrolizumab and axitinib showed antitumor activity in a phase 1b trial (Atkins 2018) involving patients with previously untreated advanced renal-cell carcinoma. Whether pembrolizumab plus axitinib would result in better outcomes than sunitinib in such patients was unclear. In an open-label, phase 3 trial, 861 patients with previously untreated advanced clear-cell renal-cell carcinoma were randomly assigned to receive pembrolizumab (200 mg) intravenously once every 3 weeks for up to 24 months plus axitinib (5 mg) orally twice daily (432 patients) or sunitinib (50 mg) orally once daily for the first 4 weeks of each 6-week cycle (429 patients). Patients who tolerated axitinib 5 mg twice daily for 2 consecutive cycles (6 weeks) could increase to 7 mg and then subsequently to 10 mg twice daily. Axitinib could be interrupted or reduced to 3 mg twice daily and subsequently to 2 mg twice daily to manage toxicity. Treatment with pembrolizumab and axitinib continued until RECIST v1.1-defined progression of disease or unacceptable toxicity. Administration of pembrolizumab and axitinib was permitted beyond RECIST-defined disease progression if the patient was clinically stable and considered to be deriving clinical benefit by the investigator. Assessment of tumor status was performed at baseline, after randomization at Week 12, then every 6 weeks thereafter until Week 54, and then every 12 weeks thereafter. The study population characteristics were: median age of 62 years (range: 26 to 90); 38% age 65 or older; 73% male; 79% White and 16% Asian; 19% and 80% of patients had a baseline KPS of 70 to 80 and 90 to 100, respectively; and patient distribution by IMDC risk categories was 31% favorable, 56% intermediate and 13% poor. The primary end points were overall survival (OS) and progression-free survival (PFS) in the intention-to-treat population. The key secondary end point was the objective response rate (ORR). All reported results are from the protocol-specified first interim analysis. After a median follow-up of 12.8 months, the estimated percentage of patients who were alive at 12 months (OS) was 89.9% in the pembrolizumab–axitinib group and 78.3%

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in the sunitinib group (hazard ratio for death, 0.53; 95% confidence interval [CI], 0.38 to 0.74; P<0.0001). Median PFS was 15.1 months in the pembrolizumab–axitinib group and 11.1 months in the

sunitinib group (hazard ratio for disease progression or death, 0.69; 95% CI, 0.57 to 0.84; P<0.001). The objective response rate (ORR) was 59.3% (95% CI, 54.5 to 63.9) in the pembrolizumab–axitinib group and 35.7% (95% CI, 31.1 to 40.4) in the sunitinib group (P<0.001). The benefit of pembrolizumab plus axitinib was observed across the International Metastatic Renal Cell Carcinoma Database Consortium risk groups (i.e., favorable, intermediate, and poor risk) and regardless of programmed death ligand 1 expression. Grade 3 or higher adverse events of any cause occurred in 75.8% of patients in the pembrolizumab–axitinib group and in 70.6% in the sunitinib group. The authors concluded that among patients with previously untreated advanced renal-cell carcinoma, treatment with pembrolizumab plus axitinib resulted in significantly longer overall survival and progression-free survival, as well as a higher objective response rate, than treatment with sunitinib.

Atkins et al (2018; NCT02853331) stated previous studies combining PD-1 checkpoint inhibitors with tyrosine kinase inhibitors of the VEGF pathway have been characterized by excess toxicity, precluding further development. The authors hypothesized that axitinib, a more selective VEGF inhibitor than others previously tested, could be combined safely with pembrolizumab (anti-PD-1) and yield antitumor activity in patients with treatment-naive advanced renal cell carcinoma. In this ongoing, open-label, phase 1b study, which was done at ten centers in the USA, patients aged 18 years or older who had advanced renal cell carcinoma (predominantly clear cell subtype) with their primary tumor resected, and at least one measurable lesion, Eastern Cooperative Oncology Group performance status 0-1, controlled hypertension, and no previous systemic therapy for renal cell carcinoma were enrolled. Eligible patients received axitinib plus pembrolizumab in a dose-finding phase to estimate the maximum tolerated dose, and additional patients were enrolled into a dose-expansion phase to further establish safety and determine preliminary efficacy. Axitinib 5 mg was administered orally twice per day with pembrolizumab 2 mg/kg given intravenously every 3 weeks. We assessed safety in all patients who received at least one dose of axitinib or pembrolizumab; antitumor activity was assessed in all patients who received study treatment and had an adequate baseline tumor assessment. The primary endpoint was investigator-assessed dose-limiting toxicity during the first two cycles (6 weeks) to estimate the maximum tolerated dose and recommended phase 2 dose. This study is registered with ClinicalTrials.gov, number NCT02133742. Between Sept 23, 2014, and March 25, 2015, we enrolled 11 patients with previously untreated advanced renal cell carcinoma to the dose-finding phase and between June 3, 2015, and Oct 13, 2015, we enrolled 41 patients to the dose-expansion phase. All 52 patients were analyzed together. No unexpected toxicities were observed. Three dose-limiting toxicities were reported in the 11 patients treated during the 6-week observation period (dose-finding phase): one patient had a transient ischemic attack and two patients were only able to complete less than 75% of the planned axitinib dose because of treatment-related toxicity. At the data cutoff date (March 31, 2017), 25 (48%) patients were still receiving study treatment. Grade 3 or worse treatment-related adverse events occurred in 34 (65%) patients; the most common included hypertension (n=12 [23%]), diarrhea (n=5 [10%]), fatigue (n=5 [10%]), and increased alanine aminotransferase concentration (n=4 [8%]). The most common potentially immune-related adverse events (probably related to pembrolizumab) included diarrhea (n=15 [29%]), increased alanine aminotransferase concentration (n=9 [17%]) or aspartate

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diarrhea (n=15 [29%]), increased alanine aminotransferase concentration (n=9 [17%]) or aspartate aminotransferase concentration (n=7 [13%]), hypothyroidism (n=7 [13%]), and fatigue (n=6 [12%]). 28 (54%) patients had treatment-related serious adverse events. At data cutoff, 38 (73%; 95% CI 59·0-84·4) patients achieved an objective response (complete or partial response). The treatment combination of axitinib plus pembrolizumab is tolerable and shows promising antitumor activity in patients with treatment-naive advanced renal cell carcinoma. Whether or not the combination works better than a sequence of VEGF pathway inhibition followed by an anti-PD-1 therapy awaits the completion of a phase 3 trial comparing axitinib plus pembrolizumab with sunitinib monotherapy.

On August 11, 2021, the U.S. Food and Drug Administration (FDA) approved the combination of Keytruda plus Lenvima for the first-line treatment of adult patients with advanced renal cell carcinoma (RCC). The approval based on supporting data from the pivotal Phase 3 CLEAR (Study 307)/KEYNOTE-581 trial (Merck, 2021b).

On November 18, 2021, the U.S. Food and Drug Administration (FDA) approved Keytruda for the adjuvant treatment of individuals with renal cell carcinoma (RCC) at intermediate-high or high risk of recurrence following nephrectomy, or following nephrectomy and resection of metastatic lesions. The FDA approval was based on supporting data from the KEYNOTE-564 study, a multicenter, randomized (1:1), double-blind, placebo-controlled, pivotal phase 3 trial. In this study, Keytruda demonstrated a statistically significant improvement in disease-free survival in individuals with RCC at intermediate-high or high risk of recurrence following nephrectomy, or following nephrectomy and resection of metastatic lesions compared with placebo. Trial continuation will assess overall survival as a secondary endpoint measure (Merck, 2021c).

Salivary Tumors

Alame et al (2020) state salivary duct carcinoma (SDC) is a rare and aggressive salivary gland cancer subtype with poor prognosis. The mutational landscape of SDC has already been the object of several studies, however little is known regarding the functional genomics and the tumor microenvironment despite their importance in oncology. This investigation was aimed at describing both the functional genomics of SDC and the SDC microenvironment, along with their clinical relevance. RNA-sequencing (24 tumors), proteomics (17 tumors), immunohistochemistry (22 tumors), and multiplexed immunofluorescence (3 tumors) data were obtained from three different patient cohorts and analyzed by digital imaging and bioinformatics. Adjacent non-tumoral tissue from patients in two cohorts were used in transcriptomic and proteomic analyses. Transcriptomic and proteomic data revealed the importance of Notch, TGF-β, and interferon-γ signaling for all SDCs. The authors confirmed an overall strong desmoplastic reaction by measuring α-SMA abundance, the level of which was associated with recurrence-free survival (RFS). Two distinct immune phenotypes were observed: immune-poor SDCs (36%) and immune-infiltrated SDCs (64%). Advanced bioinformatics analysis of the transcriptomic data suggested 72 ligand-receptor interactions occurred in the microenvironment and correlated with the immune phenotype. Among these interactions, three immune checkpoints were validated by immunofluorescence, including CTLA-4/DC86 and TIM-3/galectin-9 interactions, previously

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unidentified in SDC. Immunofluorescence analysis also confirmed an important immunosuppressive role of macrophages and NK cells, also supported by the transcriptomic data. The authors concluded that

this data significantly increase the understanding of SDC biology and open new perspectives for SDC tumor treatment. Before applying immunotherapy, patient stratification according to the immune infiltrate should be taken into account. Immune-infiltrated SDC could benefit from immune checkpoint-targeting therapy, with novel options such as anti-CTLA-4. Macrophages or NK cells could also be targeted. The dense stroma, i.e., fibroblasts or hyaluronic acid, may also be the focus for immune-poor SDC therapies, e.g. in combination with Notch or TGF-β inhibitors, or molecules targeting SDC mutations. The authors note that limited data on advanced salivary gland carcinoma and SDC response to anti-PD-1 monotherapy indicate a low response rate (11% with pembrolizumab, KEYNOTE-028 trial). This modest outcome contrasts results with other tumors such as melanoma, where anti-PD-1 monotherapies were much more successful.

Cohen 2018 (KEYNOTE-028; NCT02054806) stated that treatment options for patients with unresectable or metastatic salivary gland carcinoma (SGC) are limited. Safety and efficacy of pembrolizumab for SGC expressing programmed death ligand 1 (PD-L1) were explored in this trial. A cohort of patients with advanced, PD-L1-positive SGC was enrolled in the nonrandomized, multicohort, phase Ib trial of pembrolizumab in patients with PD-L1-positive advanced solid tumors. Key inclusion criteria included recurrent or metastatic disease, failure of prior systemic therapy, and PD-L1 expression on ≥1% of tumor or stroma cells (per a prototype immunohistochemistry assay). Patients received pembrolizumab 10 mg/kg every 2 weeks for ≥2 years or until confirmed disease progression or unacceptable toxicity. Primary end point was objective response rate per Response Evaluation Criteria in Solid Tumors version 1.1 by investigator review. Twenty-six patients with PD-L1-positive SGC were enrolled and treated; median age was 57 years, 88% were men, and 74% had received prior therapy for recurrent/metastatic disease. Confirmed objective response rate after median follow-up of 20 months was 12% (95% confidence interval, 2%-30%), with 3 patients achieving partial response; there were no complete responses. Median duration of response was 4 months (range, 4 to 21 mo). Treatment-related adverse events occurred in 22 patients (85%), resulting in discontinuation in 2 patients and death in 1 (interstitial lung disease); those occurring in ≥15% of patients were diarrhea, decreased appetite, pruritus, and fatigue. The authors concluded that pembrolizumab demonstrated promising antitumor activity and a manageable safety profile in patients with advanced, PD-L1-positive SGC.

Small Cell Lung Cancer (SCLC)

On March 1, 2021, Merck voluntarily withdrew the U.S. indication for Keytruda (pembrolizumab) for the treatment of patients with metastatic small cell lung cancer (SCLC) with disease progression on after platinum-based chemotherapy and at least one other prior line of therapy. This indication withdrawal was performed in consultation with the U.S. Food and Drug Administration (FDA), but however, did not affect other indications for Keytruda. Initially, accelerated approval for Keytruda for the treatment of metastatic SCLC was previously granted in June 2019 based on supporting data from the KEYNOTE-158 and KEYNOTE-028 studies, However, the approval was contingent upon finishing the post-

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marketing requirement showing superiority of Keytruda based on overall survival (OS). The

confirmatory Phase 3 trial, KEYNOTE-604, did not show Keytruda achieving statistical significance for another primary endpoint of OS (Merck, 2021h).

Solid Tumors - Microsatellite Instability-High (MSI-H) or Mismatch Repair Deficient (dMMR)

The U.S. Food and Drug Administration granted accelerated approval of pembrolizumab for the treatment of adult and pediatric patients with unresectable or metastatic microsatellite instability-high (MSI-H) or mismatch repair deficient (dMMR) solid tumors (Merck, 2017). This indication covers patients with solid tumors that have progressed following prior treatment and who have no satisfactory alternative treatment options.

MSI-H and dMMR tumors contain abnormalities that affect the proper repair of DNA inside the cell (Merck, 2017). Tumors with these biomarkers are most commonly found in colorectal, endometrial and gastrointestinal cancers, but also less commonly appear in cancers arising in the breast, prostate, bladder, thyroid gland and other places. Approximately 5 percent of patients with metastatic colorectal cancer have MSI-H or dMMR tumors.

Pembrolizumab was approved for this indication using the Accelerated Approval pathway, under which the FDA may approve drugs for serious conditions where there is unmet medical need and a drug is shown to have certain effects that are reasonably likely to predict a clinical benefit to patients (Merck, 2017). The FDA has stated that further study is required to verify and describe anticipated clinical benefits of pembrolizumab, and the manufacturer is currently conducting these studies in additional patients with MSI-H or dMMR tumors.

The safety and efficacy of pembrolizumab for this indication were studied in patients with MSI-H or dMMR solid tumors enrolled in one of five uncontrolled, single-arm clinical trials (Merck, 2017). In some trials, patients were required to have MSI-H or dMMR cancers, while in other trials, a subgroup of patients were identified as having MSI-H or dMMR cancers by testing tumor samples after treatment began. A total of 15 cancer types were identified among 149 patients enrolled across these five clinical trials. The most common cancers were colorectal, endometrial and other gastrointestinal cancers. The review of pembrolizumab for this indication was based on the percentage of patients who experienced complete or partial shrinkage of their tumors (overall response rate) and for how long (durability of response). Of the 149 patients who received pembrolizumab in the trials, 39.6 percent had a complete or partial response. For 78 percent of those patients, the response lasted for six months or more.

The safety and effectiveness of pembrolizumab in pediatric patients with MSI-H central nervous system cancers have not been established (Merck, 2017).

The recommended dose of pembrolizumab in adults with MSI-H or dMMR solid tumors is 200 mg

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administered as an intravenous infusion over 30 minutes every three weeks until disease progression, unacceptable toxicity, or up to 24 months in patients without disease progression (Merck, 2017). In

children, the recommended dose of pembrolizumab is 2 mg/kg (up to a maximum of 200 mg) administered as an intravenous infusion over 30 minutes every three weeks until disease progression or unacceptable toxicity, or up to 24 months in patients without disease progression.

In a multi-center phase 1 clinical trial, Brahmer et al (2012) administered intravenous anti-PD-L1 antibody (at escalating doses ranging from 0.3 to 10 mg/kg body weight) to patients with selected advanced cancers. Anti-PD-L1 antibody was administered every 14 days in 6-week cycles for up to 16 cycles or until the patient had a complete response (CR) or confirmed disease progression. As of February 24, 2012, a total of 207 patients -- 75 with non-small-cell lung cancer (NSCLC)r, 55 with melanoma, 18 with colorectal cancer, 17 with renal-cell cancer, 17 with ovarian cancer, 14 with pancreatic cancer, 7 with gastric cancer, and 4 with breast cancer -- had received anti-PD-L1 antibody. The median duration of therapy was 12 weeks (range of 2 to 111). Grade 3 or 4 toxic effects that investigators considered to be related to treatment occurred in 9 % of patients. Among patients with a response that could be evaluated, an objective response (CR or partial response [PR]) was observed in 9 of 52 patients with melanoma, 2 of 17 with renal-cell cancer, 5 of 49 with NSCLC, and 1 of 17 with ovarian cancer. Responses lasted for 1 year or more in 8 of 16 patients with at least 1 year of follow-up. The authors concluded that antibody-mediated blockade of PD-L1 induced durable tumor regression (ORR of 6 to 17 %) and prolonged stabilization of disease (rates of 12 to 41 % at 24 weeks) in patients with advanced cancers, including NSCLC, melanoma, and renal-cell cancer.

Urothelial Carcinoma (Transitional Cell Carcinoma)

Bladder cancer is the most common malignancy involving the urinary system. Urothelial (formerly called transitional cell) carcinoma is the predominant histologic type in the United States and Europe, where it accounts for approximately 90 percent of all bladder cancers. Less commonly, urothelial cancers can arise in the renal pelvis, ureter, or urethra. Urothelial bladder cancer can present as non-muscle invasive, muscle invasive, and metastatic disease. The extent of disease reflects its natural history and determines treatment and prognosis. Approximately 70 percent of new urothelial bladder cancer cases are classified as non-muscle invasive. Non-muscle invasive bladder cancer includes Ta, T1 (submucosal invasive) tumors, and Tis (carcinoma in situ [CIS]), which account for approximately 70%, 20%, and 10% of non-muscle invasive cancers, respectively (Black, 2022).

On May 23, 2017, the FDA approved pembrolizumab for the treatment of patients with locally advanced or metastatic urothelial carcinoma who are ineligible for cisplatin-containing chemotherapy. This indication is approved under accelerated approval based on tumor response rate and duration of response. Continued approval for this indication may be contingent upon verification and description of clinical benefit in the confirmatory trials.

In the second-line setting, pembrolizumab is approved for the treatment of patients with locally

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advanced or metastatic urothelial carcinoma who have disease progression during or following

platinum-containing chemotherapy or within 12 months of neoadjuvant or adjuvant treatment with platinum-containing chemotherapy.

Subsequently, in June 2018, the FDA added PD-L1 status into the label by stating pembrolizumab is indicated for the treatment of patients with locally advanced or metastatic urothelial carcinoma who are not eligible for cisplatin-containing chemotherapy and whose tumors express PD-L1 [Combined Positive Score (CPS) ≥10], or in patients who are not eligible for any platinum-containing chemotherapy regardless of PD-L1 status. The FDA decision was based on an assessment conducted by a data monitoring committee (DMC) for the phase III KEYNOTE-361 study. The DMC identified that patients with PD-L1–low status had decreased overall survival in the single-agent immunotherapy arms compared with chemotherapy.

Pembrolizumab is approved for use in these indications at a fixed dose of 200 mg every three weeks until disease progression or unacceptable toxicity, or up to 24 months in patients without disease progression.

The first-line approval was based on data from a multicenter, open-label, single-arm trial, KEYNOTE-052, investigating pembrolizumab in 370 patients with locally advanced or metastatic urothelial carcinoma who were not eligible for cisplatin-containing chemotherapy. Patients with autoimmune disease or medical conditions that required systemic corticosteroids or other immunosuppressive medication were excluded from the trial. Patients received pembrolizumab at a dose of 200 mg every three weeks until unacceptable toxicity or disease progression; patients without disease progression could be treated for up to 24 months. The major efficacy outcome measures were objective response rate (ORR), according to the Response Evaluation Criteria In Solid Tumors (RECIST) 1.1, as assessed by independent radiology review, and duration of response.

The efficacy analysis showed an ORR of 29 percent (95% CI: 24, 34), with a complete response rate of 7 percent and a partial response rate of 22 percent. The median duration of response had not been reached (range: 1.4+ to 17.8+ months) at the time of FDA approval. The median follow-up time was 7.8 months.

In KEYNOTE-052, pembrolizumab was discontinued due to adverse reactions in 11 percent of patients. The most common adverse reactions (in ≥ 20% of patients) were fatigue (38%), musculoskeletal pain (24%), decreased appetite (22%), constipation (21%), rash (21%) and diarrhea (20%). Eighteen patients (5%) died from causes other than disease progression. Five patients (1.4%) who were treated with pembrolizumab experienced sepsis which led to death, and three patients (0.8%) experienced pneumonia which led to death. Adverse reactions leading to interruption of pembrolizumab occurred in 22 percent of patients; the most common (≥1%) were liver enzyme increase, diarrhea, urinary tract infection, acute kidney injury, fatigue, joint pain, and pneumonia. Serious adverse reactions occurred in 42 percent of patients, the most frequent (≥2%) of which were urinary tract infection, hematuria, acute

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42 percent of patients, the most frequent (≥2%) of which were urinary tract infection, hematuria, acute kidney injury, pneumonia, and urosepsis.

The second-line approval was based on data from a multicenter, randomized, active-controlled trial, KEYNOTE-045, investigating pembrolizumab in patients with locally advanced or metastatic urothelial carcinoma with disease progression on or after platinum-containing chemotherapy. Patients with autoimmune disease or a medical condition that required immunosuppression were excluded from the trial. Patients were randomized to receive either pembrolizumab 200 mg every three weeks (n=270) or investigator's choice of any of the following chemotherapy regimens, all given intravenously, every three weeks (n=272): paclitaxel 175 mg/m2 (n=84), docetaxel 75 mg/m2 (n=84), or vinflunine 320 mg/m2 (n=87). Treatment continued until unacceptable toxicity or disease progression; patients without disease progression could be treated for up to 24 months. The major efficacy outcomes were overall survival (OS) and progression-free survival (PFS), as assessed by a blinded independent central review (BICR) per RECIST 1.1; additional efficacy outcome measures were ORR, as assessed by BICR per RECIST 1.1, and duration of response.

Pembrolizumab demonstrated superior OS compared to chemotherapy. Findings demonstrated that pembrolizumab resulted in a 27 percent reduction in the risk of death compared to chemotherapy – with 155 events (57%) observed in the pembrolizumab arm, compared to 179 events (66%) in the chemotherapy arm (HR, 0.73 [95% CI: 0.59, 0.91], p=0.004); the median OS was 10.3 months (95% CI: 8.0, 11.8) in the pembrolizumab arm, compared to 7.4 months (95% CI: 6.1, 8.3) in the chemotherapy arm. In October 2016, the study was stopped early at the recommendation of an independent Data Monitoring Committee following an interim analysis that showed pembrolizumab met the superiority thresholds for OS in the overall study population.

There was no statistically significant difference between pembrolizumab and chemotherapy with respect to progression-free survival (PFS). There were 218 events (81%) observed in the pembrolizumab arm, compared to 219 events (81%) in the chemotherapy arm (HR, 0.98 [95% CI: 0.81, 1.19], p=0.833). The median PFS was 2.1 months (95% CI: 2.0, 2.2) in the pembrolizumab arm, compared to 3.3 months (95% CI: 2.3, 3.5) in the chemotherapy arm.

Analysis of the ORR endpoint showed a statistically significant improvement with pembrolizumab, as compared to chemotherapy. The ORR was 21 percent (95% CI: 16, 27) in the pembrolizumab arm (with a complete response rate of 7 percent and a partial response rate of 14 percent), compared to 11 percent (95% CI: 8, 16) in the chemotherapy arm (with a complete response rate of 3 percent and a partial response rate of 8 percent) (p=0.002). At the time of FDA approval, the median duration of response for patients treated with pembrolizumab had not yet been reached (range: 1.6+ to 15.6+ months), compared to 4.3 months (range: 1.4+ to 15.4+ months) in the chemotherapy arm. The median follow-up time for this trial was 9.0 months.

In KEYNOTE-045, pembrolizumab was discontinued due to adverse reactions in eight percent of patients. The most common adverse reaction resulting in permanent discontinuation of pembrolizumab was pneumonitis (1.9%). Adverse reactions leading to interruption of pembrolizumab occurred in 20

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was pneumonitis (1.9%). Adverse reactions leading to interruption of pembrolizumab occurred in 20 percent of patients; the most common (≥1%) were urinary tract infection (1.5%), diarrhea (1.5%), and colitis (1.1%). The most common adverse reactions (≥20%) in patients who received pembrolizumab versus those who received chemotherapy were fatigue (38% vs 56%), musculoskeletal pain (32% vs 27%), pruritus (23% vs 6%), decreased appetite (21% vs 21%), nausea (21% vs 29%) and rash (20% vs 13%). Serious adverse reactions occurred in 39 percent of pembrolizumab-treated patients, the most frequent (≥2%) of which were urinary tract infection, pneumonia, anemia, and pneumonitis.

KEYNOTE-361 (NCT02853305) is an ongoing, multicenter, randomized study in previously untreated patients with metastatic urothelial carcinoma who are eligible for platinum-containing chemotherapy. The study compares KEYTRUDA with or without platinum-based chemotherapy (i.e., cisplatin or carboplatin with gemcitabine) to platinum-based chemotherapy alone. The trial also enrolled a third arm of monotherapy with KEYTRUDA to compare to platinum-based chemotherapy alone. The independent Data Monitoring Committee (iDMC) for the study conducted a review of early data and found that in patients classified as having low PD-L1 expression (CPS <10), those treated with KEYTRUDA monotherapy had decreased survival compared to those who received platinum-based chemotherapy. The iDMC recommended to stop further accrual of patients with low PD-L1 expression in the monotherapy arm, however, no other changes were recommended, including any change of therapy for patients who had already been randomized to and were receiving treatment in the monotherapy arm.

On January 8, 2020, the FDA approved pembrolizumab (Keytruda, Merck & Co. Inc.) for the treatment of patients with Bacillus Calmette-Guerin (BCG)-unresponsive, high-risk, non-muscle invasive bladder cancer (NMIBC) with carcinoma in situ (CIS) with or without papillary tumors who are ineligible for or have elected not to undergo cystectomy. Efficacy was investigated in KEYNOTE-057 (NCT02625961), a multicenter, open-label, single-arm trial that enrolled 148 patients with high-risk NMIBC, 96 of whom had BCG-unresponsive high-risk, nonmuscle invasive bladder cancer (NMIBC) with carcinoma in situ (CIS) with or without papillary tumors who were ineligible for or had elected not to undergo cystectomy. BCG-unresponsive high-risk NMIBC was defined as persistent disease despite adequate BCG therapy, disease recurrence after an initial tumor-free state following adequate BCG therapy, or T1 disease following a single induction course of BCG. Adequate BCG therapy was defined as administration of at least five of six doses of an initial induction course plus either of: at least two of three doses of maintenance therapy or at least two of six doses of a second induction course. Prior to treatment, all patients had undergone transurethral resection of bladder tumor (TURBT) to remove all resectable disease (Ta and T1 components). Residual CIS (Tis components) not amenable to complete resection was allowed. The trial excluded patients with muscle invasive (i.e., T2, T3, T4) locally advanced non-resectable or metastatic urothelial carcinoma, concurrent extra-vesical (i.e., urethra, ureter or renal pelvis) non-muscle invasive transitional cell carcinoma of the urothelium, or autoimmune disease or a medical condition that required immunosuppression.

Patients received pembrolizumab 200 mg every 3 weeks until unacceptable toxicity, persistent or recurrent high-risk NMIBC or progressive disease. Assessment of tumor status was performed every 12 weeks for two years and then every 24 weeks for three years, and patients without disease progression could be treated for up to 24 months. The major efficacy outcome measures were complete response (as

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could be treated for up to 24 months. The major efficacy outcome measures were complete response (as defined by negative results for cystoscopy [with TURBT/biopsies as applicable], urine cytology, and computed tomography urography [CTU] imaging) and duration of response. The study population characteristics were: median age of 73 years (range: 44 to 92); 44% age ≥75; 84% male; 67% White; and 73% and 27% with an ECOG performance status of 0 or 1, respectively. Tumor pattern at study entry was CIS with T1 (13%), CIS with high grade TA (25%), and CIS (63%). Baseline high-risk NMIBC disease status was 27% persistent and 73% recurrent. The median number of prior instillations of BCG was 12. The median follow-up time was 28.0 months (range: 4.6 to 40.5 months).

The complete response rate in the 96 patients with high-risk BCG-unresponsive NMIBC with CIS was 41% (95% CI: 31, 51) and median response duration was 16.2 months (0.0+, 30.4+). Forty-six percent (46%) of responding patients experienced a complete response lasting at least 12 months.

The most common adverse reactions (incidence ≥10%) in patients who received pembrolizumab in KEYNOTE-057 were fatigue, diarrhea, rash, pruritis, musculoskeletal pain, hematuria, cough, arthralgia, nausea, constipation, urinary tract infection, peripheral edema, hypothyroidism, and nasopharyngitis.

On April 3, 2023, the U.S. Food and Drug Administration (FDA) granted accelerated approval to enfortumab vedotin-ejfv (Padcev) with pembrolizumab (Keytruda) for the treatment of adult patients with locally advanced or metastatic urothelial carcinoma who are ineligible for cisplatin-containing chemotherapy. This FDA approval was based on efficacy results from the EV-103/KEYNOTE-869 study (FDA, 2023e).

In the EV-103/KEYNOTE-869 study, an open-label, multi-cohort (dose escalation cohort, Cohort A, Cohort K) study, investigators evaluated the efficacy of Padcev in combination with pembrolizumab in in patients with locally advanced or metastatic urothelial cancer who were ineligible for cisplatin-containing chemotherapy and received no prior systemic therapy for locally advanced or metastatic disease. A total of 121 patients received Padcev plus pembrolizumab. Patients in the dose escalation cohort (n=5), Cohort A (n=40), Cohort K (n=76) received Padcev 1.25 mg/kg as an IV infusion over 30 minutes on Days 1 and 8 of a 21-day cycle followed by pembrolizumab 200 mg as an IV infusion on Day 1 of a 21-day cycle approximately 30 minutes after Padcev. Treatment continued until disease progression or unacceptable toxicity. The major efficacy outcome measures were objective response rate (ORR) and duration of response (DoR). The confirmed ORR in 121 patients was 68% (95% CI: 59, 76), including 12% with complete responses. The median DoR for the dose escalation cohort + Cohort A was 22 months (range: 1+ to 46+) and for Cohort K was not reached (range: 1 to 24+) (FDA, 2023e; Merck, 2023).

Vulvar Cancer

Shields and Gordinier (2018) stated advanced vulvar cancer is associated with a very poor prognosis. Surgical resection is the mainstay of treatment, with radiation indicated for areas at high risk for

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recurrence. When surgical and radiation options have been exhausted, the effectiveness of systemic chemotherapy is poor. No biologic or targeted agents have been approved for the management of

advanced or recurrent vulvar cancer. Pembrolizumab, a humanized monoclonal antibody against programmed death 1 (PD-1), has been successfully used as a target of tumor immune therapy in small cell lung cancer and melanoma. The authors present the first case in the literature of a patient with recurrent vulvar cancer who was treated successfully with pembrolizumab. Caris next-generation testing revealed a PD-L1 and PD-1 mutation (PD-L1 positive, 2+, 100%). She attained a complete clinical remission after 2 cycles, and a CT scan after 6 cycles revealed a significant response by RECIST criteria. After completing 10 cycles, treatment was stopped due to complications of severe malnutrition related to narcotic abuse. A CT scan 10 weeks after the final treatment revealed no adenopathy. The authors concluded that pembrolizumab is a safe and effective chemotherapeutic agent to treat recurrent vulvar carcinoma.

Combined Pembrolizumab and Acalabrutinib for the Treatment of Bladder Cancer

Zhang and colleagues (2020) stated that inhibition of the PD-1 pathway has demonstrated clinical benefit in metastatic urothelial cancer (mUC); however, response rates of 15 % to 26 % highlight the need for more effective therapies. Bruton tyrosine kinase (BTK) inhibition may suppress myeloid-derived suppressor cells (MDSCs) and improve T-cell activation. The randomized phase-II clinical trial of acalabrutinib and pembrolizumab immunotherapy dual checkpoint inhibition in platinum-resistant metastatic urothelial carcinoma (RAPID CHECK; also known as ACE-ST-005) was a randomized phase-II clinical trial evaluating pembrolizumab with or without acalabrutinib for patients with platinum-refractory mUC. The primary objectives were safety and ORRs according to the RECIST, version 1.1; secondary endpoints included PFS and OS. Immune profiling was carried out to analyze circulating monocytic MDSCs and T cells. A total of 75 patients were treated with pembrolizumab (n = 35) or pembrolizumab plus acalabrutinib (n = 40). The ORR was 26 % with pembrolizumab (9 % with a CR) and 20 % with pembrolizumab plus acalabrutinib (10 % with a CR). The grade-3/4 AEs that occurred in greater than or equal to 15 % of the patients were anemia (20 %) with pembrolizumab and fatigue (23 %), increased alanine aminotransferase (23 %), urinary tract infections (18 %), and anemia (18 %) with pembrolizumab plus acalabrutinib. One patient treated with pembrolizumab plus acalabrutinib had high MDSCs at the baseline, which significantly decreased at week 7. Overall, MDSCs were not correlated with a clinical response, but some subsets of CD8+ T cells did increase during the combination treatment. The authors concluded that both treatments were generally well-tolerated, although serious AE rates were higher with the combination. Acalabrutinib plus pembrolizumab did not improve the ORR, PFS, or OS in comparison with pembrolizumab alone in mUC. Baseline and on-treatment peripheral monocytic MDSCs were not different in the treatment cohorts. Proliferating CD8+ T-cell subsets increased during treatment, especially in the combination cohort. These researchers stated that ongoing studies are correlating these peripheral immunome findings with tissue-based immune cell infiltration; and future studies are needed to carefully subtype T cells and MDSCs to examine their clinical utility as circulating, real-time biomarkers for monitoring patients treated with immune checkpoint inhibitors.

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Combined Pembrolizumab and Pelvic Chemoradiation for the Treatment of Cervical Cancer

Duska and colleagues (2020) noted that immune checkpoint inhibitors are being considered for locally advanced cervical cancer (LACC) together with standard-of-care pelvic chemoradiation (CRT). However, the safety of the combination and its optimal schedule are unknown. Defining the safety of the combination is a primary objective of a study examining concurrent and sequential schedules. These investigators presented a safety analysis that was fully accrued and met reporting requirements. Pembrolizumab was administered following CRT (arm 1) or during CRT (arm 2) according to a randomized phase-II design. Patients who were 18 years old or older and had LACC (stages IB to IVA according to the 2009 International Federation of Gynecology and Obstetrics system) were randomized 1:1 to the treatment regimens. The CRT was identical in the 2 arms. Pembrolizumab was administered every 3 weeks for 3 doses; no maintenance was allowed. All patients receiving any treatment were evaluated for safety. Safety assessments included the incidence and severity of AEs and the occurrence of protocol-defined dose-limiting toxicity (DLT) through 30 days after the last pembrolizumab infusion. As of August 2019, 52 of the 88 planned patients had completed treatment and were evaluable for toxicity. Treatment-related grade-2 or higher toxicity was experienced by 88 %; 11 had at least 1 grade-4 AE, and another 23 had at least 1 grade-3 AE. Grade-1 or higher diarrhea was reported in 34 patients (65 %; 50 % of these were grade-1), and there was no difference between arms (63 % in arm 1 versus 68 % in arm 2). Two patients experienced 3 DLTs. Most patients completed cisplatin (100 % in arm 1 versus 82 % in arm 2); 83 % in both arms completed all pembrolizumab. The authors concluded that these preliminary results support the safety and feasibility of adding pembrolizumab to pelvic CRT concurrently or sequentially. These researchers stated that future studies of the addition of immunotherapy to traditional chemoradiation are planned to determine the best way to deliver the treatment and whether any improvement is observed with the addition of immunotherapy to traditional therapy.

Combined Pembrolizumab and Radiotherapy for the Treatment of Triple-Negative Breast Cancer

In a single-arm, Simon 2-stage, phase-II clinical trial, Ho and colleagues (2020) examined the safety and efficacy of pembrolizumab-mediated programmed cell death protein 1 inhibition plus RT in patients with metastatic TNBC who were unselected for programmed death-ligand 1 expression. This study enrolled a total of 17 patients with a median age of 52 years (range of 37 to 73 years). An RT dose of 3,000 centi-grays (cGy) was delivered in 5 daily fractions. Pembrolizumab was given intravenously at a dose of 200 mg within 3 days of the 1st RT fraction, and then every 3 weeks ± 3 days until disease progression. The median follow-up was 34.5 weeks (range of 2.1 to 108.3 weeks). The primary endpoint was the ORR at week 13 in patients with unirradiated lesions measured using RECIST; version 1.1. Secondary endpoints included safety and PFS. Exploratory objectives were to identify biomarkers predictive of ORR and PFS. The ORR for the entire cohort was 17.6 % (3 of 17; 95 % CI: 4.7 % to 44.2 %), with 3 CRs, 1 case of SD, and 13 cases of progressive disease; 8 patients died prior to week 13 due to disease progression. Among the 9 women assessed using RECIST version 1.1 at week 13, 3 (33 %) achieved a CR, with a 100 % reduction in tumor volume outside of the irradiated portal. The CRs were

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durable for 18 weeks, 20 weeks, and 108 weeks, respectively. The most common grade-1 to grade-2 toxicity (assessed according to the National Cancer Institute Common Terminology Criteria for Adverse

Events, version 4.0) was dermatitis (29 %); 4 grade-3 AEs were attributed to pembrolizumab: fatigue, lymphopenia, and infection. No o grade 4 AEs or treatment-related deaths were reported. The authors concluded that the combination of pembrolizumab and RT was found to be safe and demonstrated encouraging activity in patients with poor-prognosis, metastatic, TNBC who were unselected for programmed death-ligand 1 expression. These researchers stated that these findings will inform the design of future trials of immune checkpoint inhibitor and RT combinations in patients with immunotherapy-refractory breast cancer; and larger clinical trials of checkpoint blockade plus RT with predictive biomarkers of response are needed.

The authors concluded that the concurrent administration of pembrolizumab and RT precluded the ability to determine the specific contribution of RT to checkpoint blockade in the current study. Future study designs incorporating serial research biopsies will permit the examination of conventional biomarkers as potential predictors of response to pembrolizumab alone versus pembrolizumab and RT. The small sample size constrained the analysis of factors such as the prior number of cytotoxic treatments, organ involvement, volume of liver metastases, and size and location of the target lesion with treatment response. Heterogeneity in tumor immunologic phenotypes, differential expression of PD-L1 and tumor-infiltrating lymphocytes between biopsied tumor sites, and the receipt of prior chemotherapy and RT also may have confounded the interpretation of treatment response. This was evidenced by the discrepancy between the 100 % responses observed in unirradiated tumors in the responders but the overall reductions in tumor burden being less than 100 % due to persistent tumors in the irradiated field. Biomarkers of response to treatment will be reported separately, with the caveat that the small number of responders made it unlikely that specific associations with treatment response will be observed.

Combined Selinexor and Pembrolizumab for the Treatment of Metastatic Non-Small Cell Lung Cancer

Altan et al (2023) noted that in lung cancer, over-expression of nuclear export proteins can result in inactivation of critical tumor suppressor proteins and cell-cycle regulators. Selective suppression of nuclear export proteins has immunomodulatory activities. In a prospective study, these researchers presented clinical safety and early effectiveness data on the combination of pembrolizumab and an oral selective nuclear export inhibitor, selinexor, for the treatment of metastatic NSCLC (mNSCLC). The primary objective of this investigator-initiated trial was to determine the safety and tolerability of selinexor in combination with pembrolizumab in patients with mNSCLC. Secondary objectives included determination of objective tumor response rate, disease control rate (DCR), and PFS duration. A total of 17 patients were included in the final analysis; 15 (88 %) received more than 2 lines of prior systemic therapy and 10 (59 %) had prior exposure to anti-PD-1/PD-L1 therapy. The median age was 67.5 years; 10 patients had grade-3 or higher AEs related to selinexor treatment. Responses to treatment occurred in patients who did and did not undergo previous anti-PD-1/PD-L1 therapy and in patients with

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activating driver mutations. The median OS and PFS were 11.4 months (95 % CI: 3.4 to 19.8 months) and 3.0 months (95 % CI: 1.7 to 5.7 months), respectively. The ORR was 18 % and the 6-month DCR

was 24 %. The authors concluded that selinexor in combination with pembrolizumab showed promising anti-tumor activity in patients with mNSCLC, including those who had previously received anti-PD-1/PD-L1 therapy. The therapy-related toxic effects were consistent with the prior safety data for both drugs, and no overlapping toxic effects were observed.

Combined Niraparib and Pembrolizumab for the Treatment of Locally Advanced/Metastatic Non-Small Cell Lung Cancer

Ramalingam et al (2022) stated that poly(ADP-ribose) polymerase (PARP) inhibitors may synergize with PD-1 inhibitors to enhance adaptive and innate anti-tumor immune responses. In the phase-II JASPER Trial, the PARP inhibitor niraparib was evaluated in combination with pembrolizumab in patients with metastatic and/or locally advanced NSCLC. Patients whose tumors had PD-L1 TPS of 50 % or higher (cohort 1) or 1 % to 49 % (cohort 2) received 1st-line niraparib (200 mg once-daily) plus pembrolizumab (200 mg every 3 weeks). The primary endpoint was investigator-assessed ORR; and secondary endpoints included DOR, PFS, OS, safety, and pharmacokinetics. A total of 38 patients were enrolled in cohorts 1 and 2. In cohort 1, ORR (95 % CI was 56.3 % (9 of 16 patients; 29.9 % to 80.2 %); 2 of 16 patients had CRs and 7 of 16 had PRs. In cohort 2, ORR was 20.0 % (5.7 % to 43.7 %) with 4 of 20 PRs. In cohorts 1 and 2, the median DOR was 19.7 months (95 % CI: 4.2 months to not estimable [NE]) and 9.4 months (95 % CI: 4.2 months to NE), the median PFS was 8.4 months (95 % CI: 3.9 to 22.1 months) and 4.2 months (95 % CI, 2.0 to 6.2 months), and the median OS was NE (95 % CI: 6.0 months to NE) and 7.7 months (95 % CI: 4.0 to 12.5 months), respectively. Grade-3 or higher TRAEs occurred in 88.2 % and 85.7 % of patients in cohorts 1 and 2, respectively. Safety was consistent with known profiles of single-agent niraparib and pembrolizumab. The authors concluded that niraparib plus pembrolizumab showed clinical activity in patients with advanced and/or metastatic NSCLC. Moreover, these researchers stated that despite the small number of patients in this study (n = 38), these findings suggested that niraparib plus pembrolizumab may be an active combination with no new safety signals and support further evaluation of this novel combination approach in the treatment of advanced NSCLC. These investigators stated that the combination of niraparib plus pembrolizumab is being studied as maintenance therapy in patients with advanced NSCLC who had stable disease or response to pembrolizumab plus platinum-based 1st-line induction chemotherapy in the ongoing phase-III ZEAL-1L Trial.

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